

Reconstruction-independent moment tests for quantum-computed dynamical spectra

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Abstract

We test spectral functions $A(\omega)$ reconstructed in quantum-sampled subspaces by comparing their moments with operator expectation values on the prepared ground state, never read off $A(\omega)$. A mismatch beyond a preset threshold rejects, under an assumed estimator-bias budget; a pass certifies nothing. In a self-attested, hash-sealed blinded simulation with synthetic errors of known truth (current probe of one doped $L = 6$ Hubbard ring; oracle estimate: exact moment plus Gaussian noise), the pre-registered primary endpoint was 53.7% detection (101/188, including all 56 subspace truncations and none of 71 under-converged Krylov reconstructions) at a false-positive rate of 0.9% (1/112) that is low by construction; two sealed secondary predictions failed. With m_0 – m_2 , guaranteed power against weight moved among four or more fixed poles is zero. On IBM Heron the device supplied only the sampled support; no moment was estimated from its counts. The weight Δ_0 this support misses rewards support spreading (a uniform in-sector sampler scores lower), and $\Delta_0 = 0.004$ coexists with relative L_1 error 0.35. A sealed on-device momentum-distribution test has not been run; its one permitted noise-floor relaxation failed its sealed simulated gate.

Contents

1	Introduction	2
2	Method: reconstruction-independent moment tests	7
2.1	The quantum processor and the origin of the samples	7
2.2	The independent moment estimator and the rejection test	9
3	Results I: moment tests across observables (exact simulation)	10
4	Results II: why the estimator must be independent of the reconstruction	14
5	A blinded, pre-registered calibration in simulation	17
6	Coverage residual on a device-sampled support (IBM Heron)	20
7	A closed-form bound: the Christoffel miss-distance	25
8	Detectability and its sample-complexity limit	28

29	9 Discussion and limitations	30
30	9.1 Relation to prior work	30
31	9.2 Limitations	31
32	9.3 A code-scalability check beyond exact diagonalization	33
33	9.4 The verification stack: two necessary layers	33
34	9.5 Outlook	34
35	10 Conclusions	34
36	A Derivation of the moment identities	36
37	B The reconstruction-independent moment estimator	40
38	C The Christoffel miss-distance: proof and evaluation	46
39	D Complementary state-level check: the Gauss law	48
40	E Within-sector redistributions and the deployed battery	49
41	F Device run: reference samplers, line shapes and the n_k channel	53
42	References	56
43		
44		

45 1 Introduction

46 Dynamical response functions such as the spectral function $A(k, \omega)$, the structure factors
47 $S(q, \omega)$ and $S^{zz}(q, \omega)$ and the optical conductivity are measured by photoemission, neutron
48 and resonant x-ray scattering and infrared spectroscopy [1, 2] and computed classically by
49 many-body methods [3, 4]. They are now also computed on quantum processors first used
50 for ground-state energies [5–9]. Trapped ions have produced structure factors and spectral
51 functions [10, 11] and Green’s functions built from measured Hamiltonian moments [12]; su-
52 perconducting processors have produced an impurity Green’s function for density-functional
53 plus dynamical mean-field theory (DFT+DMFT) with up to 14 qubits [13] and structure
54 factors with up to 50 qubits, compared quantitatively with neutron data [14]. Sample-based
55 quantum diagonalization (SQD), quantum-selected configuration interaction (QSCI) and
56 quantum Krylov methods for ground states [15–19] have been extended to spectra [20, 21]:
57 the device samples a subspace, and classical resources diagonalize the Hamiltonian in it and
58 assemble $A(\omega)$ [22].

59 A reconstructed spectrum can be wrong in a way that a total-weight check misses: spectral
60 weight sits at the wrong frequencies while the total weight and other integrated quantities
61 stay within tolerance. This is the error we target. Critical studies document the ground-state
62 limitations of QSCI and subspace diagonalization [23–27]; that the same mechanisms misas-
63 sign spectral weight is our extrapolation, and the same concern motivates response-targeted
64 determinant selection on the classical side [28]. Such errors matter most where no trusted
65 classical cross-check exists, a regime whose edge moves as classical methods improve and
66 demonstrations are dequantized [29–34] (for sample-based diagonalization see Ref. [35]).
67 That regime is the eventual target, not the setting of this paper: a check intended for it must
68 first be calibrated where the ground truth is known [36–40], as we do here.

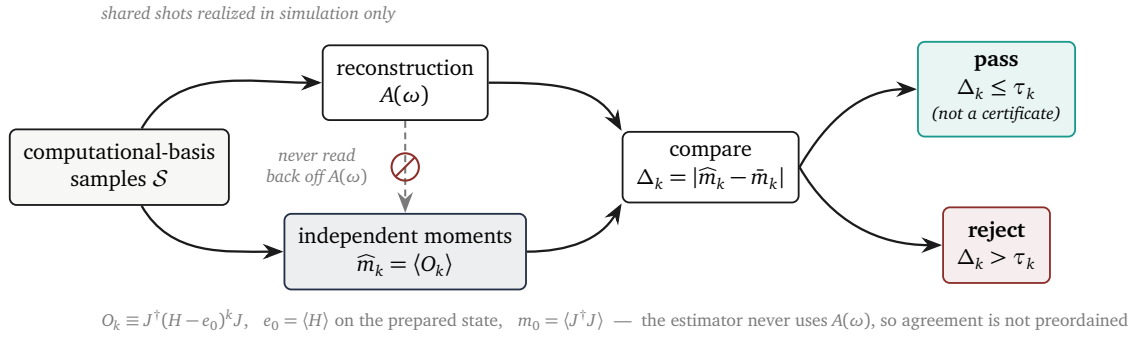


Figure 1: Two moment branches from one sample set (design). Computational-basis samples fix a subspace $\mathcal{S}$. The reconstruction branch diagonalizes H in $\mathcal{S}$, assembles $A(\omega)$ and reads off $\tilde{m}_k = \int \omega^k A(\omega) d\omega$. The estimator branch evaluates $\hat{m}_k = \langle O_k \rangle$, with $O_k \equiv J^\dagger(H - e_0)^k J$ and $e_0 = \langle H \rangle$ on the prepared state (footer); it never uses $A(\omega)$ (crossed arrow), so it can disagree with the reconstruction. On an exact eigenstate $\langle O_k \rangle$ equals m_k and the commutator form $\langle J^\dagger \text{ad}_H^k(J) \rangle$; off an eigenstate they differ (Appendix A). The reconstruction is rejected when $\Delta_k = |\hat{m}_k - \tilde{m}_k|$ exceeds a threshold τ_k set beforehand from shot noise and an assumed, unmeasured bias budget; a pass does not certify it. Here the two branches share one set of shots only in simulation; on hardware the device supplies the support $\mathcal{S}$ alone (Sec. 6).

The test compares two sets of moments (Fig. 1). Physics supplies the conditions: every $T = 0$ Lehmann spectrum obeys moment sum rules [41–43] and positivity [44], however it was computed, and the state behind it obeys its conserved-charge identities, such as the Gauss law of a lattice gauge theory [45]. For a zero-temperature Lehmann function of a probe J , the power moments $m_k = \int \omega^k A(\omega) d\omega$ are ground-state expectation values, $m_k = \langle 0 | J^\dagger (H - E_0)^k J | 0 \rangle$; for a Hermitian probe $m_0 = \langle J^2 \rangle$ and $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ [41, 44, 46]. We estimate them as $\hat{m}_k = \langle O_k \rangle$, with $O_k \equiv J^\dagger (H - e_0)^k J$ and $e_0 = \langle H \rangle$ on the prepared state, never by integrating $A(\omega)$, so $\hat{m}_k$ can disagree with the reconstruction’s moments $\tilde{m}_k$; a moment read back off $A(\omega)$ gives $\Delta_k \equiv 0$. The reconstruction is rejected when $\Delta_k = |\hat{m}_k - \tilde{m}_k|$ exceeds a threshold τ_k set beforehand from shot noise and an assumed, unmeasured bias budget of 2% of each moment. These are necessary conditions. Failing one rejects the reconstruction, provided the estimator’s bias lies within that budget; the sealed nominal family-wise level of the battery of tests on m_0, m_1, m_2 is $1 - 0.95^3 = 14.3\%$, and a realized $\pm 2\%$ bias gives a modeled false-positive rate of 8.0–11.5%, still below that level (Sec. 5). Passing does not show that the reconstruction is correct; it rules out only errors that move a tested moment past its threshold. Figure 3 shows an injected corruption of $A(k, \omega)$ that keeps the gap, the band edges and the total weight, so a total-weight check accepts it, but has the wrong first moment. The design takes both branches from one set of samples; Sec. 2.1 states how far this is realized: not on hardware, where the device supplies only the sampled support; in part, in simulation; and not in the blinded test. Appendix D adds a state-level Gauss-law identity, evaluated exactly on the state vector [45, 47].

What we find. Each item states a result together with its main limitation.

(i) *Tests across observables* (Figs. 3–6). In the doped Hubbard current response, m_1 and m_2 reject a spurious low-frequency peak that preserves $m_0 = \langle J^2 \rangle$; the negative-order moment m_{-1} moves most in relative terms (154% against 38% for m_1) but, normalized by noise at a 2% bias, rejects after m_1 and m_2 for this error model, on the ring and the open chain, unless the scanned peak lies at $0.5 t$ or below, and on the ring it needs the Kohn stiffness from a separate calculation. For $A(k, \omega)$, $S(q, \omega)$ and $S^{zz}(q, \omega)$ the construction is only illustrated, with injected

corruptions and no calibrated threshold (for $A(k, \omega)$ the residual is fixed by construction, the $L = 6$ collective channels are compared with a 5% guide, and the $L = 12$ structure factors get no moment test), all in exact simulation at $L \leq 12$.

(ii) *Independence under truncation* (Sec. 4). Under determinant-subspace truncation the independent estimate of m_1 crosses an illustrative fixed 5% line only at severe truncation (7% of the determinant support with an idealized full-state estimator, 18% with both branches on one subspace state), while a control read back off $A(\omega)$ is zero by definition. For this error the companion's zeroth-order residual $\Delta_0/\widehat{m}_0 = 1 - w$ crosses the same line earlier, between about 29% and 21% coverage on the shared subspace state, so the added reach of m_1 and m_2 lies mainly in errors that preserve m_0 : in the blinded test, 20 of the 26 instances rejected only by m_1 or m_2 were m_0 -preserving spurious features.

(iii) *A blinded, pre-registered calibration* (Sec. 5). Before any instance existed we sealed the battery, thresholds, error catalog and seed by a SHA-256 hash; the seal is self-attested (no third-party timestamp), and because the simulation is deterministic given the seed, it cannot by itself exclude iteration before sealing. On synthetic errors of known truth for the current probe of one doped $L = 6$ ring, with an oracle estimator (exact moment plus Gaussian noise), the pre-registered primary endpoint was 53.7% detection (101/188: 56/56 truncations in this sealed draw, although an exhaustive $U/t = 4$ scan finds 12 of 185 truncation depths that pass, three inside the catalog's depth range; 45/61 spurious features; 0/71 under-converged Krylov reconstructions) at 0.9% false positives (1/112), low by construction because the simulated estimator is unbiased while the threshold budgets a 2% bias; robustness to a real estimator bias is untested. Two sealed secondary predictions failed (one-node Krylov, expected to be caught, was caught in 0/15; m_0 -preserving features, expected to be caught rarely, in 20/34). The sealed Hankel-positivity branch, applied to the reconstruction moments, could reject nothing, and the generator added δ atoms where the sealed text specifies Gaussian ones.

(iv) *Exact limits* (Secs. 7 and 8). Classical Chebyshev–Markov–Stieltjes and Markov–Krein brackets, re-derived here [48–51], bound the cumulative weight of a reconstruction with exact moments, but at the order m_0 – m_2 used here they are vacuous at the mean ($\max_x W_1(x) = m_0$), and finite tolerances widen them. The first N moments are complete for weight redistributions among N distinct fixed poles, in exact arithmetic only (Theorem 2); with m_0 – m_2 and $N > 3$ the guaranteed power is zero whatever the pole separation, and on the $L = 6$, $U/t = 4$ current spectrum ($N = 5$) a redistribution that keeps m_0 – m_2 exact moves 72% of the weight and passes, as it would even with m_3 and m_4 at the same shot budget.

(v) *A device evaluation of the coverage channel* (Sec. 6). On IBM Heron (ibm_fez) the device supplied only the support S of an $L = 8$ electron-addition sector, sampled at four shot budgets by seven number-conserving circuits that are not the Hubbard propagator of the probe state; no moment was estimated from device counts, and the raw counts are deposited. The missed weight Δ_0 grows from 0.0044 to 0.37 as coverage falls from 94% to 35% but rewards support spreading, not fidelity: a uniform in-sector sampler at the device's retained count reaches a lower Δ_0 and a more accurate reconstruction at every budget, and the reconstruction with $\Delta_0 = 0.0044$ has a relative L_1 error of 0.35. The moments m_1, m_2 were not estimated on hardware: for a surrogate bond-basis probe (the $q = \pi$ density probe of an open $L = 6$ chain), with no mitigation modeled, the forecast depolarizing bias is 37–116% of the signal (local to global noise, two-qubit error 2.5×10^{-3}) at 292 controlled-Z (CZ) gates, the depth of the companion's sampler circuits rather than of the ground-state preparation this estimator needs; and a pre-registered on-device momentum-distribution (n_k) test has not been run: the committed record does not state why it was not run at its sealed noise floors, and its one permitted relaxation of those floors failed the sealed simulated gate.

Prior art. Current practice validates a quantum-computed spectrum against a reference: neutron data [14], or the validation framework of Buchs *et al.*, with experimental and classical

anchors and metrics that include low-order moments and sum-rule residuals; its asymmetric mode for the regime without full classical benchmarks, leaning more on experiment, sum rules and internal consistency, is left to future work [52]. Other verification tools, from self-verifying variational simulation to observable estimation without classical verification, certify a number, not a function, acting on states or expectation values [6, 47, 53–64]. None of these state- or number-level tools tests a reconstructed $A(\omega)$ against the Hamiltonian that produced it. Sum-rule checks do test spectra against Hamiltonian-derived moments: they are long-standing in dynamical mean-field theory and quantum Monte Carlo, most systematically in the sum-rule program of Turkowski and Freericks [65, 66], and analytic-continuation pipelines report sum-rule agreement routinely [67, 68]. We claim no novelty for testing a spectrum against sum rules; what we add is this test’s form for sample-based quantum reconstructions, with an estimator evaluated independently of the reconstruction, thresholds from shot noise and a bias budget, a blinded calibration in simulation and the exact limits of a pass. The classical moment problem, with its orthogonal polynomials and Gauss quadrature [48–50, 69–74], underlies the kernel-polynomial and recursion methods [75–78] and now bounds smeared spectral functions from Euclidean lattice correlators [79–81]; our moments come instead from ground-state operators, and our use is a rejection test on a real-frequency reconstruction. Semidefinite-programming (SDP) moment certificates [82, 83] would supply a second necessary condition on the reconstruction, not a sufficient one; we have not applied them (Sec. 9.4). Pipelines that hold the probe acting on the sampled ground state exactly, such as the Green’s-function solver of Patel *et al.* [21], never lose spectral weight; when both branches use the same ground-state vector, our coverage channel carries no signal for them, and only the higher moments test them. Table 1 compares this work with the closest prior art.

Relation to the companion study. This paper complements a companion study that reconstructs these spectra from sampled subspaces and bounds their error [20]; we adopt its terms (its μ_j is our m_j). Its version 3 already contains the captured weight w , which gives our coverage residual $\Delta_0 = m_0(1 - w)$, the $k = 0$ case of the operator leak of Eq. (B.3) with $\varphi = J|0\rangle$ built from the exact ground state; moment exactness through order $2K + 1$ for a subspace containing the order- K Krylov space, which is why Krylov reconstructions with two or more Lanczos nodes reproduce m_0 – m_2 exactly and pass our battery, while the one-node case misses m_2 by 2–4% and passes only inside the 2% bias budget (Sec. 5); the leakage identity $\mu_2 - \mu_2^S = \|Q_S H P_S \varphi\|^2$, the $k = 2$ leak on a subspace of unit captured weight; and the result that captured weight alone cannot certify a reconstruction, of which our $L = 8$ device run is an instance. Its own device run covers the whole $L = 6$ sector and shows only that the pipeline executes (Sec. 6). We add not an error bound but a rejection test with shot-noise and bias-budget thresholds, defined for any diagonal $T = 0$ Lehmann $A(\omega)$ (single-particle, current or collective; not only Rayleigh–Ritz reconstructions) but calibrated only for the current probe of one doped $L = 6$ ring; its m_1 and m_2 see weight moved in frequency at fixed total weight, which captured weight cannot, and the single-particle and collective cases are illustrations with injected corruptions and no calibrated threshold. We also add the sealed, blinded calibration with its failures reported; the exact limits of what a pass implies, including zero guaranteed power at the order tested; and a device evaluation of the coverage channel against reference samplers, which shows that it cannot distinguish a good device from one that merely spreads its samples.

Scope. The method can reject a reconstruction but cannot certify one; its conditions are necessary, not sufficient. The moment test targets determinant-subspace truncation in SQD and QSCI spectra and injected spurious features (modeled as analytic-continuation artifacts, δ atoms, whose realism for sample-based pipelines is not established); the Gauss-law identity targets gauge-symmetry violation in the prepared state. Under-converged Krylov reconstructions were in the sealed catalog and the primary endpoint (0/71 caught; low detection was

Table 1: **Contribution relative to the closest prior art.** “recon.-indep.”: compared with something the reconstruction did not produce, and with what; “same samples”: comparison quantity taken from the spectrum’s own samples; “—”: the axis does not apply. The “blinded” column records design, not quality: the prior methods were not built as blinded rejection tests. The semidefinite programs bound the true moments; read-off moments inside their intervals are a second necessary condition, not a sufficient one (Sec. 9.4). Notes *a–d* follow the table.

method	recon.-indep.	same samples	on device	blinded	role
Turkowski–Freericks sum-rule checks [65, 66]	yes (vs classical sum-rule moments)	no	no	no	consistency check
maxent / stochastic continuation [67, 68]	yes (vs classical sum-rule moments)	no	no	no	consistency check
Buchs <i>et al.</i> validation framework [52]	yes (vs experiment, classical, sum-rule targets)	no	no	no	validation metrics, incl. moments
Lee <i>et al.</i> neutron benchmark [14]	yes (vs experiment)	no	yes	no	benchmark
Greene-Diniz <i>et al.</i> cumulant Lanczos [12]	— (builds G)	—	yes	no	constructive
Wang–Acín / Mortimer SDP [82, 83]	— (bounds moments)	no	no	no	two-sided moment bounds
Abbott <i>et al.</i> / causal bootstrap [79–81]	yes (Euclidean)	no	no	no	two-sided bound
Kokail <i>et al.</i> self-verification [47]	— (certifies state)	—	yes	no	state verification
reference-free observable estimation [63]	— (certifies number)	—	yes	no	validity
companion: captured weight and leakage [20]	— (bounds the error)	—	no ^d	no	a posteriori error bound
this work	yes (vs full-operator $\langle O_k \rangle$ on the subspace state, evaluated classically; oracle moments in the blinded test)	designed^a	support only; no moment^b	yes^c	necessary condition

^aDesigned; realized only in part, and only in simulation (Sec. 2.1); same-sample estimation, blinded pre-registration and on-device execution were not realized together. ^bThe device supplied the sampled support only; no moment was estimated from device counts (Table 2). ^cBlinded simulation with an oracle-centered estimator; the seal is self-attested (SHA-256, no third-party timestamp) (Sec. 5). ^dThe companion’s $L = 6$ run covers the whole sector, so its bound is not exercised on device data.

197 predicted for two or more Lanczos nodes and high detection, which failed, for one node).
 198 By the moment exactness stated above, and for one node by the 2% bias budget, the battery
 199 cannot detect them, and we do not target them: that error class calls for a separate instru-
 200 ment, most naturally a Ritz-residual check on the Krylov basis, which we leave to future work.
 201 Neither check is a general device-noise diagnostic [63]. The test uses integrated moments
 202 of diagonal $T = 0$ Lehmann spectra [44], not pointwise line shapes; every demonstration is
 203 classically reproducible ($L \leq 12$ Hubbard chains and rings, a small lattice-Schwinger model,

a density-matrix renormalization group (DMRG) check of code scalability at $L = 24$); and the blinded test withholds a classically known truth, so it is not a field catch, and no device is in the loop. On hardware, where moments carry statistical and systematic error, interval tests must replace the exact-moment tests (Appendix B); Table 2 separates what is measured on hardware from what is simulated or deferred.

Sections 2.1 and 2.2 describe the pipeline and the test; Sec. 3 applies the test across observables; Sec. 4 shows why the estimator must not be read off $A(\omega)$; Sec. 5 reports the blinded calibration and Sec. 6 the device run; Secs. 7 and 8 derive what a pass implies and what the battery can detect; Sec. 9 collects the limitations, and Sec. 10 concludes. The appendices derive the moment identities, analyze the estimator, prove the bounds, and record the Gauss-law check, the within-sector analysis and the device run.

2 Method: reconstruction-independent moment tests

We first describe the object the screen tests, then the checks attached to it.

Notation. The hopping t is the energy unit throughout; a frequency rescaled to $[-1, 1]$ is written x (Sec. 7). J always denotes the probe operator, and the spin-exchange constant $4t^2/U$ is written J_{ex} . The moment operator O_k and its commutator form C_k are defined once, in Eq. (3). K is the battery order, the highest moment order the screen tests; in Theorem 2 it is the highest order a redistribution preserves, and in the moment-exactness result (Sec. 1, Appendix B) the order of a Krylov space.

2.1 The quantum processor and the origin of the samples

The screen tests the output of a sample-based-diagonalization pipeline. The pipeline and the $L = 6$ device spectrum of Fig. 10 are the companion study's [20]; this paper contributes the test, its blinded calibration in simulation, its exact limits and four $L = 8$ ibm_fez jobs (one per shot budget, run on 2026-08-28) that supply device-sampled supports (Sec. 6). In sample-based diagonalization [15–17] the processor does not evaluate $A(\omega)$: each computational-basis shot of a prepared state returns a bitstring (a candidate Slater determinant), and the distinct bitstrings surviving post-selection to the physical particle-number and spin sector form the subspace $\mathcal{S}$. The Hamiltonian is projected onto $\mathcal{S}$ and diagonalized classically, and the spectrum is the $T = 0$ Lehmann sum

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta_\eta(\omega - (E_n - E_0)) \quad (1)$$

over the resulting eigenpairs, with δ_η a narrow broadening [44]. In the companion's formulation the prepared state is a Trotterized real-time evolution of the $(N+1)$ -particle seed $c_p^\dagger|\psi_0\rangle$ [84]. The device circuits actually run, the companion's $L = 6$ job and our $L = 8$ jobs, are number- and S_z -conserving samplers of the addition sector, not the Hubbard propagator of the seed, as the companion's v3 discloses for its run (Sec. 6) [20]. Neither the companion's run nor ours used configuration recovery [35], and the companion's $L = 6$ run read the bitstrings in reversed qubit order during post-selection, so every determinant it retained arose from a device error [20].

The intended role of the processor appears only at scale, where the configurations carrying the weight of the probe state $|\varphi\rangle = J|0\rangle$ (on the exact ground state) cannot in general be enumerated classically: an ideal sampler of the targeted state would concentrate shots on them and fix $\mathcal{S}$ by measurement rather than by a classical selection heuristic [16]. We do not show that any device sampler does this better than a classical heuristic, and the companion claims no quantum advantage [20]. The circuits run here are not targeted at φ : for them

Table 2: **What is demonstrated on hardware, in simulation, analytically, and deferred.** The only device-derived quantity is the support set $\mathcal{S}$; no moment is estimated from device counts. Every discriminating result is classically reproducible simulation, and the beyond-classical field test is deferred.

quantity	source	status
support set $\mathcal{S}$ and coverage residual Δ_0 , $L = 8$ addition sector	ibm_fez ($\mathcal{S}$ only)	Δ_0 computed classically on $\mathcal{S}$; rewards support spreading (a uniform in-sector sampler scores lower); tests $A(\omega)$ only weakly (a counterfactual threshold passes the 5×10^4 -shot reconstruction, relative L_1 error 0.35) and does not test the device (Fig. 11)
full-sector recovery, $L = 6$ (companion)	ibm_fez	executability only; the retained determinants all came from device errors (Fig. 10)
off-diagonal moments m_1, m_2 (weight discrimination)	simulation	deferred: forecast depolarizing bias 37–116% of the signal at 292 CZ (Sec. 6; Fig. 9)
momentum distribution n_k (diagonal discriminant)	simulation (sealed gate)	gate-negative at the relaxed noise floors; not run (Appendix F)
blinded detection and false-positive rates	simulation	sealed primary endpoint: TPR $101/188 = 53.7\%$ (0/71 Krylov) at FPR $1/112 = 0.9\%$; oracle-centered, unbiased simulated estimator; current probe of one $L = 6$ ring; self-attested seal; two sealed secondary predictions failed (Sec. 5)
Hankel/Stieltjes positivity of the estimated moments	simulation	not exercised: evaluated only on reconstruction moments, where it cannot reject (Sec. 5)
fixed-pole weight redistribution	exact / analytic	zero guaranteed power with m_0 – m_2 (two-dimensional null space for the $L = 6$, $U/t = 4$ current spectrum; Sec. 8)
same-sample $\widehat{m}_k$ and $\bar{m}_k$ (one set of shots)	simulation	correlation $\rho \approx 0.43$ at 84% coverage, no power gain (Sec. 9; Appendix B)
Christoffel miss-distance bound; moment-interval layer	exact / analytic	classical bound, re-derived; Hausdorff interval shown; composition with the Wang/Mortimer SDP not implemented
field catch of an unknown-truth error (beyond-classical)	—	future work (Sec. 9)

the coverage residual Δ_0 (Sec. 6) measures the weight of φ outside the configurations they happened to reach, and at $L = 8$ a uniformly random in-sector sampler, given the device's retained shot counts, leaves less of that weight uncovered than the device at every shot budget.

The design goal is that the moment estimator and the reconstruction share the sampled support, so that $\widehat{m}_k$ inherits the shot noise and mitigation choices of the spectrum it tests. The estimator $\widehat{m}_k$ defined below applies the full operator to the subspace ground state built on $\mathcal{S}$, a classical post-processing step that reaches configurations outside $\mathcal{S}$; a device-native alternative, measuring $\langle O_k \rangle$ in qubit-wise-commuting Pauli groups (rotated bases for the hopping terms) on a separately prepared ground state, is simulated only (Appendix B). We state where the shared-sample design is realized. On hardware it is not: the device supplies only the support $\mathcal{S}$, and every moment in Sec. 6 is computed classically. In simulation it is realized in three places: the shared-state test of Fig. 8, where both moments are evaluated on one subspace state (a deterministic truncation by $|\psi_0|^2$, not a set of shots); the joint-sampling simulations of Sec. 9 and Appendix B, where one set of shots yields both $\widehat{m}_1$ and $\bar{m}_1$ (pre-registered under its own self-attested SHA-256 seal; correlation $\rho \approx 0.43$ at 84% coverage, reached with 2000 shots per replica rather than the $N_s = 5 \times 10^4$ of the deployed rule; no gain in detection power; the local estimator uses exact amplitudes and is not device-realizable); and the $L = 24$ DMRG test of Sec. 9.3 (current probe of an open chain), where both branches use one shot-sampled subspace state. The blinded test of Sec. 5 does not use it: there $\widehat{m}_k$ is an oracle-centered estimate, the exact moment plus Gaussian shot noise; that section also reports a post hoc rescoring of its 56 truncations with the shared-state estimator.

2.2 The independent moment estimator and the rejection test

Evaluated with the exact eigenpairs of H and $\eta \rightarrow 0$, Eq. (1) is the zero-temperature diagonal Lehmann spectral function of the probe J , a positive measure supported on $\omega = E_n - E_0 \geq 0$ [44]. On an exact eigenstate $|0\rangle$ its power moments are ground-state expectation values (Appendix A) [41],

$$m_k = \int \omega^k A(\omega) d\omega = \langle 0 | J^\dagger (H - E_0)^k J | 0 \rangle = \langle 0 | J^\dagger \text{ad}_H^k(J) | 0 \rangle, \quad (2)$$

with $\text{ad}_H(J) \equiv [H, J]$, so that $m_0 = \langle J^\dagger J \rangle$ and, for a Hermitian probe, $m_0 = \langle J^2 \rangle$ and $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$. For the current probe m_0 is the total current-current weight and m_1/m_0 the mean frequency; the regular part of the optical f -sum is set by the separate moment m_{-1} (on a ring the Drude weight carries the rest; Appendix A) [42, 43]. On a prepared state $|\psi\rangle$ that need not be an eigenstate we define the moment operator and its commutator form,

$$O_k \equiv J^\dagger (H - e_0)^k J, \quad C_k \equiv J^\dagger \text{ad}_H^k(J), \quad e_0 = \langle \psi | H | \psi \rangle. \quad (3)$$

On an exact eigenstate $e_0 = E_0$ and $\langle O_k \rangle = \langle C_k \rangle = m_k$ [Eq. (2)]; off an eigenstate the two in general differ for $k \geq 1$ and neither equals m_k , and on a subspace state they differ by enough to change a verdict (Appendix A). The deployed estimator is $\widehat{m}_k = \langle \psi | O_k | \psi \rangle$; C_k is used only on exact eigenstates. The reason is structural. With $\Pi_{\mathcal{S}}$ the projector onto the span of $\mathcal{S}$ and $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, a reconstruction built on the subspace ground state $|0_{\mathcal{S}}\rangle$ has the moments $\bar{m}_k = \langle 0_{\mathcal{S}} | J^\dagger (H_{\mathcal{S}} - e_0)^k J | 0_{\mathcal{S}} \rangle$ [Eq. (B.1)], so with O_k the difference $\widehat{m}_k - \bar{m}_k$ is exactly the operator leak of Eq. (B.3), which at $k = 0$ is the captured-weight deficit below; with C_k it would also contain $\langle C_k \rangle - \langle O_k \rangle$, which is not a leak. In the committed records the form is first used off an eigenstate, with this justification, in the shared-state test of Sec. 4, after the sealed blinded test (which used an oracle-centered estimate; Sec. 5) and before the off-eigenstate example of Appendix A was computed; that example is evidence that the choice matters, not

the reason for it. The post hoc shared-state rescoring of Sec. 5 rejects every truncation on m_0 , where $O_0 = C_0 = J^\dagger J$, so its 56/56 does not depend on the choice.

The screen attaches to the spectrum an estimator $\widehat{m}_k$ of each moment that is independent of the reconstruction: it is never obtained by integrating $\omega^k A(\omega)$. The reconstruction moment $\bar{m}_k = \int \omega^k A(\omega) d\omega$ is the quantity a circular check would use, and the test statistic is the discrepancy $\Delta_k = |\widehat{m}_k - \bar{m}_k|$. The true spectrum gives $\Delta_k = 0$ up to estimator error, so the reconstruction is rejected when $\Delta_k > \tau_k$, with $\tau_k = z \delta_k$, $z = 1.96$ ($\alpha = 0.05$ per moment), and $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$ the shot variance plus a bias budget b_k (Appendix B); the deployed rule uses $N_s = 5 \times 10^4$ and $b_k = 0.02 |\widehat{m}_k|$. The bias budget is assumed, not measured: the nominal level, and the attribution of a rejection to the reconstruction rather than to the estimator, both presuppose that the estimator's bias lies within b_k . For the three-moment battery the nominal family-wise level is $1 - 0.95^3 = 14.3\%$, the target sealed for the blinded test (Sec. 5). “Deployed” names these default settings (the estimator $\widehat{m}_k = \langle \psi | O_k | \psi \rangle$, z , N_s , b_k and $K = 2$ below), not use on a production pipeline: no moment in this paper is estimated from device counts (Sec. 6). A circular check compares $\bar{m}_k$ with itself, returns $\Delta_k \equiv 0$ and can never reject; Sec. 4 shows this contrast on controlled truncation. At $k = 0$, where $O_0 = C_0 = J^\dagger J$, the discrepancy is the weight the support misses,

$$\Delta_0 = \langle \varphi | (1 - \Pi_S) | \varphi \rangle = \|\varphi\|^2 (1 - w), \quad w = \|\Pi_S \varphi\|^2 / \|\varphi\|^2,$$

with w the companion's captured weight [20]: for the shared-state estimator $|\varphi\rangle = J|0_S\rangle$ and $\|\varphi\|^2 = \widehat{m}_0$ [Eq. (B.3)]; for the device coverage residual $|\varphi\rangle = J|0\rangle$ on the exact ground state, with the addition probe $J = c_{0\uparrow}^\dagger$, and $\|\varphi\|^2 = m_0$ (Sec. 6). The companion's moment-exactness theorem and m_2 leakage identity [Eq. (B.3) at $k = 2$] bound what the test can see, and what this paper adds is stated in Sec. 1.

The screen is a battery, not a single test, and we use the two words interchangeably. One comparison $\Delta_k > \tau_k$ is a moment test; the battery applies the moment tests for $k = 0, 1, \dots, K$ (deployed: $K = 2$) and rejects the reconstruction when any of them rejects; a reconstruction that no moment test rejects passes. Separately, the battery checks that $\{\widehat{m}_k\}$ is itself a moment sequence of a positive measure: the Hankel matrix $H_n = [\widehat{m}_{i+j}] \succeq 0$ (Fig. 2) and, because a diagonal $T = 0$ Lehmann measure is supported on $\omega = E_n - E_0 \geq 0$, also the shifted Stieltjes matrix $H_n^{(1)} = [\widehat{m}_{i+j+1}] \succeq 0$ (one-sided spectra only; Appendix A) [49, 50, 69]. A violation rejects the estimate—device, mitigation or estimator—independently of $A(\omega)$. It cannot reject a non-negative reconstruction, whose own moments satisfy these conditions by construction, and it has not been applied to estimated moments in this paper: in the blinded test and the interval battery it was evaluated at $\bar{m}_k$ and could not reject (Sec. 5, Appendix B). The battery can also include an exactly known operator identity, such as a conserved charge $[H, G] = 0$ with a known physical eigenvalue, whose violation rejects the prepared state; Appendix D illustrates this with a Gauss law evaluated exactly on the state vector [45]. The construction transports across models and observables because it needs only (a) a positive spectral measure and (b) operators whose ground-state expectations can be estimated without the reconstruction [48, 75]; its thresholds have been calibrated only for the current probe of one doped $L = 6$ ring (Sec. 5).

3 Results I: moment tests across observables (exact simulation)

Figure 3 shows $A(k, \omega)$ of the half-filled Hubbard ring, a one-dimensional correlated insulator with Hubbard bands separated by the Mott gap [3, 85–88]. The single-particle gap approaches U only in the atomic limit $t \rightarrow 0$, and at the size ($L = 12$) and broadening of the figure the spinon and holon branches are only partly resolved. An injected corruption flattens the band-

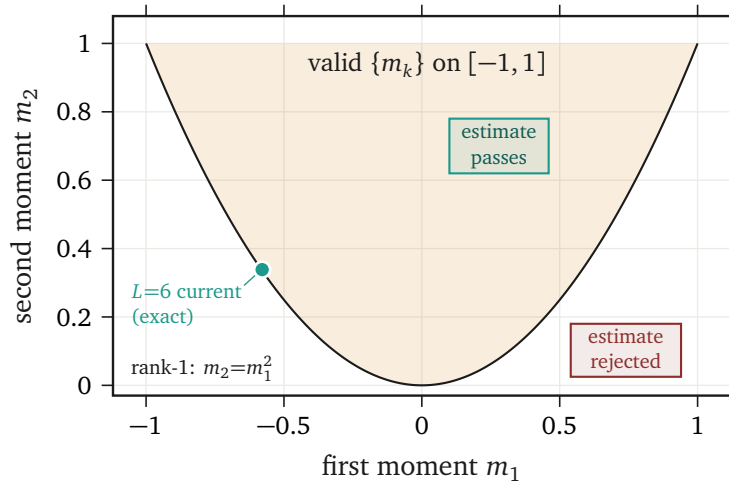


Figure 2: **Hankel-positivity region in the (m_1, m_2) plane.** With $m_0 = 1$ and frequencies rescaled to $x \in [-1, 1]$ as in Fig. 13 (Appendix C), moments lie in the shaded body $m_1^2 \leq m_2 \leq 1$, bounded below by the Hankel (rank-one) edge of a single δ and above by the interval (Hausdorff) localizing condition. The shifted (Stieltjes) condition refers to the unrescaled $\omega \geq 0$ frame and is not drawn. Schematic shot-budget boxes: the dark-red one (“estimate rejected”), wholly Hankel-negative, would reject the estimate without reference to $A(\omega)$; the teal one (“estimate passes”) would pass. A non-negative reconstruction’s moments always lie inside; the check was not applied to estimated moments here. Teal dot: exact moments of the doped $L = 6$, $U/t = 8$ current measure; one pole carries 78% of its weight, so the dot lies just above the rank-one edge [49, 69].

centroid dispersion by moving weight between the Hubbard bands at each momentum, leaving the Mott gap and the per- k total weight intact. For this probe m_0 is pinned to 1, so the first informative moment is m_1 [44, 66]: from the anticommutator sum rule for the non-Hermitian probe $c_{k\sigma}$, measured from $U/2$, it is the Hartree centroid $\epsilon_k + U(\langle n_{-\sigma} \rangle - \frac{1}{2}) = \epsilon_k$ at half filling (Appendix A), and it exposes the flattening (right panel). The corruption moves the centroid by a fixed fraction of ϵ_k and the moment is closed-form, so the residual is set by construction and no threshold is calibrated for this probe: the panel illustrates the check rather than testing it.

Figure 4 shows the collective response: the charge structure factor $S(q, \omega)$, gapped by a particle-hole onset Δ_c , and the spin structure factor $S^{zz}(q, \omega)$, whose gapless spinon continuum [89] signals spin-charge separation [3]. The onset Δ_c is a finite-size quantity distinct from the single-particle Mott gap ($\approx 4.97 t$); both are set by U and coincide only in the thermodynamic limit. The charge (spin) data, on the scale $\sim U$ ($\sim J_{\text{ex}}$, with $J_{\text{ex}} = 4t^2/U$ the exchange constant), are computed to $11 t$ ($2.6 t$), displayed to $10 t$ ($2.4 t$) and linearly interpolated in q . Each channel is a positive $T = 0$ Lehmann measure with its own moment sum rules [41], so the screen applies to each; we apply it at $L = 6$ (Fig. 6), not to these $L = 12$ spectra. Such spectra are standard output of classical many-body methods [67, 90–93]; at these sizes we compute them exactly.

We now apply the screen to a wrong distribution of correct total weight. In the doped one-dimensional Hubbard model the regular ($\omega > 0$) current spectral function $A_J(\omega)$ is dominated by a mid-infrared/Mott band of total weight $m_0 = \langle J^2 \rangle$ [2, 42, 43, 96]; on the finite ring the charge stiffness is a separate zero-frequency Drude term, absent from $A_J(\omega)$. A wrong reconstruction moves a fraction f of the weight above $6 t$ into a spurious Drude-like pole at $2.5 t$, keeping the total fixed, so a total-weight check passes it (Fig. 5; Lanczos ground state,

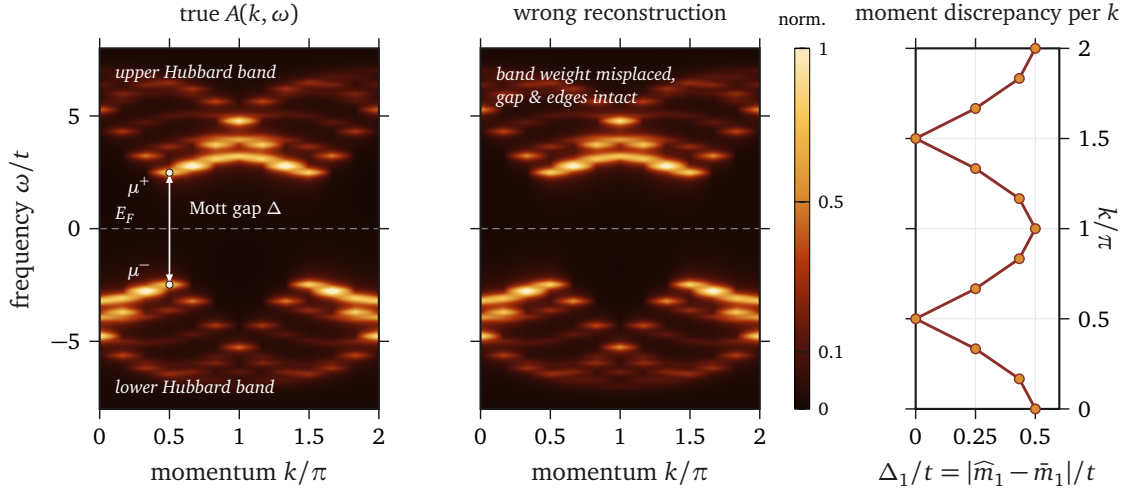


Figure 3: **First-moment check on the single-particle spectral function $A(k, \omega)$ of the Hubbard ring.** Left: exact $A(k, \omega)$ at half filling, frequencies measured from $E_F = U/2$ (dashed); the band edges $\mu^\pm = \pm 2.484 t$ at $k_F = \pi/2$ (dots, arrow) give the Mott gap $\Delta = \mu^+ - \mu^- = 4.97 t$. Center: an injected corruption rebalances intensity between the Hubbard bands so that the band centroid becomes $(1 - \alpha)\epsilon_k$, $\alpha = 0.25$. The gap stays free of poles (only the Lorentzian tail reaches E_F), $\mu^\pm$ are unchanged and $m_0 = 1$ per k , so a total-weight check accepts it. Right: $\Delta_1 = |\widehat{m}_1 - \bar{m}_1|$, with $\widehat{m}_1(k) = \langle [c_{k\sigma}, H], c_{k\sigma}^\dagger \rangle$ the closed-form Hartree centroid, not a sampled estimate; $\Delta_1 = \alpha|\epsilon_k|$ by construction ($0.50 t$ at $k = 0, \pi$; zero at $k = \pi/2, 3\pi/2$). $L = 12$ (12 momenta; the $k = 2\pi$ column repeats $k = 0$), $U/t = 8$, $\eta = 0.16 t$; color $\propto (A/A_c)^{0.8}$, saturated at A_c , the 99.6th percentile (colorbar ticked at true A/A_c); Lanczos with Haydock Green's functions [74, 76].

Haydock/Ritz current poles [74, 76], every moment exact, no sampling). The first moment $m_1 = \frac{1}{2} \langle [J, [H, J]] \rangle$ [41, 42] differs from the reconstruction's by up to 38% of m_1 at $f = 0.5$ and, under the deployed threshold rule, rejects from $f \approx 0.057$. Here the rule takes $z = 1.96$, an assumed 2% bias budget and a shot term at $N_s = 5 \times 10^4$ with the local-estimator variance of each estimated operator on the exact $L = 12$ ground state (Appendix B); its blinded calibration covers only the $L = 6$ catalog of Sec. 5. Integrated weight cannot see this error: m_0 is preserved by construction.

The misplaced weight also moves $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega = \sum_n |\langle n|J|0 \rangle|^2/\omega_n$, whose kernel $1/\omega$ weights low frequencies, unlike the kernel ω^k of the positive moments (Fig. 5). Since $J = i[H, P]$, this optical f -sum moment is a ground-state kinetic energy: $m_{-1} = \frac{1}{2} \langle -\hat{T} \rangle$ on open chains and $\frac{1}{2} \langle -\hat{T} \rangle - \mathcal{D}$ on the ring, with $\mathcal{D}$ the finite-size Drude weight (derived and verified in Appendix A) [41–43]. On this doped ring only 1.4% of m_0 lies below $6 t$, and $m_{-1} = 0.324$ is the small difference of $\frac{1}{2} \langle -\hat{T} \rangle = 5.025$ and $\mathcal{D} = 4.701$: the stiffness cancels 94% of the kinetic term. The spurious peak therefore moves m_{-1} by up to 154% at $f = 0.5$, $4.0\times$ the shift of m_1 (38%) and $3.3\times$ that of m_2 (47%).

A larger relative shift does not mean earlier detection. Because of the cancellation, a 2% bias on $\frac{1}{2} \langle -\hat{T} \rangle$ sets $\tau_{-1} = 0.198$, which is 61% of m_{-1} , whereas τ_1 and τ_2 are 4.3% of m_1 and m_2 . So m_2 rejects first ($f \approx 0.046$), then m_1 (0.057), and m_{-1} only at $f \approx 0.20$. The open chain gives the same order, and across spurious-pole positions of $0.25\text{--}4 t$ the moment m_{-1} rejects first only for poles at $0.25 t$ and $0.5 t$, on both geometries (Appendix A). It is thus a complementary check for weight very close to $\omega = 0$, not the most sensitive check for this error model. On the ring, $\mathcal{D}$ requires ground states at twisted flux and is not a same-sample

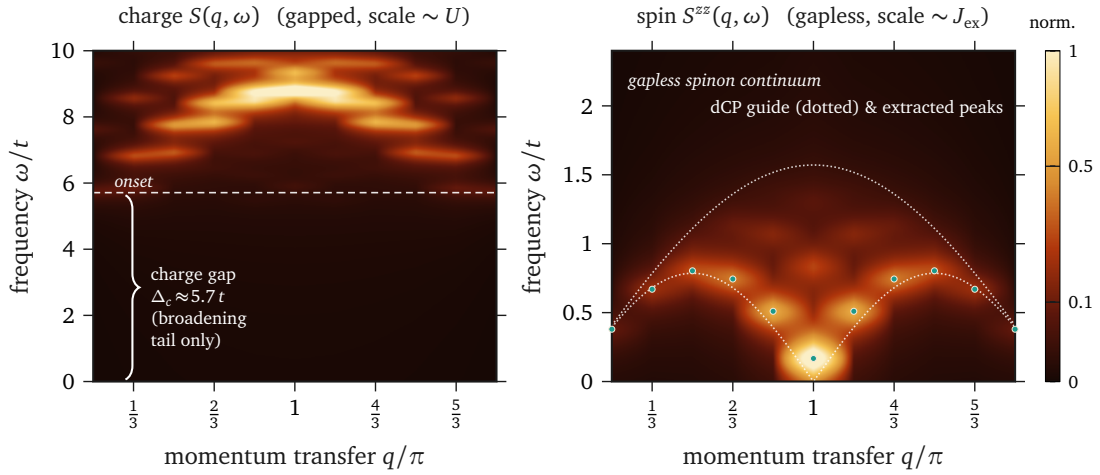


Figure 4: **Charge and spin structure factors of the half-filled Hubbard ring.** Left: $S(q, \omega)$ is gapped by a particle–hole onset $\Delta_c \approx 5.7t$ (dashed: lowest frequency above 5% of the peak intensity); below it only the broadening tail appears. Right: $S^{zz}(q, \omega)$ carries gapless spinon weight. White dotted: des Cloizeaux–Pearson (dCP) two-spinon boundaries $\omega_L = (\pi J_{\text{ex}}/2)|\sin q|$ [94] and $\omega_U = \pi J_{\text{ex}}|\sin(q/2)|$ [95], $J_{\text{ex}} = 4t^2/U = 0.5t$, an ideal large- U guide, not a strict bound (finite U , size and broadening put some weight below ω_L). Teal points: dominant spinon peak at $q/\pi \in \{1/6, \dots, 11/6\}$, within $0.17t$ of ω_L (largest offset at $q = \pi$). No moment test is applied here; Fig. 6 applies it at $L = 6$. $L = 12$, $U/t = 8$, $\eta = 0.20t$ (charge), $0.12t$ (spin); color $\propto (S/S_c)^{0.62}$, saturated at S_c , the 99.5th percentile of each panel, with one colorbar for both panels, ticked at true S/S_c ; Lanczos, Haydock evaluation [74, 76].

380 quantity. We claim no novelty for the identity, the textbook f -sum [42, 43].

381 On the collective channels (Fig. 6), for each probe $J \in \{\rho_q, S_q^z\}$, with $\rho_q = \sum_j e^{-iqj} n_j$ and
 382 $S_q^z = \sum_j e^{-iqj} S_j^z$, both moments are ground-state expectations, $m_0(q) = \langle J^\dagger J \rangle_c$ and $m_1(q) =$
 383 $\langle 0 | J^\dagger (H - E_0) J | 0 \rangle = \frac{1}{2} \langle [J^\dagger, [H, J]] \rangle$. On a half-filled $L = 6$ ring at $U/t = 8$ the operator mo-
 384 ments reproduce the exact Lehmann moments of $S(q, \omega)$ and $S^{zz}(q, \omega)$ to better than 10^{-14}
 385 at every momentum, and the two forms of m_1 , equal on the exact ground state (Appendix A),
 386 agree to the same precision. A reconstruction that moves band weight into a spurious low-
 387 frequency peak at fixed total weight passes the m_0 check (residual $\sim 10^{-15}$), but its first
 388 moment is off by 29–36% once 40% of the band weight is moved, for both observables at
 389 $q/\pi = 1/3, 2/3, 1$; this is read against a fixed 5% guide, not a calibrated threshold. The mean
 390 excitation frequency m_1/m_0 registers the two energy scales, set by U for charge ($\approx 8.5t$ at
 391 $q = 2\pi/3$) and by $J_{\text{ex}} = 4t^2/U$ for spin ($\approx 1.0t$). These are exact simulations at $L = 6$, not
 392 the $L = 12$ spectra of Fig. 4, and not a hardware result.

393 The battery also admits exactly known operator identities that screen a different error class.
 394 The $U(1)$ lattice-Schwinger Gauss law $\langle \sum_x G_x^2 \rangle = 0$ is violated by gauge-symmetry breaking;
 395 its operator is diagonal in the computational basis and so estimable from samples in principle,
 396 but here it is evaluated exactly on the state vector, and it checks the state, not a reconstruction
 397 (Appendix D) [45, 47].

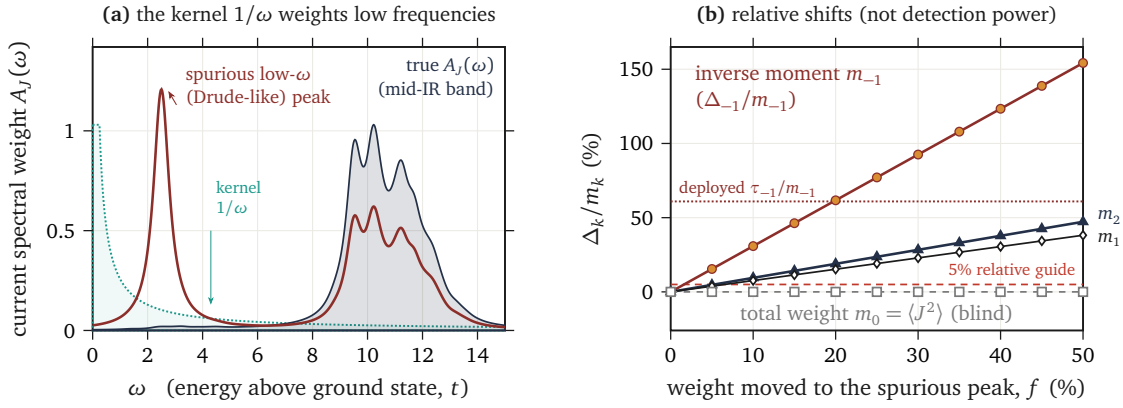


Figure 5: **Current response: misplaced weight at fixed total.** Doped Hubbard ring ($L = 12$, $U/t = 8$, $2/3$ electrons per site; $\text{Re } \sigma \propto A_J/\omega$). A fraction f of the weight above $6t$ moves to a spurious pole at $2.5t$ at fixed m_0 . **(a)** True spectrum (dark blue; 1.4% of m_0 below $6t$) and wrong reconstruction at $f = 0.4$ (dark red: spurious pole plus depleted band); Lorentzian broadening $\eta = 0.35t$. Teal: kernel $1/\omega$, rescaled, capped below $0.25t$. **(b)** Exact Δ_k/m_k versus f : $\Delta_0/m_0 < 10^{-13}$ (grey, blind); m_1 (black, open diamonds), m_2 (dark blue) and $m_{-1} = \frac{1}{2}\langle -\hat{T} \rangle - \mathcal{D} = 0.324$ (dark red) move by up to 38%, 47% and 154%. Dashed: common 5% guide. Deployed thresholds: 4.3% of m_1 and m_2 but 61% of m_{-1} (dotted), since the Kohn stiffness $\mathcal{D}$ (from the flux curvature, not the same samples) cancels 94% of $\frac{1}{2}\langle -\hat{T} \rangle$; rejection at $f \approx 0.046$ (m_2), 0.057 (m_1), 0.20 (m_{-1}).

4 Results II: why the estimator must be independent of the reconstruction

A moment read back off the reconstruction agrees with it by construction, so every moment in the screen must be estimated independently of $A(\omega)$. We illustrate this on the native failure mode of sample-based diagonalization, determinant-subspace truncation, in which the sampled subspace $\mathcal{S}$ omits configurations that carry spectral weight and the reconstruction is internally consistent with an incomplete basis [15, 16]. Two checks run on the same truncated data (Fig. 7). The back-computed control integrates the reconstructed $A(\omega)$, which lives inside $\mathcal{S}$: its first moment is $\bar{m}_1 = \langle 0_{\mathcal{S}} | J_{\mathcal{S}}^{\dagger} (H_{\mathcal{S}} - e_0) J_{\mathcal{S}} | 0_{\mathcal{S}} \rangle$, with $|0_{\mathcal{S}}\rangle$ the ground state of $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, e_0 its energy and $J_{\mathcal{S}} = \Pi_{\mathcal{S}} J \Pi_{\mathcal{S}}$, obtained from one subspace ground state without full diagonalization [74]. The independent estimator $\hat{m}_1 = \langle O_1 \rangle$ applies the full operators J and H , not their restriction to $\mathcal{S}$; in Fig. 7 it is evaluated on the exact full-sector ground state (where $e_0 = E_0$), an idealization removed below. The circular control compares $\bar{m}_1$ with itself and is zero by definition, so the contrast with it illustrates the principle rather than testing it; what the data show is whether, and at what depth, the independent discrepancy crosses a threshold. On the idealized state Δ_1 grows with truncation and crosses an illustrative fixed 5% line only at severe truncation (7% coverage).

Because the independent estimator applies the full O_k to the prepared state, Δ_k detects weight that leaks *outside* the retained determinant support; a redistribution among the retained poles contributes no leak and is seen only through the shift it causes in a tested moment, so one that preserves the tested moments passes. In exact arithmetic this blind spot closes at order $N - 1$ on N distinct poles (Theorem 2, Appendix E). At the deployed order ($m_0 - m_2$), however, the guaranteed power against such redistributions is zero, by dimension count rather than conditioning: every redistribution in the $(N - 3)$ -dimensional null space of

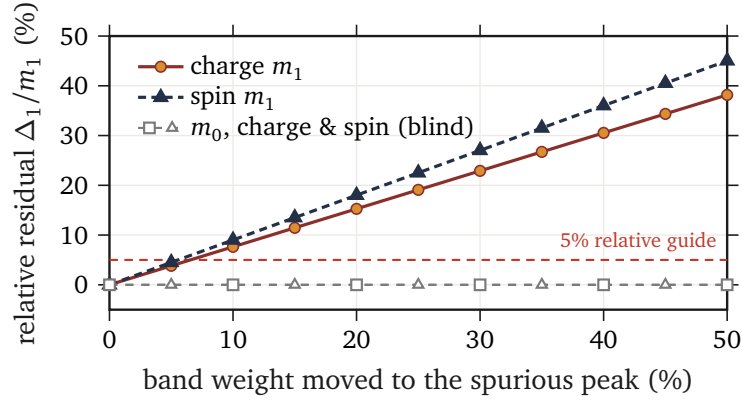


Figure 6: **Moment test on the charge and spin channels.** The comparison of Fig. 5 applied to $S(q, \omega)$ and $S^{zz}(q, \omega)$ of a half-filled Hubbard ring ($L = 6$, $U/t = 8$; exact simulation). A wrong reconstruction moves band weight into a spurious peak at $\omega = 2.0t$ (inside the charge gap) or $0.10t$ (spin), conserving m_0 exactly. At $q/\pi = 2/3$ the total-weight residuals (grey; squares charge, triangles spin, both $\sim 10^{-15}$) stay blind, while the relative first-moment residual of the charge (ρ_q , dark red) and spin (S_q^z , dark blue) probes grows linearly and crosses a fixed 5% guide, not a calibrated τ_1 . Operator and exact Lehmann moments agree to better than 10^{-14} at every $q/\pi \in \{1/3, 2/3, 1\}$. $L = 12$ spectra: Fig. 4.

the moment rows 0, 1, 2 passes, whatever its size. For the $U/t = 4$ current measure of the blinded test (Sec. 5; $N = 5$ weighted poles, a two-dimensional null space) one such redistribution moves 72% of the weight, empties two poles and passes, also staying below the m_3 and m_4 thresholds at the deployed budget (Sec. 8). The screen is thus a necessary-condition detector of support leakage and of weight misassignment that shifts the low moments; redistributions that preserve the tested moments are a disclosed blind spot, not a claimed error class.

Figure 7 is an idealization: a sample-based pipeline has only the subspace state built from the sampled support, and both branches must be computed from it. We therefore evaluate both moments on the *same* state $|0_S\rangle$, with shared reference $e_0 = \langle 0_S | H | 0_S \rangle$ (Fig. 8). The independent estimator $\widehat{m}_k = \langle 0_S | O_k | 0_S \rangle$, with $O_k = J^\dagger (H - e_0)^k J$ (Sec. 2.2), applies the *full* sparse operators, reaching determinants outside $\mathcal{S}$; the reconstruction moment $\bar{m}_k = \langle 0_S | J_S^\dagger (H_S - e_0)^k J_S | 0_S \rangle = \int \omega^k A(\omega) d\omega$ stays within $\mathcal{S}$. We use this form throughout because with it $\widehat{m}_k - \bar{m}_k$ is exactly the operator leak of Eq. (B.3), which at $k = 0$ is the missed weight $\widehat{m}_0(1 - w)$; off an eigenstate it differs, by enough to change a verdict, from the commutator form $\langle C_k \rangle$, $C_k = J^\dagger \text{ad}_H^k(J)$; both equal m_k on an exact eigenstate, and in general neither does off one (Appendix A). The form is not part of the sealed protocol, whose estimator was the exact moment plus Gaussian noise (Sec. 5); it was adopted afterwards, for these shared-state runs. The shared-state estimator needs only the sampled support and is evaluated classically; it is not a moment measured on a device. The $k = 0$ and $k = 2$ forms of the leak are the companion's captured-weight and boundary-coupling results, stated there for the probe on the exact ground state [20], and their use as a thresholded test against the circular control is this paper's (Sec. 1).

The shared-state first-moment discrepancy rises by about five orders of magnitude, from 5×10^{-6} at full support to $\approx 100\%$ at deep truncation (non-monotone near both ends: below 10^{-4} near full support, and at saturation). It crosses the 5% line at $d \approx 4.4 \times 10^4$ (18% coverage); the idealized curve crosses at $d \approx 1.6 \times 10^4$ (7%) after a dip at moderate truncation, where the full-state reference and the truncated reconstruction happen to track each other. With the

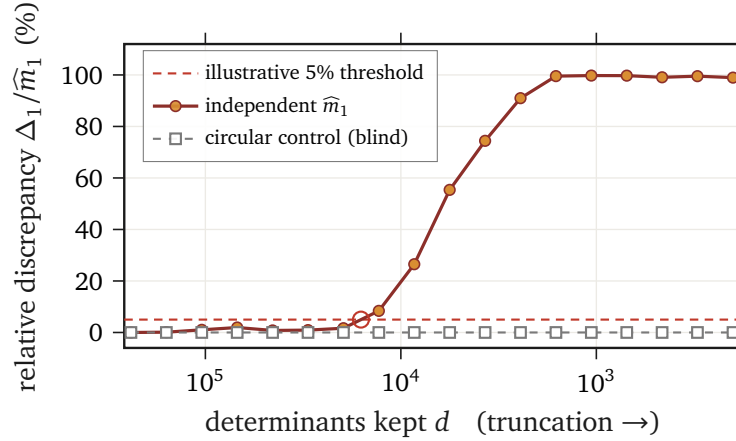


Figure 7: **Independent vs. back-computed discrepancy under determinant truncation (idealized estimator).** The determinant subspace of the doped $L = 12$ Hubbard ring ($N_\uparrow = N_\downarrow = 4$, $U/t = 8$) is truncated to its top d configurations by $|\psi_0|^2$ (the sample-based-diagonalization ordering [15, 16]; full support $n_{\text{sup}} = 2.4 \times 10^5$), and $\bar{m}_1$ is recomputed within it. The independent discrepancy $\Delta_1/\hat{m}_1 = |\hat{m}_1 - \bar{m}_1|/\hat{m}_1$ (dark red), with $\hat{m}_1$ on the exact full-sector ground state, stays below 2% down to 8% coverage, crosses an illustrative fixed 5% line (dashed) at $d \approx 1.6 \times 10^4$ (7% coverage; open ring, log-interpolated between computed d) and saturates near 100%. The circular control (grey) compares $\bar{m}_1$ with itself and is zero by definition. This first-moment channel catches only severe truncation; Fig. 8 removes the full-sector idealization and adds the zeroth-order residual, which crosses first.

449 shared state the first moment thus crosses at milder truncation than in the idealization, which
 450 at this threshold does not inflate the result. Neither branch stays exact under truncation: $\hat{m}_1$
 451 rises from 35.1 at full support to 60.1 at the deepest cut, while $\bar{m}_1$ falls toward zero. The
 452 discrepancy measures the leak, not the error of either branch against the exact moment.
 453 The first moment is not, however, the channel that responds first to truncation. On
 454 the same state the zeroth-order residual $\Delta_0/\hat{m}_0 = 1 - w$, the companion's captured-weight
 455 deficit [20] and the $k = 0$ leak of Eq. (B.3) (here $w = \|\Pi_S J|0_S\rangle\|^2/\|J|0_S\rangle\|^2$), exceeds $\Delta_1/\hat{m}_1$
 456 at every computed truncation depth from 73% down to 0.9% coverage (for example 1.6%
 457 against 0.37% at 39% coverage) and lies below it only at full support, where both are below
 458 10^{-5} , and where both have saturated (for example 93.8% against 99.4% at $d = 337$). It
 459 crosses the 5% line between $d = 6.9 \times 10^4$ (29% coverage, 4.0%) and $d = 5.0 \times 10^4$ (21%,
 460 8.3%), at milder truncation than Δ_1 (Fig. 8). The blinded simulation agrees: in the post
 461 hoc shared-state rescoring every one of its 56 truncations is rejected on m_0 , where the two
 462 operator forms coincide, so this does not depend on the choice of form, and none is rejected
 463 only by m_1 or m_2 ; under the sealed oracle-centered estimator m_0 exceeded its threshold
 464 for 55 of the 56 (Sec. 5). For determinant truncation the operative channel is therefore the
 465 companion's Δ_0 ; what this paper adds there is the thresholded test form and its contrast with
 466 the circular control, which holds for m_0 as for m_1 . The moments m_1 and m_2 add reach for
 467 errors that preserve m_0 : weight moved in frequency at fixed total (Sec. 3), and 20 of the 34
 468 m_0 -preserving spurious features of the blinded simulation, scored with its oracle-centered
 469 estimator, an outcome the pre-registration did not anticipate rather than a designed capability
 470 (Sec. 5).

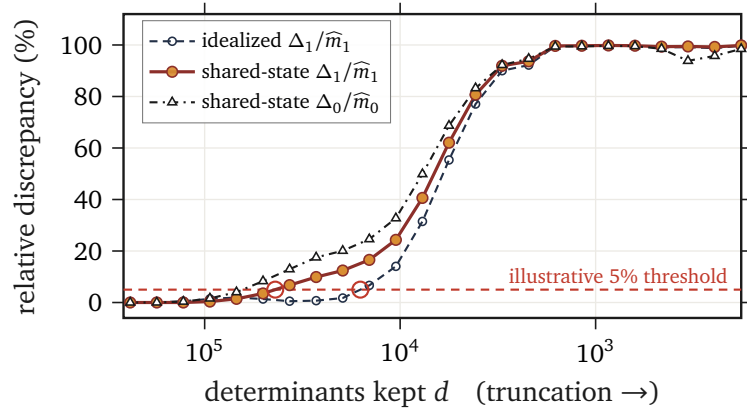


Figure 8: **Shared-state discrepancies on one subspace state.** All curves but the idealized one use the *same* ground state $|0_S\rangle$ of $H_S = \Pi_S H \Pi_S$, with $e_0 = \langle 0_S | H | 0_S \rangle$ (Appendix B); the evaluation is classical. Dark red, filled dots: $\Delta_1/\widehat{m}_1$, with $\widehat{m}_1 = \langle 0_S | O_1 | 0_S \rangle$ from the *full* operators and $\bar{m}_1$ the within- S reconstruction moment. Black, dash-dot, open triangles: $\Delta_0/\widehat{m}_0 = 1 - w$, the fraction of the weight of $J|0_S\rangle$ outside S , the companion’s captured-weight deficit [20]. Dark blue, dashed, open circles: the idealized curve of Fig. 7. Against the illustrative 5% line, $\Delta_0/\widehat{m}_0$ crosses first (between $d = 6.9 \times 10^4$ and 5.0×10^4 , 29–21% coverage; not marked), then $\Delta_1/\widehat{m}_1$ ($d \approx 4.4 \times 10^4$, 18%), then the idealized curve (1.6×10^4 , 7%); open rings mark these two log-interpolated Δ_1 crossings. At full support, and for $d < 2 \times 10^3$ where all curves have saturated, $\Delta_0/\widehat{m}_0$ lies below $\Delta_1/\widehat{m}_1$. Doped $L = 12$ ring ($N_\uparrow = N_\downarrow = 4$, 2/3 electrons per site), $U/t = 8$.

471 5 A blinded, pre-registered calibration in simulation

472 To probe the screen on errors unknown to the classifier, we ran a blinded, pre-registered sim-
 473 ulation. Before any instance existed we sealed by SHA-256 hash (0e4c3368 . . .) the battery
 474 (m_0, m_1, m_2 plus an interval-Hankel condition), the decision rule and thresholds, the error
 475 catalog, the primary endpoint, the secondary predictions and the random seed. The sealed
 476 rule rejects when any $|\widehat{m}_k - \bar{m}_k| > \tau_k$, $k \leq 2$ (the thresholds of Sec. 2.2, $z = 1.96$), when the
 477 interval-Hankel condition on the estimates fails, or when the m_1 interval lies entirely below
 478 zero (tolerance 10^{-6}). Its inputs are $N_s = 5 \times 10^4$, the per-circuit budget of the companion’s
 479 `ibm_fez` run [20] (also the largest of our later $L = 8$ runs); an assumed bias $b = 2\%$; and
 480 $\text{Var}(O_k)$, the local-estimator variance on the prepared state (the truncated-subspace ground
 481 state for truncations, the exact ground state otherwise), which the implementation evaluates
 482 with H shifted by the exact E_0 (Appendix B). Generator and classifier re-verify the hash and
 483 refuse to run if it is broken. The seal is self-attested: it carries no third-party timestamp, the
 484 blinding is enforced in code, and because the simulation is deterministic given the sealed seed,
 485 the seal cannot by itself exclude iteration before sealing. The departures of the implementation
 486 from the sealed text are collected at the end of this section.

487 The generator produced $G = 300$ instances: 112 faithful controls and three *algorithmic*
 488 error causes, determinant-subspace (SQD) truncation (56), under-converged Krylov recon-
 489 struction (71), and an analytic-continuation artifact adding a spurious spectral feature (61; a
 490 modeled artifact whose realism for sample-based pipelines is not established); the classifier
 491 saw no labels. The calibrated scope is narrow: every instance uses the current probe J of the
 492 doped $L = 6$ Hubbard ring ($N_\uparrow = N_\downarrow = 2$, sector dimension 225) at $U/t \in \{3, 4, 5, 6\}$. Trunca-
 493 tions keep the top $d = 29$ –138 of the 186 ground-state configurations by $|\psi_0|^2$ (16–74% of the

support) and are mild by weight (75.1–98.7% of $|\psi_0|^2$); in four, $J|0_S\rangle$ has no component inside S and the reconstruction is empty. Krylov instances are n_l -node Lanczos representations, $n_l \in \{1, 2, 3, 4\}$. A spurious feature is a point (δ) atom of weight 2–25% of m_0 at a frequency uniform between the sector’s lowest and highest excitation energies; with probability 1/2 the measure is rescaled to restore m_0 , and in this m_0 -preserving subclass the atom carries 2–20% of the reconstructed weight. The independent estimate is the exact full-sector moment plus unbiased Gaussian shot noise of variance $\text{Var}(O_k)/N_s$ (the idealized estimator of Fig. 7), drawn independently of $\bar{m}_k$, so the test has no same-sample covariance (a joint-sampling simulation finds a same-sample correlation but no power gain; Sec. 9).

The sealed primary endpoint was the pooled detection rate, $\text{TPR} = 101/188 = 53.7\%$ (Wilson 95% interval [46.6, 60.7]%), at a false-positive rate (FPR) of $1/112 = 0.9\%$ ([0.2, 4.9]%). The pooled rate is low because 71 of the 188 corrupted instances are under-converged Krylov reconstructions, which the battery misses (0/71). All 56 truncations were caught ([93.6, 100]%), and 74% of the spurious features (45/61): 93% (25/27) of the weight-changing ones, which the diagonal m_0 channel can reach, and 59% (20/34, [42, 74]%) of the m_0 -preserving ones, which only the off-diagonal moments can reach (a failed sealed prediction; see below). The truncation result is a property of the sampled cuts, not a guarantee: an exhaustive scan of the $U/t = 4$ ring, with exact moments, a deterministic tie order and the same thresholds (Appendix B), finds 12 truncated reconstructions that pass the full battery, three at mid coverage ($d = 86, 87, 95$ of 186) and nine near-complete ($d \geq 160$), plus three narrow windows in which m_1 alone, at the shot-only threshold, has power only at the false-positive level. The near-complete cuts lie outside the catalog’s range ($d \leq 138$), no $U/t = 4$ instance of the catalog fell on the mid-coverage ones, and the other U values were not scanned.

The Krylov misses follow from Gauss quadrature: an n_l -node Lanczos representation reproduces $m_0, \dots, m_{2n_l-1}$ exactly [48, 72], and, as the companion study proves, a subspace containing the order- K Krylov space reproduces the moments through order $2K + 1$ [20]. For $n_l \geq 2$ the reconstruction matches m_0 – m_2 to below 10^{-15} relative and is indistinguishable from a faithful one at this order. For $n_l = 1$ it misses m_2 by 2.4–4.2% and passes only because the threshold budgets a 2% bias (largest residual $0.99 \tau_2$). At the shot-only threshold (no bias term) 6 of the 15 one-node instances would be rejected on m_2 , but 5 of the 56 exact $n_l \geq 2$ instances would also be rejected, on some moment, from noise alone.

Two sealed secondary predictions failed. The one-node Krylov case, expected to be caught, was caught 0/15. The m_0 -preserving spurious-feature subclass, sealed as designed-blind with low expected detection, was caught 20/34, because the rescaling restores only m_0 and still shifts m_1 and m_2 . We report this 59% as an outcome the pre-registration did not anticipate, not as a designed capability; the register was not revised. The other secondary predictions held: high detection of truncations (56/56) and weight-changing artifacts (25/27), low detection of $n_l \geq 2$ Krylov reconstructions (0/56).

The false-positive rate is low by construction: the simulated estimator is unbiased, while the frozen threshold budgets a 2% bias comparable to the shot term. Redrawing the noise for the 112 faithful instances under the frozen rule gives a modeled FPR of 1.3%, far below the sealed nominal level of the battery, the family-wise $1 - 0.95^3 = 14.3\%$ of three comparisons at 5% each; one or fewer false positives occur in more than half of the redraws, so the sealed 1/112 is typical. A realized bias of +2% (–2%) on all three moments, at the edge of the budget, gives a modeled FPR of 8.0% (11.5%; the threshold scales with $|\widehat{m}_k|$), still below the nominal 14.3%. Within the budget the modeled rate thus stays below the nominal level; no larger bias was modeled, and the bias of a real estimator has not been measured. The 0.9% therefore confirms the implementation, not robustness to real mitigation bias, which remains untested.

Post hoc, after unblinding, we rescored the 56 sealed truncations with the shared-state estimator of Fig. 8, after a rebuild that reproduces every sealed reconstruction moment, variance and noise draw to rounding error (the rebuild depends on the stored eigensolver tie order inside degenerate $|\psi_0|^2$ shells). With $\widehat{m}_k = \langle 0_S | J^\dagger (H - e_0)^k J | 0_S \rangle$ plus the sealed noise draw and variances, the frozen rule rejects all 56, every one on m_0 (smallest $r_0 = 5.9$, where $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$); before noise, $\widehat{m}_0 - \bar{m}_0 = \widehat{m}_0(1 - w)$ is exactly the weight of $J|0_S\rangle$ outside $\mathcal{S}$. This count does not depend on the estimator form, which was chosen after the sealed test (Appendix A): the m_0 comparison alone rejects all 56, and at $k = 0$ the two forms coincide, $O_0 = C_0 = J^\dagger J$. The m_1, m_2 thresholds reuse the sealed variances, which are centered on E_0 rather than e_0 ; this leaves the m_0 verdicts unchanged. The idealized sealed estimator rejected on m_0 in 55 of the 56; the exception, id 162 ($r_0 = 0.62$, rejected only off-diagonally), is an artifact of the oracle-centered estimate and has $r_0 = 15.8$ under the shared-state estimator. That estimate is itself far from the exact moment (id 162: $\widehat{m}_0 = 1.04$ against $m_0 = 0.37$): the test registers the leak, not the distance to the truth.

Deviations from the sealed protocol. Three departures of the implementation from the sealed text, and one post hoc analysis, qualify the result. (i) The sealed catalog calls the spurious feature a Gaussian atom and gives no width; the generator adds a point atom. (ii) The sealed rule applies the interval-Hankel condition and the m_1 -sign clause to the estimates. The classifier evaluated both on the reconstruction moments, as positivity of the 2×2 Hankel matrix H_1 of $\bar{m}_0, \bar{m}_1, \bar{m}_2$ and as $\bar{m}_1 \geq 0$, which cannot fail on a catalog of positive measures on $\omega \geq 0$ (smallest eigenvalue of H_1 over the 300 instances -3.6×10^{-15} , rounding error). That branch rejected 0/300 by construction: every rejection came from the moment comparisons, and the sealed battery acted as the three comparisons alone. (iii) The sealed shot model sets the noise standard deviation to z times the local-estimator standard error; the generator draws it with the standard error itself, $\sqrt{\text{Var}(O_k)/N_s}$, the shot term inside τ_k . The literal sealed version, with 1.96 times larger noise, was not scored. (iv) The shared-state rescoring above was not pre-registered and covers only the 56 truncations; the 112 faithful controls were not rescored, so no shared-state false-positive rate exists. The sealed files were not modified, and the reported endpoint is the sealed one.

Within this catalog the screen detects support truncation and most added spectral features, and misses reconstructions that reproduce the tested moments within the thresholds. The ground truth is classically known and merely withheld from the classifier: this is a blinded test, not a field catch of an error whose correctness is unknown, and no device is in the loop; both remain future work.

Figure 9 shows how the two channels divide the work. Determinant truncation moves the total-weight sum rule $m_0 = \langle J^\dagger J \rangle$, so the diagonal channel catches it (55/56 idealized, 56/56 shared-state), as it catches most weight-changing spurious features (21 of the 25 caught). For truncation the operative channel is thus m_0 , the companion's captured-weight deficit [20]: under the shared-state estimator m_1 and m_2 add no rejection to it. The m_0 -preserving subclass keeps the total weight and the determinant support fixed by construction, so neither the m_0 residual nor a support-coverage check can see it (cf. Fig. 5); only the off-diagonal moments reject it (20/34, a post hoc reading, since low detection was pre-registered), for example id 222 with $(r_0, r_1, r_2) = (0.16, 3.3, 9.1)$. In all, 26 sealed instances have $r_0 < 1$ but an off-diagonal residual above threshold (m_1 on 16, m_2 alone on 10): the 20 above, 4 weight-changing artifacts whose weight shift fell below τ_0 , truncation id 162 and false positive id 184. The legend counts plotted points: its m_0 -preserving (20) and m_0 -changing (4) entries count only the shaded region, and the other 14 preserving and 23 changing artifacts form “spurious, unshaded” (37). Theorem 2 does not apply: the artifacts add a pole, leaving its fixed-pole class. This off-diagonal reach holds so far only in simulation. The noise analysis of Sec. 6 defers it to lower-error hardware; its forecast is for a surrogate probe at 292 CZ, a reference

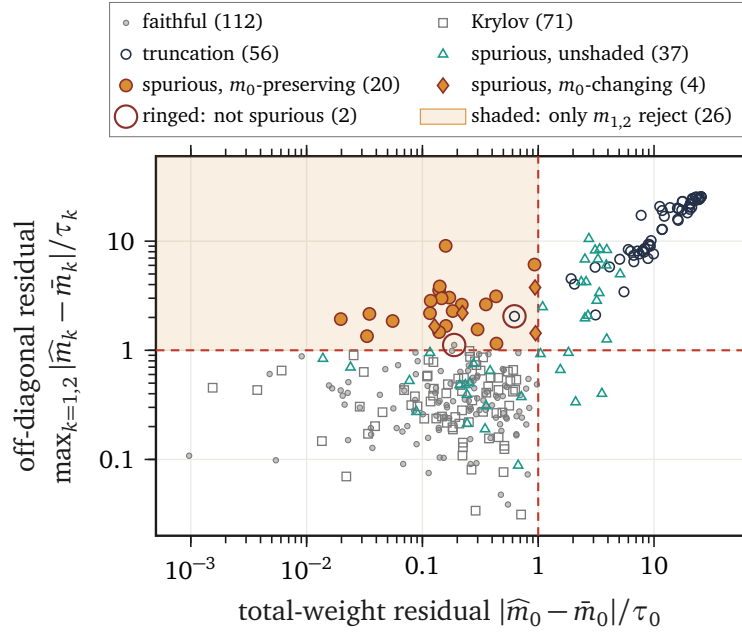


Figure 9: **The diagonal and off-diagonal moments have complementary reach** (sealed blinded simulation, current probe of the doped $L = 6$ ring; SHA-256 0e4c3368... , self-attested). $r_k = |\widehat{m}_k - \bar{m}_k|/\tau_k$ with the frozen thresholds ($z = 1.96$, $N_s = 5 \times 10^4$, assumed 2% bias); $r_k > 1$ rejects. All 300 instances are plotted at their sealed r_k ; none is clipped. Truncations: 56/56 rejected, $r_0 > 1$ in 55. Shaded: the 26 rejected only off-diagonally: 20 m_0 -preserving and 4 m_0 -changing spurious features, false positive id 184 and truncation id 162 (both ringed; id 162 has shared-state $r_0 = 15.8$). Reading the 20/34 m_0 -preserving catches as a capability is post hoc; low detection was pre-registered. Krylov (0/71): rejected by neither channel. The thresholds omit device gate-error bias, forecast for a surrogate probe on current hardware at 37–116% of m_1 (Sec. 6). A structural separation in simulation, not a device-realizable rejection.

depth rather than that of a ground-state preparation, and the deposited data give no trace ratio $\text{Tr}(O_1)/\dim \mathcal{H}$ for this current probe. On `ibm_fez` only the sampled support came from the device, and the coverage residual was evaluated on it with exact classical amplitudes.

6 Coverage residual on a device-sampled support (IBM Heron)

No moment in this paper is estimated from device counts. On hardware the device supplies only the sampled support $\mathcal{S}$; the moments $\bar{m}_0, \bar{m}_1$ read off the reconstruction are computed classically on $\mathcal{S}$, and the reference values $\widehat{m}_0, \widehat{m}_1$ used below are exact classical numbers. Same-sample estimation, in which $\widehat{m}_k$ and $\bar{m}_k$ come from one set of shots, has been realized only in simulation (Sec. 9).

The companion study ran its sample-based diagonalization pipeline at $L = 6$ on 12 qubits of `ibm_fez` (3.5×10^5 shots, 5×10^4 on each of seven circuits) [20]; its job ID is `d9s16avpemt73ct6g8g`. Its retained support is the whole addition sector ($|\mathcal{S}| = 300/300$), so the reconstruction equals exact diagonalization by coverage (Fig. 10). The bitstrings were post-selected after being read in reversed qubit order, which swaps the spin labels and moves the circuits' own output into the discarded $S_z = -1$ sector, so all 300 retained determinants

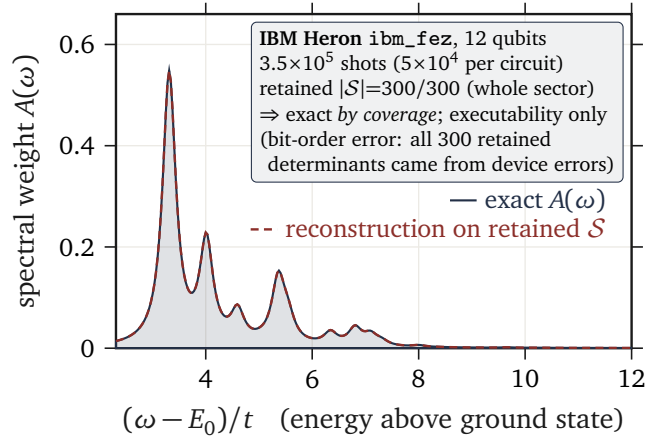


Figure 10: **Executability run of the companion study ($L = 6$, full-sector support).** Exact $A(\omega)$ (dark blue, shaded) and reconstruction on the device-retained support (dark red, dashed) against $\omega - E_0$ in units of t ; $L = 6$ Hubbard model, $U/t = 4$, addition sector $4\uparrow, 3\downarrow$ (300 configurations), $\eta = 0.15t$. `ibm_fez`, 12 qubits, 3.5×10^5 shots (5×10^4 on each of seven circuits; job `d9s16avpemts73ct6g8g`); measurement twirling, Pauli gate twirling and dynamical decoupling, no readout-error mitigation [20]; raw counts not deposited. The retained support is the whole sector ($|S| = 300/300$), so the reconstruction equals exact diagonalization by coverage. The bitstrings were read in reversed qubit order before post-selection, so all 300 retained determinants came from device errors. The run shows only that the pipeline executes; it says nothing about accuracy at incomplete coverage (Fig. 11).

came from device errors [20]. The raw counts are not deposited. The run shows only that the pipeline executes; no moment residual can be nonzero on a complete support.

To evaluate the coverage residual on an incomplete support we ran four jobs of our own at $L = 8$ (16 qubits, periodic chain, $U/t = 4$; one job per shot budget, job IDs under Data availability) and deposit their raw counts; we raised the size because at $L = 6$ even noiseless sampling of the companion's circuits covers the whole sector at 5×10^4 shots per circuit [20]. The probe is $J = c_{0\uparrow}^\dagger$ and the reference state is $|\varphi\rangle = J|0\rangle$, with $|0\rangle$ the exact half-filled ground state ($E_0 = -4.6035t$); here φ is built from the exact ground state, whereas Eq. (B.3) defines it on the subspace state $|0_S\rangle$. It lies in the addition sector $N_\uparrow = 5$, $N_\downarrow = 4$ of dimension 3920. Its weight is $m_0 = \langle J^\dagger J \rangle = \langle c_{0\uparrow}^\dagger c_{0\uparrow} \rangle = 1 - \langle n_{0\uparrow} \rangle = 1/2$ exactly, because translation invariance fixes $\langle n_{j\uparrow} \rangle = 4/8$ on every site (operator and Lehmann values agree to 6×10^{-17}). The seven circuits belong to the companion's family [20]: each prepares a computational-basis determinant with X gates (five up and four down electrons) and applies three Trotter steps of on-site R_{zz} and nearest-neighbor $R_{xx}R_{yy}$ hopping, without Jordan–Wigner strings, up to times $0, 0.5, \dots, 3.0$ in units of $1/t$. They conserve particle number and S_z , so without noise they sample only the addition sector; they do not apply the Hubbard propagator to $c_p^\dagger|0\rangle$ and are not targeted at $|\varphi\rangle$. We ran them at 5×10^4 , 3×10^4 , 1.6×10^4 and 4×10^3 shots per circuit (labeled 50k, 30k, 16k and 4k below; 3.5×10^5 shots in total at 50k) with the `SamplerV2` primitive of Qiskit Runtime, using measurement twirling, Pauli gate twirling and `XpXm` dynamical decoupling (pairs of π pulses about $+x$ and $-x$ on idle qubits); no readout-error mitigation was applied (`SamplerV2` returns twirled raw counts). Each pooled record is read in the correct bit order and post-selected to the sector, with no configuration recovery, which keeps 23.5–24.0% of the shots (82,697 of 350,000 at 50k: 92% of the shots of the zero-time circuit and 11–13% of each of the six evolved circuits).

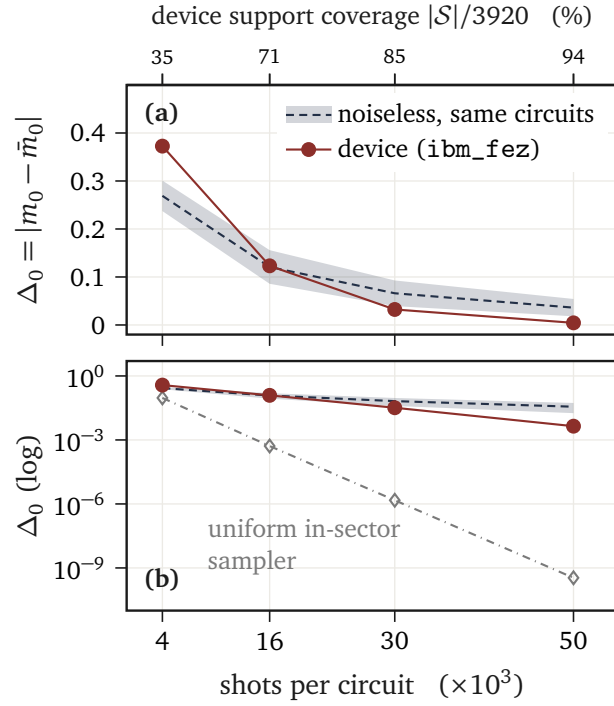


Figure 11: **Coverage residual on device-sampled supports** (ibm_fez, $L = 8$, 16 qubits, $U/t = 4$, sector dimension 3920, seven number-conserving sampler circuits). Dark red dots: Δ_0 , exact on the device-retained support $\mathcal{S}$; top axis of (a): coverage $|\mathcal{S}|/3920$. Dark blue dashed line and grey-blue band: mean and ± 1 sd (right-skewed, not a 68% interval) of 2000 noiseless multinomial replicas of the same circuits at the device’s raw shots. Grey dash-dot line with open diamonds, (b) only: uniform in-sector sampler at the device’s retained count N_{ret} , $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^{N_{\text{ret}}}$. (a) Linear, (b) log scale. The device lies below all noiseless replicas at 50k and above all at 4k (percentiles in text); the uniform sampler, which uses no device, scores lower at every budget. Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic.

The retained bitstrings define $\mathcal{S}$, on which the coverage residual $\Delta_0 = |m_0 - \bar{m}_0| = \langle \varphi | (\mathbb{1} - \Pi_{\mathcal{S}}) | \varphi \rangle$ [Eq. (B.3) at $k = 0$, with this exact-state φ] is evaluated exactly. As the budget falls from 50k to 4k, the configuration coverage $|\mathcal{S}|/3920$ falls from 94% to 35% and the residual grows: $\Delta_0 = 0.0044, 0.0323, 0.1232$ and 0.3726 (Fig. 11). In the companion’s notation $\Delta_0 = m_0(1 - w)$, with w the captured weight of $\mathcal{S}$ [20]; coverage and captured weight differ, and at 50k 94% of the configurations carry 99.1% of the weight ($1 - w = 0.009$). The first-moment residual $\Delta_1 = |\hat{m}_1 - \bar{m}_1|$ (0.019, 0.057, 0.024, 0.869) is not monotone in the budget; like Δ_0 , it combines a device-sampled support with exact classical amplitudes.

Reference distributions. We compare the device with four reference samplers and report empirical percentile ranks (the fraction of replicas at or below the device value), not Gaussian z -scores, because the weight of $|\varphi\rangle$ is concentrated (1% of the configurations carry 58% of m_0) and the distributions of Δ_0 are skewed.

(i) *Noiseless sampling at the device’s raw shots.* Two thousand multinomial replicas of the exact output distributions of the same seven circuits give $\Delta_0 = 0.036 \pm 0.018, 0.066 \pm 0.027, 0.121 \pm 0.035$ and 0.269 ± 0.032 (mean $\pm$ sd) at 50/30/16/4k. The device lies below all 2000 replicas at 50k, at the 1.7th percentile at 30k, at the 57.9th at 16k and above all 2000 at 4k. Its support is larger than in every replica at 50k and 30k ($|\mathcal{S}| = 3682$ against $3379 \pm 14; 3330$

against 3183 ± 16) and smaller than in every replica at 16k and 4k (2798 against 2877 ± 18 ; 1379 against 2027 ± 21). Noiseless sampling cannot reach $\Delta_0 = 0$: the seven output distributions together cover 3864 of the 3920 configurations, leaving 8.8×10^{-5} at infinite shots. The device's Δ_1 lies at the 35th, 82nd and 27.5th percentiles at 50/30/16k and above all replicas at 4k.

(ii) *Noiseless sampling at the device's retained counts* puts the device below every replica at 50, 30 and 16k under both count allocations, and at the 0.6th or 5.8th percentile at 4k, with a margin that shrinks as the budget falls (Appendix F).

(iii) *Uniform noise* mixed into the circuits' output with fraction p lowers the expected Δ_0 at 50k from 0.036 at $p = 0$ to 0.0024 at $p = 1$; at the p that reproduces the device's retention the mixture's mean lies below the device at every budget, although the device lies inside the mixture's replica distribution at 50–16k (Appendix F).

(iv) *Uniform in-sector sampler*. Drawing configurations uniformly from the sector at the device's retained count N_{ret} , with no device and no circuit, gives $\mathbb{E}[\Delta_0] = m_0(1 - 1/3920)^{N_{\text{ret}}}$, which is 3.4×10^{-10} , 1.5×10^{-6} , 5.2×10^{-4} and 0.093, below the device at every budget. At 50k and 30k this sampler covers essentially the whole sector, and in all ten replicas checked its reconstruction is exact to the Lanczos floor (relative L_1 error below 4×10^{-9}).

What the device run shows. Δ_0 rewards support spreading, not fidelity. The noisy device leaves less weight uncovered than all 2000 noiseless replicas of its own circuits at 50k and than 98% of them at 30k, where its support is the larger one; at 16k and 4k, where its support is smaller, it is typical or worse. Uniform noise lowers Δ_0 at 50k, and a sampler with no quantum content does best, because the 3920-configuration sector is small enough for uniform sampling to cover all or most of it at these budgets (81% of the configurations at 4k); a related effect has been reported for ground-state quantum-selected configuration interaction, where sampling noise can enlarge the configuration space beyond what the ideal ansatz supports [27]. Δ_0 grows as the budget falls for every sampler considered. The run thus shows that the coverage residual can be evaluated on a device-sampled support; it cannot distinguish a good device from a noisy one and does not certify the reconstruction. Relative to the companion's captured-weight results [20] (Sec. 1), this section adds the evaluation on incomplete, device-sampled supports at four budgets with deposited counts, and the reference distributions above.

A device-data false negative. The relative L_1 error of each reconstruction on the job's 600-point frequency grid at $\eta = 0.15t$, $\int |A_S - A| d\omega / \int A d\omega$, is 0.35, 0.56, 0.90 and 1.12 at 50/30/16/4k. At 50k both residuals, $\Delta_0 = 0.0044$ and $\Delta_1 = 0.019$, lie below the bias terms that the pre-registered rule of Sec. 5 would set ($\tau_k \geq z b m_k$ with $z = 1.96$ and $b = 2\%$: $\tau_0 \geq 0.020$, $\tau_1 \geq 0.082$), yet the relative L_1 error is 0.35. These thresholds are counterfactual: the rule was sealed for the $L = 6$ current probe, and here $\widehat{m}_0, \widehat{m}_1$ are exact. Under the same counterfactual rule the other three reconstructions, with large L_1 errors (0.56, 0.90, 1.12), would be rejected on Δ_0 (0.0323, 0.1232, 0.3726), since the shot term cannot raise τ_0 that far ($O_0 = J^\dagger J = 1 - n_{0\uparrow}$ is a projector, with variance at most $1/4$). On these device data the rule would thus reject three poor reconstructions and pass the 50k one, a false negative: it tests $A(\omega)$ only weakly and does not test the device. This is consistent with the companion's result that captured weight alone cannot certify a reconstruction: its bound places the error above $1 - w = 0.009$ and needs the boundary coupling for an upper bound [20]. In L_1 error the device beats all 80 noiseless replicas at 50k and loses to all of them at 4k, and the uniform in-sector sampler beats the device at every budget (Appendix F, which also records a superseded noise model and a nonparametric bootstrap, neither used as a test).

The discriminating off-diagonal estimator (m_1, m_2 from rotated, qubit-wise-commuting bond-basis settings, Fig. 12) is verified only in simulation (Appendix B); no off-diagonal moment has been measured on a device in this work, and on present hardware its forecast bias is a large fraction of the signal. Under global depolarizing noise of strength p the estimate

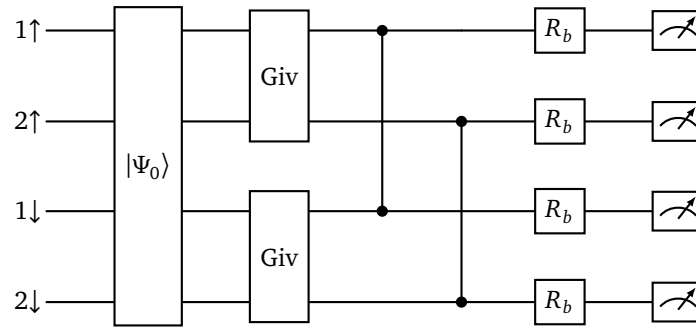


Figure 12: **Measurement circuit of the bond-basis moment estimator (schematic; not run on hardware).** One circuit, drawn for a two-site unit cell (4 qubits: two orbitals per spin). The first blocks stand for state preparation: reference state $|\Psi_0\rangle$, number-conserving hopping per spin species (Giv), on-site interaction $U\hat{n}_\uparrow\hat{n}_\downarrow$ (controlled up-down pairs). A per-qubit rotation R_b (H or $S^\dagger H$ for X or Y , identity for Z) precedes a computational-basis measurement; each qubit-wise-commuting Pauli group needs one circuit. The all- Z setting gives m_0 ; the hopping content of m_1, m_2 needs rotated settings. The verification of Appendix B samples the exact ground state in each basis instead of running a preparation. On present hardware the off-diagonal estimate would carry the bias forecast in Sec. 6.

of $m_1 = \langle 0|O_1|0\rangle$, with $O_1 = J^\dagger(H - E_0)J$, becomes $(1 - p)m_1 + p \text{Tr}(O_1)/\dim \mathcal{H}$, a bias of $p[\text{Tr}(O_1)/\dim \mathcal{H} - m_1]$. For the $L = 8$ device Hamiltonian and $J = c_{0\uparrow}^\dagger$, $\text{Tr}(O_1)/\dim \mathcal{H} = 6.80$ against $m_1 = 2.10$ (ratio 3.24), so the bias is upward. The forecast uses as a surrogate a bond-basis probe with nearly the same ratio, the $q = \pi$ density probe of the open $L = 6$ chain at $U/t = 4$ ($m_1 = 10.6$, $\text{Tr}(O_1)/\dim \mathcal{H} = 34.3$, ratio 3.23), with no error mitigation modeled. An exact density-matrix simulation with local two-qubit depolarizing noise at $\varepsilon = 2.5 \times 10^{-3}$ puts its first-moment bias at 13% of the signal at 100 controlled- Z (CZ) gates and 37% at 292 CZ; global depolarizing noise at the same depths gives 49% and 116%, and a coherent- ZZ model 47% at 292 CZ. The local value is exact for a static noise model that omits the operator spreading of a real circuit, leakage and readout error, so it is expected to underestimate, but is not proven to bound, the bias of a real circuit. The depth of 292 CZ is that of the companion's sampler circuits and serves as a reference: the bond-basis estimator needs a ground-state preparation, whose depth is not established here. The particle-number post-selection that protects the diagonal channel does not survive the bond-basis rotations. A device-side off-diagonal test therefore awaits lower gate error; no device datum here bears on off-diagonal content (Table 2).

The off-diagonal limitation concerns the bond-basis estimator of m_1 , not every discriminating on-device channel. A diagonal one exists: the momentum distribution n_k , the occupied spectral weight at momentum k , with the sum rules

$$n_k = \int_{\omega < 0} A(k, \omega) d\omega = \langle c_{k\sigma}^\dagger c_{k\sigma} \rangle, \quad \int A(k, \omega) d\omega = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1.$$

The total weight is pinned at every k and is blind to how the weight splits between the occupied and empty sides; n_k is not. Read out diagonally after a number-conserving fermionic Fourier transform, n_k keeps the particle-number post-selection that the bond-basis rotations lose. It is not immune to noise: depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own (forecast and scope in Appendix F).

A pre-registered on-device n_k test was sealed on 2026-09-02 after a simulated Stage-0 gate passed, in its fourth version and with little margin (upper confidence bound of its post-mitigation residual check 0.0260, against the sealed threshold 0.0267). Stage-0 is a noisy-simulation check of the design, not itself a pre-registered test, and it was revised the same day: the first version passed but was later ruled a model artifact; the second returned a stop verdict; the third, run after an adjudicated redesign of the gate (simulated device layout, the form of the residual check, and relaxation modeled at the floors $T_1 \geq 150 \mu\text{s}$ and $T_2 \geq 100 \mu\text{s}$, which the sealed day-of rule requires on every qubit of the chain), returned a stop verdict on one row; and the fourth replicated that row over ten seeds under a decision rule fixed before any replica ran, and passed. The test has not been run, and the committed records do not state why it was not run at these floors. Its single permitted amendment, which would have relaxed them to $T_1 = 100 \mu\text{s}$ and $T_2 = 70 \mu\text{s}$, failed its sealed simulated gate: the worst-admissible row (two-qubit error parameter $\varepsilon = 3.5 \times 10^{-3}$, a compound knob that moves the depolarizing and coherent- ZZ components together, not a physical two-qubit error rate; the forecasts above assume 2.5×10^{-3}) reached an upper confidence bound of 0.0287 against the same threshold, and the amendment was denied. The amendment's verdict file was re-sealed on 2026-09-05 after its free-text finding was corrected by hand and its first digest failed verification over line endings; the verdict is unchanged (full record in Appendix F). Like the off-diagonal test, the n_k test awaits lower-error hardware.

7 A closed-form bound: the Christoffel miss-distance

We now state what a *passing* screen implies. The bound below is classical (moments and orthogonal polynomials); our contribution is its evaluation as a diagnostic for a reconstructed spectrum, not the mathematics, and the Christoffel–Darboux kernel is an established data-analysis tool [48, 71, 99, 100]. Let $F(x) = \int_{-\infty}^x A(\omega) d\omega$ be the cumulative weight of a diagonal $T = 0$ Lehmann measure whose moments $m_0, \dots, m_{2n}$ are known exactly. Here x is a frequency; Fig. 13 plots it after an affine map onto $[-1, 1]$, which does not change W_n (Appendix C), and the hopping t remains the energy unit. The moments fix the orthonormal polynomials $p_0, \dots, p_n$ via the Hankel system [69], and the Christoffel function is $W_n(x) \equiv \lambda_n(x) = 1/K_n(x, x) = 1/\sum_{j=0}^n p_j(x)^2$, with $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ the Christoffel–Darboux kernel (construction from the moments in Appendix C) [48, 71].

Theorem 1 (Cumulative-weight miss-distance bound; classical, re-derived). *Let $A(\omega)$ be a diagonal $T = 0$ Lehmann spectrum, so that $d\mu = A(\omega) d\omega$ is a positive measure; let it have compact support and a positive-definite Hankel matrix $H_n = [m_{i+j}]_{i,j=0}^n$, and suppose the moments $m_0, \dots, m_{2n}$ are known exactly. For $x \in \mathbb{R}$ define the cumulative-weight miss-distance as the extremal spread $\sup |F_{\nu_1}(x) - F_{\nu_2}(x)|$ over all positive measures ν_1, ν_2 on $\mathbb{R}$ sharing those moments. Then for every $x \in \mathbb{R}$ this spread equals the Christoffel function $W_n(x)$, which is also the supremum of the mass a moment-matching measure can place at x (the Gauss–Radau weight $\lambda_n(x)$, attained except at the n zeros of the orthonormal polynomial p_n). In particular, any reconstruction whose moments $m_0, \dots, m_{2n}$ equal the true ones obeys $|F_{\text{rec}}(x) - F_{\text{true}}(x)| \leq W_n(x)$ at every $x \in \mathbb{R}$ (Appendix C) [51, 101].*

Remarks. No convex-hull hypothesis is needed: for $H_n \succ 0$ the Gauss–Radau rule with a node at x has positive weights for every real x except the n zeros of p_n , and its atom at x is $W_n(x)$ (Appendix C). The bound is monotone, $W_{n+1} \leq W_n$. For N distinct poles, $H_n \succ 0$ only for $n \leq N - 1$, and at a pole $W_n(\omega_i) \geq w_i > 0$ (the residue), with equality at $n = N - 1$: W_n does not vanish on the support. Pointwise vanishing $W_n(x) \rightarrow 0$ is a thermodynamic-limit statement that needs regularity of the limiting measure (Appendix C) [97, 98].

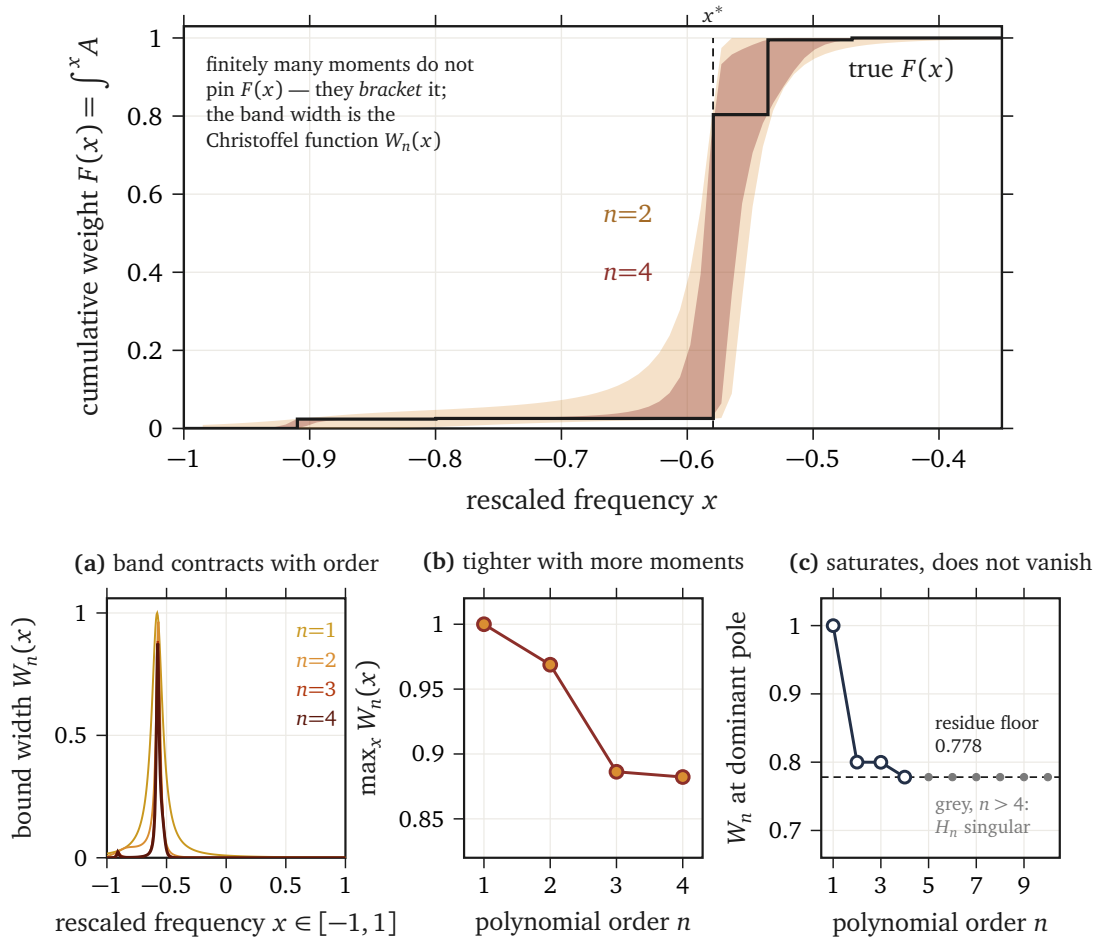


Figure 13: **The Christoffel miss-distance bound at exact moments.** *Top.* The moments $m_0, \dots, m_{2n}$ bracket the cumulative weight $F(x)$ between extremal Gauss–Radau measures, in a band of Christoffel width $W_n(x)$; the $n=4$ band (dark red) nests in the $n=2$ band (amber); x^* (dashed) marks the dominant atom. **(a)** $W_n(x)$ contracts as n grows. **(b)** $\max_x W_n = 1.000, 0.969, 0.886, 0.882$ for $n = 1, \dots, 4$; at the deployed order $n = 1$ it is $W_1 = m_0$, at the mean: the bracket is vacuous. **(c)** At the dominant atom ($\omega = 11.09 t$, residue 0.778) W_n reaches the residue (dashed) at $n = N - 1 = 4$, the last order with $H_n > 0$ (five weighted atoms); grey points ($n > 4$, H_n singular) lie outside the theorem’s hypothesis. All curves assume exact moments; a reconstruction passing within tolerances is bracketed less tightly. Doped $L = 6$, $U/t = 8$ Hubbard current measure, $m_0 = 1$, frequency rescaled to $x \in [-1, 1]$ (Appendix C) [48, 49, 69, 71, 97, 98].

771 The equality is the classical content of the Chebyshev–Markov–Stieltjes inequalities and
 772 Gauss–Christoffel moment theory [48, 49, 69]. We say “bound”, not “certify”: the screen never
 773 certifies that $A(\omega)$ is correct. The bounded object is cumulative weight, not the pointwise value
 774 of $A(\omega)$ or a peak position [48], and the contraction *rate* holds only under that regularity [97,
 775 98]. The top panel of Fig. 13 shows the mechanism: the true $F(x)$ lies between the extremal
 776 Gauss–Radau envelopes, a bracket of width $W_n(x)$.

777 The theorem assumes exact moments; the deployed screen passes a reconstruction whose
 778 residuals lie within tolerances, $|\Delta_k| \leq \tau_k$, and only through order $2n = 2$. We evaluate the
 779 consequences on the measure of Fig. 13, the doped $L = 6$, $U/t = 8$ current measure. It
 780 lies outside the blinded catalog ($U/t \in \{3, 4, 5, 6\}$, Sec. 5) and is not the $U/t = 4$ measure of

Sec. 8; the record does not state why it was chosen, and the thresholds used below are those the deployed rule would give on this state. There are two consequences. First, at $n = 1$ the exact-moment bracket is the one-sided Chebyshev (Cantelli) bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with mean $\bar{\omega} = 11.10 t$ and variance $\sigma^2 = 2.10 t^2$; its maximum m_0 lies at the mean, $0.01 t$ from the dominant atom ($11.09 t$), so the bracket is vacuous at the peak, and just above the top pole, at $\omega = 14 t$, it is still 0.20 . Second, tolerances loosen it further: the largest atom that a positive measure with $|m_k - m_k^{\text{true}}| \leq \tau_k$ ($k \leq 2$, the deployed threshold rule evaluated on this state) can place at ω is 3.2 – 7.7 times W_1 at $\omega = 14$ – $20 t$ (0.63 against $W_1 = 0.20$ at $14 t$), 10 – 11 times W_1 far outside the band ($\omega \approx 34$ – $50 t$), and more than m_0 at the mean ($1.045 m_0$). On the $U/t = 4$ measure of Sec. 8 the same computation gives a similar inflation, 3.6 – 5.7 times W_1 at $\omega = 12$ – $20 t$, above its top pole at $10.66 t$. At exact-moment orders $n \leq 4$ ($m_0, \dots, m_8$) the global maximum falls only to $\max_x W_4 = 0.882$, and at $n = N - 1 = 4$, W_4 at the dominant atom equals its residue 0.778 , below which no order can go (panel c; Appendix C). No positive-order moment beyond m_2 is estimated in this work. At the deployed order, passing the screen therefore places no constraint on the cumulative weight at the peak. This is the moment-level counterpart of the companion's finding that captured weight alone cannot certify a sample-based spectral function [20].

Refutation and bracketing are separate statements. Rejection ($|\Delta_k| > \tau_k$ for some k) does not involve W_n , and its reach is set by the thresholds (Sec. 8); a Hankel/Stieltjes violation of the estimated moments rejects the estimate, not a non-negative reconstruction. W_n governs only how tightly a passing reconstruction's cumulative weight is bracketed. The looseness above therefore limits what passing implies; it does not limit the power to reject.

At a frequency the measure leaves empty, however, $W_n(x)$ bounds the weight any moment-matching reconstruction can place there; far above the band ($\omega \approx 34$ – $50 t$), $W_1 \sim 10^{-3}$ and the higher orders are far smaller (Appendix C). These small values reflect the distance from the band, not resolution: at $n = 1$ they are the Cantelli tail. Nearer the band the bracket is wider ($W_1 = 0.20$ at $\omega = 14 t$), and the deployed tolerances raise the far-out-of-band $n = 1$ values about tenfold. This is a computed property of this measure, not a theorem, and we do not extrapolate it to the two-sided single-particle Mott gap, a distinct measure not tested here.

Proposition 1 (Two-sided Markov–Krein window-mass bound). *Let $d\mu = A(\omega) d\omega$ be a positive measure (a diagonal $T = 0$ Lehmann spectrum) with $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$, and suppose $m_0, \dots, m_{2n}$ are known exactly. For real x with $p_n(x) \neq 0$, let $F_-(x)$ be the mass that the $(n+1)$ -point Gauss–Radau rule with a node fixed at x places strictly left of x , and let $F_+(x) = F_-(x) + W_n(x)$, where $\lambda_n(x) = W_n(x)$ is that rule's atom at x ; at the n zeros of p_n , where the rule does not exist, both are taken as limits. These are the lower and upper ends of the Chebyshev–Markov–Stieltjes bracket on $F(x)$ (the ordinate-gap Lemma 3 of Appendix C). For any frequency window $[c, d]$ with $c \leq d$ real, the mass $\mu([c, d]) = \int_c^d A(\omega) d\omega$ of every positive measure sharing those moments obeys*

$$\max\{0, F_-(d) - F_+(c)\} \leq \mu([c, d]) \leq \min\{m_0, F_+(d) - F_-(c)\}. \quad (4)$$

The bracket is two-sided. Before intersection with the trivial bounds $[0, m_0]$ its width is exactly $W_n(c) + W_n(d)$; after it, the width is still at least $\max\{W_n(c), W_n(d)\}$, because $F_{\pm}$ are non-decreasing and bounded by m_0 . The bracket is therefore tight exactly when W_n is small at both window edges, and it inherits the monotone contraction $W_{n+1} \leq W_n$. The sharp Krein–Nudelman extremum, attained by a principal representation with prescribed nodes at c and d (equivalently, the optimum of the scalar truncated-moment problem over positive measures), is no wider. Equation (4) follows directly from the Chebyshev–Markov–Stieltjes inequalities and extends the one-sided point bound $|F_{\text{rec}} - F_{\text{true}}| \leq W_n$ to a two-sided bound on integrated weight [48, 50, 51, 101].

Evaluated on the same $L = 6$, $U/t = 8$ current measure, again at exact moments (Appendix C, evaluation of the Markov–Krein window bound), the bracket (4) is narrow at $n = 4$

for a window whose two edges both lie where this measure has no weight: a peak-containing window gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744; at the deployed order $n = 1$ the same window gives the wider $[0.848, 1]$. It is as loose as W_n when an edge sits on the peak, and the deployed tolerances loosen it further. The narrowness at weight-free edges is a computed property of this measure, on which W_n is small at those edges; it is not part of Proposition 1. In general W_n need not be small where a measure has no weight: $W_1 = m_0$ at the spectral mean, which for two equal atoms is the empty midpoint between them. The bound is the real-axis, ground-state-moment counterpart of the Euclidean-time two-sided spectral bounds obtained from moment-problem semidefinite programs [79–81]; its moments are ground-state operator expectations, not lattice correlators. As with the one-sided bound, we do not extrapolate it to the two-sided single-particle Mott gap.

8 Detectability and its sample-complexity limit

The Christoffel bound of Sec. 7 concerns what passing implies; rejection has a matching characterization: which errors the moment battery detects in exact arithmetic, and what it guarantees at finite thresholds, where the Vandermonde structure of the moment map sets the shot cost. The limiting error class is the *within-sector redistribution*: a corruption that keeps the retained determinant support and the reconstructed poles and only reweights those poles. Because the support is unchanged, it contributes no leak term to the estimator discrepancy of Sec. 4; it is visible only through the moment shifts that Theorem 2 characterizes. We study this class because no support-based check can signal it; the record documents no mechanism by which a sample-based pipeline produces it. The statement below is not a new theorem: part (a) restates the classical Prony/Vandermonde uniqueness of an atomic measure from its moments [69, 102], and part (b) turns it, with an elementary shifted-power bound, into a detectability-versus-shot-budget statement.

Theorem 2 (Detectability and completeness of the moment battery). *Let*

$$A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n), \quad w_n = \sum_{a: E_a - E_0 = \omega_n} |\langle a | J | 0 \rangle|^2 \geq 0,$$

be a diagonal $T = 0$ Lehmann measure on N distinct poles $\{\omega_n\}$, with w_n the total weight of the eigenstates at ω_n , and let a within-sector redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ ($\boldsymbol{\delta} \neq 0$, keeping $w_n + \delta_n \geq 0$) act on the same poles. Write

$$\mathbf{m} = V \mathbf{w}, \quad V_{kn} = \omega_n^k, \quad \Delta_k = |(V \boldsymbol{\delta})_k| = \left| \sum_n \delta_n \omega_n^k \right|, \quad \rho = \|\boldsymbol{\delta}\|_1.$$

Then $\|\boldsymbol{\delta}\|_2 \leq \rho \leq \sqrt{N} \|\boldsymbol{\delta}\|_2$, and $\rho/2$ is the weight moved when m_0 is preserved.

(a) (Completeness in exact arithmetic.) If the redistribution preserves $m_0, \dots, m_K$ with $K < N - 1$, then $\Delta_j > 0$ for at least one j with $K < j \leq N - 1$; quantitatively

$$\max_{K < j \leq N-1} \Delta_j \geq \frac{\sigma_{\min}(V_{0:N-1})}{\sqrt{N-1-K}} \|\boldsymbol{\delta}\|_2, \quad (5)$$

with $V_{0:N-1}$ the $N \times N$ node-Vandermonde on rows $0, \dots, N - 1$ and both sides in one fixed frequency unit. Hence the order- $(N - 1)$ battery is complete for fixed-pole redistributions: no nonzero within-sector redistribution preserves all of $m_0, \dots, m_{N-1}$. If the poles may also move (two measures of at most N atoms each), completeness holds at order $2N - 1$ by Prony counting [69, 102]. The only alteration invisible to every moment is weight traded among degenerate eigenstates at one frequency, which leaves $A(\omega)$ unchanged and is not an error.

(b) (Finite thresholds.) At N_s shots, with $\text{Var}(O_k)$ the per-shot variance of the estimator operator O_k and b_k the bias budget (Sec. 2.2), each moment carries the two-sided threshold $\tau_k = z_{1-\alpha/2} [\text{Var}(O_k)/N_s + b_k^2]^{1/2}$, that is, $z_{1-\alpha/2}$ times its noise floor (the square root). We call a given δ exposed at order j when its noise-free shift exceeds the threshold, $\Delta_j > \tau_j$; whether a finite-shot test then rejects it is a question of power (Corollary 1). Some redistribution preserving $m_0, \dots, m_K$ is exposed at order $K+1$ iff $\Delta_{K+1}^{\max} > \tau_{K+1}$, with $\Delta_{K+1}^{\max}$ the linear program (E.1). If rows $0, \dots, K+1$ of V have a nontrivial null space ($N - K - 2 \geq 1$), the smallest Δ_{K+1} over such redistributions is zero for every size ρ up to a limit set by positivity: the guaranteed power at order $K+1$ is zero, however well separated the poles. Two conditions bracket detection.

(i) Sufficient. If δ preserves $m_0, \dots, m_K$, it is exposed at some order $j \leq N - 1$ whenever the right-hand side of (5) exceeds $\max_{K < j \leq N-1} \tau_j$.

(ii) Necessary (cluster bound). If δ preserves $m_0, \dots, m_{s-2}$ and is supported on poles within an interval of diameter D and midpoint ω_c , then $(V\delta)_{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, so $\Delta_{s-1} \leq \rho (D/2)^{s-1}$; every higher shift obeys $\Delta_j \leq \rho \sum_{i=s-1}^j \binom{j}{i} |\omega_c|^{j-i} (D/2)^i$, which is also $O(\rho D^{s-1})$. Exposure at order $s-1$ therefore requires

$$\rho > \frac{\tau_{s-1}}{(D/2)^{s-1}}, \quad \text{i.e.} \quad N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2} \quad (6)$$

when the denominator is positive, and no N_s suffices when $\rho (D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost to detect a redistribution of fixed size at its first exposable order grows at least as $D^{-2(s-1)}$ as the cluster contracts.

Proof. Appendix E.

Remarks. Super-resolution bounds for clustered nodes [103–106] are proved for Fourier nodes, and part (b)(ii) does not rely on them; for real nodes the same $\sigma_{\min} \propto D^{s-1}$ scaling, at fixed relative geometry of the cluster, follows from Gautschi's bound [107, 108], so for an isolated cluster the sufficient condition (i) weakens with the same power as the necessary bound (ii) (Appendix E, which also gives the shot-cost scaling under uniform cluster contraction). In part (b), K is the highest preserved order and $K+1$ the tested one, so a battery through m_{K+1} faces a null space of dimension $N - K - 2$; elsewhere in the text K is the highest order of the battery itself (Sec. 2.2), and the same null space has dimension $N - K - 1$. The fixed-pole class excludes under-converged Krylov reconstructions, whose poles move; their misses are explained in Sec. 5.

Corollary 1 (Detection power vs. cluster size and shot count). *Under the Gaussian estimator validated in Fig. 14 (skewness -0.04 , excess kurtosis -0.05), the test reaches power $1 - \beta$ at level α when $\Delta_j \gtrsim z_{\alpha,\beta} \tau_j / z_{1-\alpha/2}$, that is, when the shift exceeds $z_{\alpha,\beta}$ noise floors, with $z_{\alpha,\beta} = z_{1-\alpha/2} + z_{1-\beta}$ and the bias budget counted in the noise floor as in the deployed rule. By Theorem 2(b)(ii) this requires $N_s \gtrsim z_{\alpha,\beta}^2 \text{Var}(O_{s-1}) / [\rho^2 (D/2)^{2(s-1)} - z_{\alpha,\beta}^2 b_{s-1}^2]$ at the first exposable order $s-1$, and no budget suffices below the bias-limited cluster diameter $D_\star = 2(z_{\alpha,\beta} b_{s-1} / \rho)^{1/(s-1)}$. Without bias the cost is finite for every $D > 0$ and diverges only as $D \rightarrow 0$.*

For the deployed $L = 6$, $U/t = 4$ battery the $N = 5$ poles are well separated (minimal separation $\Delta\omega_{\min} = 0.87t$; on nodes rescaled to $[-1, 1]$, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$), yet its guaranteed power against fixed-pole redistribution is zero. The reason is the null-space dimension, not conditioning: rows $0, 1, 2$ of V leave a two-dimensional null space on five poles. Over m_0, m_1 -preserving redistributions the linear program (E.1) puts the signed shift of m_2 in $[-0.92, 7.09]$ (Table 3): up to $3.8 \tau_2$ ($\tau_2 = 1.848$), but the range contains zero. Three explicit redistributions pass the deployed battery. CE1, in the null space, preserves m_0, m_1, m_2 exactly,

moves 72% of the weight ($\rho = 1.43 m_0$), empties two of the five poles and cuts the dominant residue from 0.393 to 0.037. CE2 preserves m_0, m_1 and sets $\Delta_2 = \tau_2$ while moving 86% of the weight (dominant residue $0.393 \rightarrow 0.073$); it sits on the threshold and passes because rejection requires $\Delta_k > \tau_k$. A third, CE3, also passes, as does the random redistribution committed with the original analysis (Appendix E). Orders 3 and 4 would catch some of these, but not all: with the same rule ($\tau_3 = 15.0$, $\tau_4 = 122$; the exact-state local-estimator variance behind them is expected to underestimate the true per-shot variance, so realistic thresholds would be wider), CE2 and CE3 give $\Delta_3 \approx 3.3 \tau_3$ and $\Delta_4 \approx 7.1\text{--}7.2 \tau_4$, whereas CE1 gives $0.23 \tau_3$ and $0.70 \tau_4$; over all redistributions that keep m_0, m_1, m_2 exactly, $\Delta_3 \leq 0.35 \tau_3$, and Δ_4 reaches $1.19 \tau_4$ at the one that maximizes Δ_3 , a marginal case. Neither m_3 nor m_4 is estimated in this work. The residual blind spot thus has two sources: the battery order, since with $m_0, \dots, m_K$ every fixed-pole redistribution in the $(N - K - 1)$ -dimensional null space passes, regardless of pole separation; and pole clustering, which by (6) raises the shot cost of detection at least as $D^{-2(s-1)}$. Injectivity guarantees that some m_j separates any fixed-pole redistribution; the thresholds decide whether that separation is observable at a given budget.

9 Discussion and limitations

This paper contributes a reconstruction-independent moment test for sample-based quantum spectral functions; estimators that do not read the moments off $A(\omega)$ (classical full-operator evaluation on the sampled subspace state, or bond-basis measurements on a prepared ground state, simulated only; Appendix B); a self-attested, hash-sealed blinded calibration in simulation, with an oracle-centered estimator and a classically known truth; and classical bounds on what a pass implies. The test is defined for any diagonal $T = 0$ Lehmann spectrum but calibrated only for the current probe of one doped $L = 6$ ring. It is a necessary condition. A failed comparison shows that the reconstruction and the estimate disagree; it rejects the reconstruction provided the estimator's bias lies within the assumed, unmeasured 2% budget, and a realized bias of $\pm 2\%$ still gives a modeled false-positive rate (8.0–11.5%) below the sealed nominal joint level of 14.3% (Sec. 9.2). Passing excludes only the tested failure modes. Independence from $A(\omega)$ is what allows the two branches to disagree; it does not turn agreement into a statement about the line shape.

9.1 Relation to prior work

The companion study. The companion [20] already contains the captured weight w (our Δ_0 is $m_0(1 - w)$), the moment-exactness theorem that explains the Krylov passes, the leakage identity [Eq. (B.3) at $k = 2$], and the result that captured weight alone cannot certify, of which the 5×10^4 -shot run of Sec. 6 ($\Delta_0 = 0.0044$, relative L_1 error 0.35 at $\eta = 0.15 t$) is a device-data instance. What this paper adds is stated in Sec. 1.

Positioning by mechanism. Self-verifying variational simulation certifies a prepared state through an energy-variance bound measured on the device [47]; observable-estimation validation certifies error-mitigated *expectation values* by validating the noise model and the mitigation [63]. Our test certifies neither: it tests a reconstructed spectral function for weight misassignment, documented by the companion for sample-based spectral functions [20] and, for subspace methods in general, our extrapolation from the limitations documented for their ground-state energies [23–27]. Cross-checking the classical diagonalizer across independent LAPACK paths [109] catches solver errors but not sampling incompleteness or weight misassignment. The approaches are complementary: a hardware spectrum could be produced under observable-estimation validation from a self-verified input state and then tested with our moments. We claim only the third role.

Constructive alternatives. Moment information can instead be built into the reconstruction: Nevanlinna continuation imposes the Pick/positivity structure of a causal response [110], maximum-entropy and stochastic continuation take moments and sum rules as prior information [67, 68], and Prony, minimal-pole and stochastic-pole-expansion fits can pin low-order moments to their operator values [111, 112]. Where available this is preferable, but such a spectrum is consistent only with the moments it was given, within the method that enforced them, for the data it was built from. Irmejs and Santos propose building dynamical correlation functions on hardware without real-time evolution, from weight- $O(k)$ moments measured on a prepared eigenstate, prove exponential accuracy away from the real axis, and demonstrate the method in classical simulations of small lattices [113]; a circuit-differentiation framework builds the retarded Green's function from derivatives of real-time-evolution circuits under realistic noise [114]. These programs construct the spectrum. Our test is their diagnostic counterpart for a consumer handed a finished $A(\omega)$ by a pipeline it does not control, on the real axis, where the constructive guarantees are weakest. Building the moments in and testing the result are two layers of one defense, not alternatives.

9.2 Limitations

The limits of the test are the following.

Bounds on a passing reconstruction. The Christoffel bound (Sec. 7) bounds cumulative weight, not pointwise $A(\omega)$ or peak positions; its contraction rate needs regularity [97, 98]; with m_0 – m_2 it is vacuous at the spectral mean, and a reconstruction passing with tolerances is bracketed only by a tolerance-inflated extremum (3.2 – $7.7 W_1$ at $\omega = 14$ – $20 t$ on the measure of Fig. 13). A passing test does not resolve a line shape [48, 71, 75].

Necessary, not sufficient; zero guaranteed power with m_0 – m_2 . Infinitely many measures share any finite set of moments, so passing can never certify correctness [49, 69]. On the five weighted poles of the $L = 6$, $U/t = 4$ current measure, m_0, m_1, m_2 leave a two-dimensional null space ($N - 3$ for N poles), so the guaranteed power against fixed-pole redistributions is zero whatever the pole separation; the cause is the dimension of the null space, not conditioning, and it holds for any spectrum with four or more distinct poles. Redistributions moving 72% of the weight (m_0 – m_2 kept exactly) and 86% (m_0, m_1 kept, m_2 moved by exactly τ_2 , on the boundary of the strict rejection rule) both pass (Sec. 8). The blind spot closes at order $N - 1$ in exact arithmetic (Theorem 2), which needs moments beyond m_2 (up to m_4 here), not estimated in this work; m_3 at the same shot budget would not close it, since no redistribution keeping m_0, m_1, m_2 exact shifts it by more than $0.35 \tau_3$. Truncations also escape: at a 2% bias the full battery passes 12 of 185 truncated reconstructions of the $U/t = 4$ ring (Appendix B). Device data show the same limitation: the 5×10^4 -shot $L = 8$ reconstruction has $\Delta_0 = 0.0044$ and a relative L_1 error of 0.35 (Sec. 6).

Error classes. The moment test addresses weight and subspace misassignment; the Gauss-law check addresses gauge-symmetry violation at the state level. Neither is a general noise diagnostic; device-noise trust remains the domain of observable-estimation validation [63]. Reconstructions that preserve the estimated moments pass. Under-converged Krylov reconstructions are the clearest case: they were part of the sealed catalog and count in the primary endpoint, and none was caught (0/71). With two or more Lanczos nodes they match m_0 – m_2 exactly, as the companion's moment-exactness theorem predicts [20], and the one-node case, sealed as expected to be caught, passes only within the 2% bias budget (Sec. 5).

Positivity. The positivity that underwrites the moment bounds is that of a diagonal $T = 0$ Lehmann measure. Off-diagonal and finite-temperature spectral functions are not positive measures in general and need separate treatment [44].

Estimator bias and the calibration. The blinded test simulated the estimate as the exact full-sector moment plus unbiased Gaussian shot noise, with no same-sample covariance and no

mitigation bias, while the threshold budgets a 2% bias. Its 0.9% false-positive rate is therefore low by construction: the modeled rate is 1.3% against the sealed nominal joint level of 14.3% ($\alpha = 0.05$ on each of three moments), and a realized +2% (−2%) bias raises it to 8.0% (11.5%), still below that level; no rate was computed for a bias beyond the budget, and the bias of a real estimator has not been measured (Sec. 5). A rejection therefore establishes an inconsistency, and it is attributed to the reconstruction only under the assumed budget. The post hoc shared-state rescoring rejects all 56 truncations; it covers the truncations only, and the 112 controls were not rescored, so no shared-state false-positive rate is available. That estimator is $\widehat{m}_k = \langle O_k \rangle$ with $O_k \equiv J^\dagger (H - e_0)^k J$ and $e_0 = \langle H \rangle$ on the prepared state. Off an eigenstate the commutator form $\langle C_k \rangle$, $C_k \equiv J^\dagger \text{ad}_H^k(J)$, differs from it by enough to change a verdict (Appendix A), so the form has to be chosen before a test that uses it is scored. Here it was chosen after the sealed blinded test had been scored (that test’s oracle-centered estimate does not use it) and first applied in the shared-state analysis of Sec. 4. The choice rests on the leak identity: with O_k , $\widehat{m}_k - \bar{m}_k$ is exactly the operator leak of Eq. (B.3), which at $k = 0$ is the captured-weight deficit. The 56/56 does not depend on the choice, because every truncation is rejected on m_0 , where $O_0 = C_0$.

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation, with a local estimator that uses exact ground-state amplitudes and is not device-realizable, in a study sealed before it was run (SHA-256 8bf87b5a. . . , self-attested; Appendix B). One set of shots gives both $\widehat{m}_k$ and $\bar{m}_k$, with a correlation of about 0.43 at 84% coverage, which that study reaches at 2000 shots per replica, not at the 5×10^4 of the blinded test. Treating the two estimates as independent inputs, as one would for an interval from separate data [83], makes the test over-conservative (realized false-positive rate 1.5% at a nominal 5%), and at matched false-positive rate the correlation-aware test gains no detection power.

Hardware. The device supplies only the sampled support; no moment has been estimated from device counts. The coverage residual rewards support spreading rather than fidelity: at 5×10^4 shots per circuit the device lies below all 2000 noiseless-sampling replicas of its circuits (at 4×10^3 above all), and a uniform in-sector sampler scores lower at every budget. The discriminating moments m_1, m_2 remain simulation-only. For a surrogate bond-basis probe with nearly the same trace ratio as the $L = 8$ device probe (the $q = \pi$ density probe of an open $L = 6$ chain), with no error mitigation modeled, the forecast depolarizing bias of m_1 at 292 CZ and two-qubit error 2.5×10^{-3} is 37% of the signal under a static local noise model and 116% under global depolarizing noise (Sec. 6). The local model omits operator spreading, leakage and readout error, so it is expected to underestimate the bias of a real circuit; it is not a proven lower bound. The depth of 292 CZ is that of the companion’s sampler circuits, a reference value: the ground-state preparation that the estimator needs is not constructed in this work, and its depth is not known. The sealed on-device n_k test passed its simulated gate at the manifest’s noise floors, in the gate’s fourth form (see below), but has not been run; the record does not state why it was not run at those floors, and the single permitted amendment that would have relaxed them failed its sealed simulated gate (Sec. 6; Appendix F).

Synthetic errors and the seal. Every error rejected here is synthetic: an injected corruption or a controlled truncation of otherwise exact data. The labels are withheld from the classifier, but the truth is classically known and the calibrated scope is narrow: the current probe of the doped $L = 6$ ring at $U/t \in \{3, 4, 5, 6\}$, truncations keeping 15.6–74.2% of the support configurations but 75.1–98.7% of the ground-state weight, and spurious point atoms carrying 2–25% of m_0 , modeled as analytic-continuation artifacts whose realism for sample-based pipelines is not established. Two implementation details deviate from the sealed text: the spurious feature is a δ atom rather than the sealed Gaussian atom, and the Hankel branch acts on the reconstruction moments, where it cannot fail (Sec. 5). The seal is self-attested: it

has no third-party timestamp, the blinding is implemented in code, and since the simulation is deterministic given the sealed seed, the seal cannot by itself exclude iteration before sealing. For the n_k test such iteration is on record: on the day of sealing its simulated Stage-0 gate was revised after unfavorable verdicts (a first pass later ruled a model artifact, then two versions ending in a stop) and passed in a replication of the stopping row under a rule fixed before any replica ran; the amendment's verdict file was re-sealed after its prose finding was corrected by hand and its first digest failed verification over line endings, with its pass/fail outcome unchanged (Appendix F). No error has yet been caught in a spectrum whose correctness was unknown in advance; that field test is future work.

Scale. One-dimensional Hubbard rings at $L \leq 12$ are solved outright by density-matrix renormalization group (DMRG) and exact diagonalization, so no quantum resource is needed for their spectra. This is deliberate: calibration requires exact moments, and the test is characterized against these methods, not in competition with them. Its intended regime, where no controlled classical spectral reference exists, is real-time dynamics of two-dimensional and frustrated lattices and *ab initio* molecular spectra. The only run beyond exact diagonalization quoted here is a code-scalability check (Sec. 9.3).

9.3 A code-scalability check beyond exact diagonalization

On an open Hubbard chain at $L = 24$, half filling and $U/t = 8$ ($S_z = 0$ sector of dimension $\approx 7.31 \times 10^{12}$), with the total current J as probe, S is taken from Born-rule samples of a TeNPY DMRG ground state [115]. The subspace ground state $|0_S\rangle$ of $H_S = \Pi_S H \Pi_S$ on the sampled configurations is found by sparse diagonalization, and both branches are evaluated on it. From 500 to 15 000 shots Δ_0 falls from 44.2 to 24.5 (DMRG $m_0 = 55.0$). It tracks the reconstruction error measured against DMRG, the relative error of the reconstruction's total weight $\bar{m}_0$, which falls from 0.95 to 0.55. Even at 15 000 shots the subspace energy $e_0 = -2.90 t$ is far above the DMRG $E_0 = -7.66 t$, so a large residual is expected. The check shows that the code runs at this scale; it is not a measure of power and not a beyond-classical claim.

9.4 The verification stack: two necessary layers

Evaluated as full-operator expectation values on the sampled subspace state, the battery is classical post-processing; a device-native estimate of m_1, m_2 needs separate bond-basis settings on a prepared ground state (Appendix B; simulated only). Classical shadows [61, 116] could supply such device-side estimates of $\langle O_k \rangle$; their spectral specialization extracts gaps without forming $A(\omega)$ [117], and their fermionic variant constructs dynamical correlation functions [118]. Those methods build the quantity; our test checks a finished $A(\omega)$ it did not produce.

The moment vector of every positive measure lies in the moment body: its Hankel matrices, and its localizing matrices on any interval known to contain the support, are positive semidefinite [49, 69]. A violated Hankel condition on the estimated moments rejects the estimate without reference to $A(\omega)$ (Fig. 2); it cannot reject a non-negative reconstruction. In the blinded test this branch was applied to the reconstruction moments, where it cannot fail, and rejected none of the 300 instances; it has not been applied to estimated moments anywhere in this paper.

Wang *et al.* turn the same machinery into certified two-sided bounds on many-body ground-state properties [82], and Mortimer *et al.* extend it to finite measurement statistics [83]. Applied to the operators O_k on the ground state, these semidefinite programs (SDPs) would bound the true moments, $m_k \in [\ell_k, u_k]$; this is possible in principle but not done in the cited works or here. Membership of the read-off moments $\bar{m}_k$ in such intervals would be a second necessary condition, of the same logical type as ours; their conjunction is at least as strict as either

alone and is not a composition with a sufficient condition. No finite set of moment conditions certifies $A(\omega)$.

The classical moment-body conditions (not the Wang/Mortimer programs) illustrate such bounds at exact moments on the $L = 6$, $U/t = 8$ current measure of Sec. 7 (outside the blinded catalog's $U/t \in \{3, 4, 5, 6\}$), normalized to $m_0 = 1$ (true $m_2 = 125.4$). Positivity gives $m_2 \geq m_1^2/m_0 = 123.3$. For a support contained in $[a, b]$, the Hausdorff condition $\int (\omega - a)(b - \omega) d\mu \geq 0$ gives $m_2 \leq (a+b)m_1 - ab m_0$: 147.6 for the weighted support $[2.49, 13.92]t$, which is oracle knowledge requiring the exact spectrum; a priori, 243.3 for the excitation range $[0, 21.91]t$ of the probe's four-electron, $S_z = 0$ sector and 579.8 for the full Fock range $[0, 52.21]t$ (573.7 with the lower edge at the smallest positive Fock excitation, $0.148t$). An under-dispersed moment vector ($m_2 = 122.1$) violates the positivity bound; an over-dispersed one ($m_2 = 148.8$) violates only the oracle Hausdorff bound and passes both a priori bounds. Neither condition has been applied to device-estimated moments.

The two layers could meet on integrated window weight: over all positive measures that share given moments, the window mass $\mu([c, d]) = \int_c^d A d\omega$ ranges over an interval, given in closed form at exact moments by Proposition 1 (the Markov–Krein envelope, of width at most $W_n(c) + W_n(d)$), and an SDP fed with interval-valued moments [83] would give a finite-statistics counterpart against which a reconstruction's window mass could be checked. We do not implement this. Nor do we attempt frequency-resolved weights, a limitation rather than a stylistic choice: a weight $|\langle n|J|0\rangle|^2$ at a resolved frequency is an excited-state matrix element, outside the ground-state moment hierarchy that makes these SDPs tractable; the resolved line shape almost certainly remains beyond this machinery.

9.5 Outlook

Several extensions follow. The interval-moment formulation is demonstrated on the exact state (Appendix B); a device-side use needs the mitigation bias measured and bounded rather than assumed, for example from error-model violation [119] or by validating the noise model and the mitigation [63]. Device estimation of the discriminating moments awaits lower gate error, and the sealed n_k device test has not been run (Sec. 9.2, *Hardware*). The Gauss-law check, evaluated here exactly rather than from samples, illustrates how an exactly known operator identity can serve as a test, inviting a catalog of conservation laws, symmetry sectors and sum rules, each testing a different error class [45, 75]. Extending positivity-based bounds to off-diagonal and finite-temperature spectra, where Lehmann positivity fails, is the main theoretical opening [44]. Built on kernel-polynomial, recursion and quadrature machinery [48, 75, 76], the test attaches to the sample-based diagonalization and Krylov pipelines that generate the spectra it tests [15–17, 20, 120, 121], although with m_0 – m_2 it cannot detect Krylov reconstructions with two or more Lanczos nodes and caught none of the 71 under-converged ones in the blinded test.

10 Conclusions

We have developed and calibrated, building on the companion's captured-weight and moment-exactness results [20], a reconstruction-independent, necessary-condition test for sample-based quantum spectral functions. It compares the low-order moments read off a reconstructed $A(\omega)$ with ground-state operator moments evaluated without reference to $A(\omega)$; in this paper these are always computed classically (exact moments plus simulated noise in the blinded test, full-operator evaluation on the subspace state in the shared-state and $L = 24$ checks, exact values in the device evaluation). A failed comparison rejects the reconstruction, provided

the estimator’s bias lies within the assumed, unmeasured budget; a passed one excludes only the tested failure modes and certifies nothing. The test is defined for any diagonal $T = 0$ Lehmann spectrum but calibrated only for the current probe of one doped $L = 6$ Hubbard ring; the single-particle and collective cases are illustrations with injected corruptions.

The calibration is a self-attested, hash-sealed blinded simulation of 300 instances with synthetic errors of classically known truth, in which the estimate was the exact moment plus unbiased Gaussian noise (an oracle-centered estimator); the seal cannot by itself exclude iteration before sealing. Its pre-registered primary endpoint was a detection rate of 53.7% (101/188; Wilson 95% interval [46.6, 60.7]%) at a false-positive rate of 0.9% (1/112; [0.2, 4.9]%), low by construction, because the simulated estimator is unbiased while the threshold budgets a 2% bias. The pooled rate is low because the test caught none of the 71 under-converged Krylov reconstructions (0/71). It caught every determinant truncation in the sealed draw (56/56; an exhaustive depth scan of the $U/t = 4$ ring finds 12 of 185 that pass) and 74% of the spurious features (45/61). Two sealed secondary predictions failed, and the sealed Hankel branch, applied to the reconstruction moments, could not reject. For determinant truncation the operative channel is m_0 , the companion’s captured-weight deficit: all 56 truncations are rejected on m_0 in the shared-state rescoring, and in the $L = 12$ shared-state sweep behind Fig. 8 the relative residual $\Delta_0/\widehat{m}_0$ crosses the illustrative 5% line between 29% and 21% coverage, before $\Delta_1/\widehat{m}_1$ does (at 18%). What this paper adds for truncation is the thresholded test against a circular control; m_1 and m_2 add reach mainly for errors that keep m_0 within its threshold (20 of 34 weight-preserving spurious features and 4 weight-changing ones rejected only off-diagonally, in simulation). No error has been caught in a spectrum whose correctness was unknown. With m_0 – m_2 the guaranteed power against fixed-pole weight redistributions is zero for any spectrum with four or more distinct poles; in exact arithmetic this blind spot closes only at order $N - 1$ for N poles.

The hardware run shows less. On IBM `ibm_fez` the device supplied only the sampled support of an $L = 8$ electron-addition sector; no moment was estimated from device counts. The coverage residual Δ_0 grows from 0.004 to 0.37 as support coverage falls from 94% to 35%, but it rewards support spreading rather than device fidelity: at 5×10^4 shots per circuit the device lies below every noiseless-sampling replica of the same circuits, a uniform in-sector sampler does better at every budget, and the 5×10^4 -shot reconstruction has a relative L_1 error of 0.35 while its residual is small. The run shows only that the coverage residual can be evaluated, classically and with exact amplitudes, on a device-sampled support. It does not validate the device, the reconstruction or the discriminating moments.

Several problems remain open: estimating m_1 and m_2 on hardware, where the forecast depolarizing bias for a surrogate probe at the companion’s sampler depth is 37–116% of the signal with no mitigation modeled; running the planned on-device n_k test, which has not been run (the single permitted relaxation of its noise floors failed its sealed simulated gate; Sec. 9.2); calibrating the test with a measured mitigation bias and a same-sample estimator realized on a device (so far realized only in simulation); closing the within-sector blind spot with higher moments at an affordable shot cost; applying the semidefinite moment bounds of Refs. [82, 83] to the same moment sequence; and a field test on a spectrum whose correctness is not known in advance.

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Data availability The code and data that support the findings of this study will be archived on Zenodo (DOI to be added at submission) and are openly available in the GitHub repository <https://github.com/nicolasbonilla/reconstruction-independent-moment-tests>. The repository contains the exact sector-Lanczos/Haydock engine; the interval-moment, blinded-test, within-sector and hardware-analysis scripts; the pre-registrations with their SHA-256 seals and the sealed records; the per-figure data; and Makefile targets and a notebook that run the generators. The file docs/REPRODUCE.md maps each figure and the key quoted numbers to their scripts and data files and lists the known gaps. Among them, two items behind quoted results do not regenerate exactly from source: a rerun of the $L = 12$ shared-state sweep of Fig. 8 reproduces both of its 5% crossings but moves its three smallest- d points (by up to 6.2% in $\bar{m}_1$), so the committed record is kept; and the blinded truncation instances regenerate from the seed only with a stored tie order (below). The raw measurement counts of our four $L = 8$ ibm_fez jobs, one per shot budget, are included: da983qse74ec73ajfr2g (5×10^4 shots per circuit), da98k16rbfbs73chckvg (3×10^4), da98icsjbipc73fej78g (1.6×10^4) and da98cksjbipc73fej1tg (4×10^3). So are the exporter src/export_device_dat.py of Fig. 11 and the reference-distribution generator src/delta0_reference_mc.py with its output data/2026-09-27_delta0_reference_mc.json. The raw counts of the companion's $L = 6$ run (Fig. 10) are not deposited; that run, post-selected in reversed bit order, is an executability check only (Sec. 6). The sealed blinded files are the canonical record: regenerating the truncation instances from the seed depends on tie-breaking inside degenerate $|\psi_0|^2$ shells, and the tie order that reproduces them is stored with the blinded-test addenda (data/2026-09-27_blind_addenda.json).

A Derivation of the moment identities

We derive the operator form of the spectral moments used throughout. Each m_k is a ground-state expectation value computable without the dynamics, and we state where the derivation needs an exact eigenstate. Let $H = H^\dagger$ have a complete orthonormal eigenbasis $\{|n\rangle\}$, $H|n\rangle = E_n|n\rangle$, and let $|0\rangle$ be an eigenstate of lowest energy E_0 . The derivation uses only that $|0\rangle$ is an eigenstate. If the lowest level is degenerate, $A(\omega)$ and its moments refer to the chosen member; for the $L = 6$, $U/t = 8$ current measure, whose lowest level is a spin triplet, the measure is the same for all three members because J is a spin scalar (Appendix C). The probe J (a current or density) is Hermitian unless stated otherwise; we keep $J^\dagger$ on the bra side so that the same steps also cover the non-Hermitian probes treated below. The diagonal zero-temperature Lehmann spectral function

$$A(\omega) = \sum_n |\langle n|J|0\rangle|^2 \delta(\omega - (E_n - E_0)) \quad (\text{A.1})$$

1233 is a non-negative measure supported on $\omega = E_n - E_0 \geq 0$, and its power moments are

$$m_k = \int \omega^k A(\omega) d\omega = \sum_n (E_n - E_0)^k |\langle n|J|0\rangle|^2.$$

1234 Writing $|\langle n|J|0\rangle|^2 = \langle 0|J^\dagger|n\rangle\langle n|J|0\rangle$, using $\langle n|(H - E_0)^k = (E_n - E_0)^k\langle n|$, and closing with
1235 completeness $\sum_n |n\rangle\langle n| = \mathbb{1}$,

$$\begin{aligned} m_k &= \sum_n \langle 0|J^\dagger|n\rangle (E_n - E_0)^k \langle n|J|0\rangle \\ &= \langle 0|J^\dagger (H - E_0)^k J|0\rangle = \langle 0|O_k|0\rangle, \end{aligned} \quad (\text{A.2})$$

1236 where

$$O_k \equiv J^\dagger (H - e_0)^k J, \quad e_0 = \langle H \rangle,$$

1237 is the deployed moment operator, with e_0 the energy of the state on which it is evaluated
1238 ($e_0 = E_0$ on the exact eigenstate $|0\rangle$). Because $|0\rangle$ is annihilated by $H - E_0$, acting to the right
1239 gives $(H - E_0)J|0\rangle = [H, J]|0\rangle$, and iterating,

$$(H - E_0)^k J|0\rangle = \text{ad}_H^k(J)|0\rangle, \quad \text{ad}_H(J) \equiv [H, J], \quad (\text{A.3})$$

1240 where ad_H^k denotes the k -fold nested commutator $[H, [H, \dots [H, J]]]$. Substituting Eq. (A.3)
1241 into Eq. (A.2) gives the commutator form

$$m_k = \langle 0|J^\dagger \text{ad}_H^k(J)|0\rangle = \langle 0|C_k|0\rangle, \quad C_k \equiv J^\dagger \text{ad}_H^k(J). \quad (\text{A.4})$$

1242 Both O_k and C_k are built solely from H and J , with no reference to the spectrum or to the
1243 reconstructed $A(\omega)$, and $\langle C_k \rangle = \langle O_k \rangle = m_k$ on an exact eigenstate; off an eigenstate they differ
1244 (below).

1245 The lowest moments specialize immediately. For $k = 0$, $O_0 = C_0 = J^\dagger J$ gives
1246 the total weight $m_0 = \langle J^\dagger J \rangle$, which is $\langle J^2 \rangle$ for a Hermitian probe. For $k = 1$ and
1247 $J = J^\dagger$, $m_1 = \langle 0|J[H, J]|0\rangle$; expanding the symmetric double commutator $[J, [H, J]]$
1248 $= 2JHJ - J^2H - HJ^2$ and using $H|0\rangle = E_0|0\rangle$,

$$\langle 0|[J, [H, J]]|0\rangle = 2\langle 0|J(H - E_0)J|0\rangle = 2m_1, \quad (\text{A.5})$$

1249 so that

$$m_1 = \frac{1}{2} \langle 0|[J, [H, J]]|0\rangle. \quad (\text{A.6})$$

1250 This is the first-moment (double-commutator) sum rule of A_J ; for the current operator J it
1251 fixes the mean frequency of the spectral weight. It is distinct from the optical-conductivity
1252 f -sum rule $\int_0^\infty \sigma_1 d\omega = \frac{\pi}{2} \langle -\hat{T} \rangle$ (a ground-state kinetic energy), whose regular part is πm_{-1} :
1253 all of it on open chains, while on the ring the Drude weight takes the rest (see below) [41–43].
1254 On an exact eigenstate the symmetrized form (A.6) is manifestly real and independent of the
1255 additive constant E_0 , which the one-sided expression $\langle 0|J[H, J]|0\rangle$ obscures.

1256 *Negative-order moment.* The negative-order moment $m_{-1} = \int_0^\infty A_J(\omega)/\omega d\omega$ of Fig. 5 car-
1257 ries the kernel $1/\omega$ and weights low frequencies; on the ring of that figure it rejects after
1258 m_1 and m_2 . It carries the regular part of the optical f -sum, $\int_0^\infty \text{Re } \sigma_{\text{reg}} d\omega = \pi m_{-1}$, and it
1259 too is a ground-state expectation: since the current is the polarization velocity $J = i[H, P]$,
1260 $m_{-1} = \frac{1}{2} \langle [P, [H, P]] \rangle = \frac{1}{2} \langle -\hat{T} \rangle$, a kinetic energy. On open chains we verify this identity against
1261 the Lehmann sum over the computed poles, to a relative 2×10^{-15} at $L = 6$ and 8 and 2×10^{-14}
1262 at $L = 12$ [41–43]. On the ring the polarization is multivalued and the identity acquires Kohn's
1263 stiffness correction, $m_{-1} = \frac{1}{2} \langle -\hat{T} \rangle - \mathcal{D}$ with $\mathcal{D} = \frac{1}{2} \partial_\theta^2 E_0$ the finite-size Drude weight (verified
1264 at $L = 12$ to a relative 2×10^{-6} , with $\mathcal{D}$ from the flux curvature). The open chain ($\mathcal{D} = 0$)

does not restore an advantage for the corruption of Fig. 5, because 41% of its regular weight already lies below $6t$ and $m_{-1} = 4.81$ is large: there the thresholds are crossed at $f \approx 0.17$ for m_{-1} , 0.059 for m_1 and 0.045 for m_2 . When the spurious pole is moved across $0.25-4t$, m_{-1} rejects first only for poles at $0.25t$ and $0.5t$, on both geometries (at $0.25t$ on the ring, $f \approx 0.015$ against 0.044 for m_1 and 0.043 for m_2). On open chains m_{-1} is also cheap: $\langle -\hat{T} \rangle$ is a one-body operator, measured in two qubit-wise-commuting settings that are a subset of those the m_0, m_1 estimator already requires (Appendix B).

Off an eigenstate the two forms differ, and the choice matters. Equations (A.2), (A.4) and (A.6) agree because $|0\rangle$ is an exact eigenstate of H . On any other state $|\psi\rangle$, with $e_0 = \langle \psi | H | \psi \rangle$, for example the ground state $|0_S\rangle$ of a truncated subspace Hamiltonian (Appendix B), $\langle \psi | O_k | \psi \rangle$ and $\langle \psi | C_k | \psi \rangle$ are no longer equal for $k \geq 1$, and in general neither equals the exact moment m_k . At $k = 1$, $C_1 - O_1 = -J^\dagger J (H - e_0)$, so

$$\langle \psi | C_1 | \psi \rangle - \langle \psi | O_1 | \psi \rangle = -\langle \psi | J^\dagger J (H - e_0) | \psi \rangle,$$

a term set by the state's own residual $(H - e_0)|\psi\rangle$, which vanishes on an eigenstate (for Hermitian J the real part of $\langle \psi | C_1 | \psi \rangle$ is $\frac{1}{2}\langle \psi | [J, [H, J]] | \psi \rangle$). At $k = 0$ the two forms coincide.

We evaluate O_k , for a structural reason (Appendix B). With O_k and the reference e_0 shared with the reconstruction, the residual $\widehat{m}_k - \bar{m}_k$ between the estimate and the reconstruction moment is exactly the operator leak of Eq. (B.3), the part of the probe state that H carries outside the retained subspace: it vanishes whenever that subspace contains enough of the Krylov space of the probe state, and at $k = 0$ it is the companion's captured-weight deficit $\widehat{m}_0(1 - w)$ [20]. With C_k the residual would also contain the state-residual term above, which need not vanish in that case, so it would not isolate the leak. The choice was made after the sealed blinded test of Sec. 5, whose estimate is the exact moment plus Gaussian noise, so its value does not depend on it (its sealed shot widths are local variances of $J^\dagger (H - E_0)^k J$ on the prepared state, `src/blind_generate.py`): O_k was adopted, on this argument, for the shared-state analysis of Fig. 8 (`data/2026-08-26_teeth_shared_state.json`). The post hoc shared-state rescoring of the 56 sealed truncations (Sec. 5) does not depend on the choice either, because every one of them is rejected at $k = 0$, where $O_0 = C_0$.

The following example is evidence that the choice matters, not the reason for it; it was computed afterwards, by the script `src/estimator_form_offeigenstate.py` with output in `data/2026-09-27_theory_numerics.json`. Take the current probe of the doped $L = 6$, $U/t = 4$ ring and truncate its sector to the $d = 100$ determinants of largest $|\psi_0|^2$ (44% of the 225-dimensional sector, 54% of the 186-determinant ground-state support). With the determinants ordered deterministically (descending $|\psi_0|^2$, ties broken by sector index, NumPy `lexsort`), the subspace ground state gives $\langle O_1 \rangle = 4.558$ but $\frac{1}{2}\langle [J, [H, J]] \rangle = 3.717$, against the reconstruction moment $\bar{m}_1 = 3.743$ and the exact $m_1 = 4.227$. At the deployed threshold $\tau_1 = 0.229$ (Appendix B) the O_1 form rejects the truncated reconstruction and the commutator form does not; with the truth known here, the rejection is the correct verdict, since $\bar{m}_1$ is 11% below the exact value. At the six other cuts of the same scan ($d = 30, 50, 75, 150, 186$ and 200) the two forms give the same verdict. This cut falls inside a 24-fold degenerate shell of $|\psi_0|^2$, so these values depend on how ties are broken: NumPy `argsort`, which does not fix the order of ties, gave 4.605, 3.707 and 3.810 on our machine, with the same two verdicts. In sum, the estimator is $\widehat{m}_k = \langle \psi | O_k | \psi \rangle$ with $e_0 = \langle \psi | H | \psi \rangle$ on the prepared state, and the commutator forms are used only on exact eigenstates.

Two features of these identities make them the operational core of the screen. First, O_k is an explicit operator polynomial in H and J ; mapped to qubits it is a weighted sum of Pauli strings, whose diagonal terms are read from computational-basis counts and whose hopping terms need rotated bases (Appendix B). Its expectation can therefore be evaluated on a prepared state without forming $A(\omega)$; this is the estimator $\widehat{m}_k$ of Appendix B. Besides the idealized

evaluation on the exact ground state, it is evaluated here in three ways, none on hardware: classically on the subspace state that also yields the reconstruction (Fig. 8, a deterministic $|\psi_0|^2$ truncation; the $L = 24$ DMRG test of Sec. 9.3, a shot-sampled subspace); as a local estimator from the same simulated shots, which needs the exact amplitudes (joint sampling, Appendix B); and from simulated Pauli-grouped measurements of a separately prepared ground state (Appendix B). Second, the derivation used only completeness of the eigenbasis and that $|0\rangle$ is an exact eigenstate, so Eqs. (A.2)–(A.4) hold for any probe and Hamiltonian, which is why the screen transports across models. Positivity of the measure (A.1) is not needed for the identities; it enters only in the positivity conditions and the bounds built on them.

Because the steps above never used $J = J^\dagger$, Eqs. (A.2)–(A.4) hold unchanged for a non-Hermitian probe, with $m_0 = \langle J^\dagger J \rangle$; only the symmetrized form (A.6) needs a Hermitian J . The single-particle spectral function $A_{k\sigma}(\omega)$ of Fig. 3 is the two-sided (photoemission plus inverse-photoemission) spectrum of the fermionic probe $c_{k\sigma}$, and its moments close under anticommutators rather than the double commutator:

$$\begin{aligned} \int A_{k\sigma} d\omega &= \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1, \\ \int \omega A_{k\sigma} d\omega &= \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle = \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A.7})$$

the Hartree-shifted band centroid [66]. The Hermitian double-commutator map above is the special case $J = J^\dagger$; Eq. (A.7) is the version used for the single-particle screen of Fig. 3, and it is likewise a ground-state expectation of explicit operators.

Collective probes. For the non-Hermitian probes ρ_q and S_q^z of Fig. 6 the double commutator is the symmetrized form $\frac{1}{2}[m_1(q) + m_1(-q)]$, equal to $m_1(q)$ by the inversion symmetry of the ring; both forms are evaluated on the exact ground state, where they coincide. Only the kinetic term contributes to m_1 , since the Hubbard interaction is diagonal in the occupation basis and commutes with ρ_q and S_q^z .

Derivation of the anticommutator moment identities. Equation (A.7) is not a fresh sum rule but the non-Hermitian specialization of Eqs. (A.2)–(A.4), derived here for completeness. Write $d \equiv c_{k\sigma}$, $d^\dagger \equiv c_{k\sigma}^\dagger$, with $\{|n\rangle\}$ the complete eigenbasis of H over all particle-number sectors, $H|n\rangle = E_n|n\rangle$, ground state $|0\rangle$, energy E_0 . The two-sided single-particle spectral function is the sum of an electron-addition ($N+1$) and an electron-removal ($N-1$) branch,

$$\begin{aligned} A_{k\sigma}(\omega) &= \sum_n |\langle n|d^\dagger|0\rangle|^2 \delta(\omega - (E_n - E_0)) \\ &\quad + \sum_n |\langle n|d|0\rangle|^2 \delta(\omega + (E_n - E_0)), \end{aligned} \quad (\text{A.8})$$

a non-negative measure whose support straddles $\omega = 0$ (addition poles at $\omega = E_n - E_0 > 0$, removal poles at $\omega = -(E_n - E_0) < 0$). Its power moments split accordingly,

$$\begin{aligned} \mu_p \equiv \int \omega^p A_{k\sigma} d\omega &= \sum_n (E_n - E_0)^p |\langle n|d^\dagger|0\rangle|^2 \\ &\quad + (-1)^p \sum_n (E_n - E_0)^p |\langle n|d|0\rangle|^2. \end{aligned} \quad (\text{A.9})$$

Applying to each branch the steps of Eqs. (A.2)–(A.4), namely completeness together with $(E_n - E_0)^p \langle n|X|0\rangle = \langle n|\text{ad}_H^p(X)|0\rangle$ for $X = d^\dagger$ and $X = d$, gives the closed operator form, the commutator form C_p of each branch probe:

$$\mu_p = \langle 0|d \text{ad}_H^p(d^\dagger)|0\rangle + (-1)^p \langle 0|d^\dagger \text{ad}_H^p(d)|0\rangle, \quad (\text{A.10})$$

1345 built solely from H , d , and $d^\dagger$. For $p = 0$ this is the canonical anticommutator,

$$\mu_0 = \langle 0 | d d^\dagger + d^\dagger d | 0 \rangle = \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle = 1.$$

1346 For $p = 1$, using $H|0\rangle = E_0|0\rangle$ and $\langle 0|H = E_0\langle 0|$ on the outer factors one verifies term by term
1347 that

$$\mu_1 = \langle 0 | d[H, d^\dagger] | 0 \rangle - \langle 0 | d^\dagger[H, d] | 0 \rangle = \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle, \quad (\text{A.11})$$

1348 the anticommutator first-moment identity, manifestly real and independent of the additive
1349 constant E_0 . For the Hubbard Hamiltonian $H = H_0 + H_U$ with

$$H_0 = \sum_{k'\sigma'} \epsilon_{k'} c_{k'\sigma'}^\dagger c_{k'\sigma'}, \quad H_U = U \sum_i n_{i\uparrow} n_{i\downarrow},$$

1350 and $c_{k\sigma} = L^{-1/2} \sum_j e^{-ikj} c_{j\sigma}$, the commutators evaluate in closed form,

$$[c_{k\sigma}, H_0] = \epsilon_k c_{k\sigma}, \quad [c_{k\sigma}, H_U] = \frac{U}{\sqrt{L}} \sum_j e^{-ikj} c_{j\sigma} n_{j,-\sigma},$$

1351 and since $\{c_{j\sigma} n_{j,-\sigma}, c_{j'\sigma}^\dagger\} = \delta_{jj'} n_{j,-\sigma}$,

$$\begin{aligned} \langle \{[c_{k\sigma}, H], c_{k\sigma}^\dagger\} \rangle &= \epsilon_k \langle \{c_{k\sigma}, c_{k\sigma}^\dagger\} \rangle + \frac{U}{L} \sum_j \langle n_{j,-\sigma} \rangle \\ &= \epsilon_k + U \langle n_{-\sigma} \rangle, \end{aligned} \quad (\text{A.12})$$

1352 which is the centroid quoted in Eq. (A.7) and evaluated per momentum in Fig. 3.

1353 *Scope: which checks transport to the single-particle channel.* The zeroth- and first-moment
1354 checks carry over intact: μ_0 and μ_1 are ground-state expectation values of explicit operators
1355 [Eqs. (A.10)–(A.11)], so they give reconstruction-independent residuals Δ_0, Δ_1 at each k . So
1356 does the two-sided Hamburger–Hankel positivity $H_n = [\mu_{i+j}] \geq 0$, since a positive measure
1357 on $\mathbb{R}$ has a positive semidefinite Hankel matrix whatever the sign of its support; it tests the
1358 estimated moments, not a non-negative reconstruction (Appendix B). The shifted-Stieltjes con-
1359 dition $H_n^{(1)} = [\mu_{i+j+1}] \geq 0$ does *not* transport: it presupposes support on $[0, \infty)$, and $A_{k\sigma}$ is
1360 supported on both signs of ω . The Christoffel bracket of Sec. 7 applies to any positive mea-
1361 sure on $\mathbb{R}$, but we have evaluated it only for the one-sided current measure and do not deploy
1362 it here. The single-particle screen is therefore Δ_0, Δ_1 and Hamburger–Hankel positivity; the
1363 shifted-Stieltjes condition belongs to the one-sided current and density responses.

1364 B The reconstruction-independent moment estimator

1365 We give the explicit estimator $\widehat{m}_k$ and show why it is independent of the reconstruction and,
1366 unlike the spectral moment $\bar{m}_k$, detects subspace truncation. Sample-based diagonalization re-
1367 turns a set $\mathcal{S}$ of computational-basis determinants $\{|x\rangle\}$ and diagonalizes the projected Hamil-
1368 tonian $H_{\mathcal{S}} = \Pi_{\mathcal{S}} H \Pi_{\mathcal{S}}$, with $\Pi_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |x\rangle\langle x|$, on a classical computer [15, 16]. This gives
1369 eigenpairs $\{E_n^{\mathcal{S}}, |n_{\mathcal{S}}\rangle\}$, among them the subspace ground state $|0_{\mathcal{S}}\rangle = \sum_{x \in \mathcal{S}} c_x |x\rangle$ with energy
1370 $e_0 = \langle 0_{\mathcal{S}} | H | 0_{\mathcal{S}} \rangle$. Unless $\mathcal{S}$ contains the support of the exact ground state, $|0_{\mathcal{S}}\rangle$ is an eigenstate
1371 of $H_{\mathcal{S}}$ but in general not of H , so O_k and C_k of Appendix A differ on it; we use O_k , for the
1372 reason given there and made explicit by Eq. (B.3) below. The probe J and the Hamiltonian H
1373 are sparse in this basis, with known matrix elements $J_{yx} = \langle y | J | x \rangle$ and $H_{yx} = \langle y | H | x \rangle$.

The moment $\langle 0_S | O_k | 0_S \rangle$ admits two evaluations, which coincide only when $\mathcal{S}$ is complete for the operators involved. Inserting the *subspace* resolution Π_S in place of the identity yields the reconstruction (spectral) moment,

$$\begin{aligned}\bar{m}_k &= \sum_n (E_n^S - e_0)^k |\langle n_S | J | 0_S \rangle|^2 \\ &= \langle 0_S | J^\dagger [\Pi_S (H - e_0) \Pi_S]^k J | 0_S \rangle \\ &= \langle 0_S | J^\dagger (H_S - e_0)^k J | 0_S \rangle,\end{aligned}\tag{B.1}$$

with the sum over the eigenpairs of H_S . It is exactly $\int \omega^k A(\omega) d\omega$ and lives entirely inside $\mathcal{S}$. Here $H_S - e_0$ acts on $\text{span } \mathcal{S}$, so the $k = 0$ term carries the convention $[\Pi_S (H - e_0) \Pi_S]^0 \equiv (H_S - e_0)^0 \equiv \Pi_S$, and $\bar{m}_0 = \langle 0_S | J^\dagger \Pi_S J | 0_S \rangle$ is the weight captured by $\mathcal{S}$ rather than the full $\langle J^\dagger J \rangle$. The reconstruction-independent estimator instead applies the sparse operator $O_k = J^\dagger (H - e_0)^k J$ directly,

$$\begin{aligned}\widehat{m}_k &= \langle 0_S | J^\dagger (H - e_0)^k J | 0_S \rangle \\ &= \sum_{x, x' \in \mathcal{S}} c_x^* c_{x'} \langle x | J^\dagger (H - e_0)^k J | x' \rangle,\end{aligned}\tag{B.2}$$

where each matrix element $\langle x | O_k | x' \rangle$ is a finite product of sparse operators summing over intermediate determinants that need *not* lie in $\mathcal{S}$. This is the operational meaning of independence: $\widehat{m}_k$ probes the action of J and H on $|0_S\rangle$ beyond the sampled subspace, whereas $\bar{m}_k$ is confined to it. Evaluated on $|0_S\rangle$, Eq. (B.2) is the shared-state estimator of Fig. 8. Evaluated instead on the exact ground state $|0\rangle$ (with $e_0 \rightarrow E_0$), it gives the exact moment m_k , where O_k and C_k coincide; this idealized version, more independent than a device allows, is used in Fig. 7, in the sealed blinded test (Sec. 5, with Gaussian shot noise added), and in the Monte-Carlo and interval-battery demonstrations below.

The two evaluations agree when the states that H generates from $\varphi_S \equiv J|0_S\rangle$ lie in $\text{span } \mathcal{S}$: if $\text{span } \mathcal{S}$ contains the Krylov space $\text{span}\{\varphi_S, H\varphi_S, \dots, H^K\varphi_S\}$, then $[\Pi_S (H - e_0) \Pi_S]^j \varphi_S = (H - e_0)^j \varphi_S$ for $j \leq K$ and $\widehat{m}_k = \bar{m}_k$ for every $k \leq 2K + 1$. This moment exactness, and its failure at order $2K + 2$, are proved and measured in the companion study [20]. It explains why the under-converged Krylov reconstructions of Sec. 5 pass the low-order battery: an n_l -step Lanczos reconstruction started from the probe state projects onto its Krylov space with $K = n_l - 1$ (the argument holds for any orthogonal projector, not only a determinant subspace) and so reproduces $m_0, \dots, m_{2n_l-1}$. The one-step case reproduces only m_0 and m_1 ; its m_2 is 2–4% low and passes only through the 2% bias budget (Sec. 5). Under determinant-subspace truncation the Krylov space is not contained, and

$$\widehat{m}_k - \bar{m}_k = \langle \varphi_S | (H - e_0)^k | \varphi_S \rangle - \langle \varphi_S | [\Pi_S (H - e_0) \Pi_S]^k | \varphi_S \rangle\tag{B.3}$$

measures the spectral weight the truncation leaks outside $\mathcal{S}$. Equation (B.3) follows directly from Eqs. (B.1) and (B.2), because both branches are built on the same probe state and the same reference e_0 . It is why the estimator uses O_k (Appendix A): with O_k the residual is exactly this leak, whereas the commutator form would add the term $\langle 0_S | (C_k - O_k) | 0_S \rangle$, which in general does not vanish off an eigenstate (Appendix A gives it at $k = 1$).

At $k = 0$, with the convention above, Eq. (B.3) reduces to $\Delta_0 = \langle \varphi_S | (1 - \Pi_S) | \varphi_S \rangle = \widehat{m}_0(1 - w) \geq 0$, with $w = \|\Pi_S \varphi_S\|^2 / \|\varphi_S\|^2$ the companion's captured weight (Π_S is its P_S) [20]. Evaluated instead on the exact probe state $\varphi_0 \equiv J|0\rangle$ (for the device, $J = c_{0\uparrow}^\dagger$), with only $\mathcal{S}$ taken from the device, the same expression is the coverage residual of Sec. 6. When $w = 1$ the $k = 1$ term vanishes as well, and the first term that can be nonzero is the companion's boundary leakage $\Delta_2 = \|(\mathbb{1} - \Pi_S)H\varphi_S\|^2$. The full-support reconstruction of Fig. 14 is an instance: with $\mathcal{S}$ the whole 186-determinant support of the $L = 6$, $U/t = 4$

ground state (so that $|0_S\rangle = |0\rangle$ and $\varphi_S = \varphi_0$), $w = 1$, Δ_0 and Δ_1 vanish to machine precision, and $\Delta_2 = 0.62$ (1.9% of $m_2 = 33.39$); it captures all of φ_S and is still not exact. These identities are the companion's [20], as are the bounds on the reconstruction error by $1 - w$ from below and by the boundary coupling from above and the result that captured weight alone cannot certify; their use as a thresholded rejection test is this paper's (Sec. 1). For the shared-state estimator, $\Delta_k = |\widehat{m}_k - \bar{m}_k|$ is nonzero only if S misses part of φ_S or of the states $H^j \varphi_S$ that order k probes; the independent estimator can then reject, while the circular control ($\bar{m}_k$ against itself) returns $\Delta_k \equiv 0$ (Sec. 4). Weight moved *within* S contributes no leak term; it is visible only through the shift of the reconstruction moments that it causes, which Theorem 2 and Appendix E characterize.

Equation (B.2) is equivalently an importance-weighted average over S ,

$$\widehat{m}_k = \sum_{x \in S} |c_x|^2 O_k^{\text{loc}}(x), \quad O_k^{\text{loc}}(x) = \frac{\langle x | O_k | 0_S \rangle}{\langle x | 0_S \rangle}, \quad (\text{B.4})$$

the local-operator estimator familiar from variational Monte Carlo [123]; the local moment $O_k^{\text{loc}}(x)$ again sums over determinants connected to x by O_k and so reaches outside S . Drawing N_s configurations from $|c_x|^2$ and averaging O_k^{loc} gives an unbiased Monte-Carlo estimate with variance $\text{Var}(O_k)/N_s$, where $\text{Var}(O_k) = \sum_x |c_x|^2 |O_k^{\text{loc}}(x)|^2 - \widehat{m}_k^2$ is the local-estimator variance. This form needs the amplitudes c_x , which a device does not return, so we use it only in simulation (Fig. 14 and the joint-sampling study below). On a device the same expectation would be estimated from Pauli-grouped measurements, described below, with a variance of its own.

Same-sample statistics. Strict same-shot estimation of both branches has been realized only in simulation, in a study sealed before it was run (SHA-256 8bf87b5a... , 2026-09-01, self-attested; data/2026-09-01_lever1_prereg_seal.json). For the current probe of the $L = 6$, $U/t = 4$ ring it drew 2000 faithful and 2000 corrupted replicas at five coverages of the ground-state support, 84, 90, 95, 99 and 100%, reached with 2000, 3200, 5000, 12000 and 40000 shots per replica; the sealed corruptions are m_0 -preserving reweightings of each replica's reconstruction within its support (a tilt and a satellite, of both signs, at seven target shifts of m_1 from 2% to 24%). The run in the record applied one sign only, and its realized mean shifts of Δ_1 are 0.4–20% of m_1 , not the targets. One set of shots per replica gives both $\widehat{m}_k$ and $\bar{m}_k$; here $\widehat{m}_k$ is the local estimator above, which uses the exact ground-state amplitudes, so it is not device-realizable. The sealed primary endpoints were the realized false-positive rate at the nominal z and the detection-power difference at matched false-positive rate. The variance of the residual is $\text{Var}(\Delta_k) = \text{Var}(\widehat{m}_k) + \text{Var}(\bar{m}_k) - 2 \text{Cov}(\widehat{m}_k, \bar{m}_k)$, with a same-sample correlation $\rho \approx 0.43$ at 84% coverage, that is at 2000 shots per replica rather than the 5×10^4 of the deployed rule; the correlation decays towards zero as coverage approaches 100%. Treating the two estimates as independent inputs, as one would when a moment interval comes from separate data, for example from the finite-statistics certificate of Mortimer *et al.* [83], over-estimates this variance, and the test becomes over-conservative in the under-sampled regime (realized false-positive rate 1.5% at a nominal 5%). At matched false-positive rate the correlation-aware test gives no detection-power advantage (power differences between -5.6 and $+0.05$ percentage points across the ladder of corruptions and coverages that was run), so the covariance is a calibration correction, not a power gain. That comparison used the covariance estimated from the Monte Carlo itself, an oracle. The committed record (data/2026-09-01_lever1_oracle.json) contains only this oracle stage, as point estimates without confidence intervals, and records no verdict on the six sealed negative rules; no single-sample plug-in stage is in the record (the script's decision gate calls for one only if the oracle stage shows a gain). This structure is a quantum instance of Hausman-type specification testing [124]: $\widehat{m}_k$ is the consistent, model-free estimator and $\bar{m}_k$ the model-implied one. The independent-input variance $\text{Var}(\widehat{m}_k) + \text{Var}(\bar{m}_k)$ is exact only at full coverage, where

$\rho \rightarrow 0$; Hausman's efficiency limit, $\text{Cov}(\widehat{m}_k, \bar{m}_k) = \text{Var}(\bar{m}_k)$, would instead reduce the residual variance to $\text{Var}(\widehat{m}_k) - \text{Var}(\bar{m}_k)$, and our correlation lies between the two limits.

The exact-moment test takes its finite-shot form here. We model the estimator uncertainty as $\delta_k^2 = \text{Var}(O_k)/N_s + b_k^2$, a shot term over N_s samples plus a bias budget b_k for mitigation and device error, and reject the reconstruction at level α when

$$\Delta_k = |\widehat{m}_k - \bar{m}_k| > \tau_k, \quad \tau_k \equiv z_{1-\alpha/2} \delta_k.$$

The deployed rule, the single threshold rule of this paper (used here only in simulation and on classically computed moments), takes $\alpha = 0.05$ per moment ($z_{0.975} = 1.96$), $N_s = 5 \times 10^4$ and $b_k = 0.02|\widehat{m}_k|$; for the three-moment battery the sealed nominal joint level is $1 - 0.95^3 = 14.3\%$ (Sec. 5). For the current probe of the doped $L = 6$, $U/t = 4$ ring, with $\text{Var}(O_k)$ and $\widehat{m}_k$ taken on the exact ground state (data/2026-09-27_interval_battery.json), it gives $\tau_0 = 0.029$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$. The level holds only while the estimator's bias stays within b_k , which is assumed, not measured: a rejection shows an inconsistency between estimate and reconstruction, and attributing it to the reconstruction rests on that assumption. A second, separate check concerns the estimate alone: whether some moment vector in the box $[\widehat{m}_\ell - \tau_\ell, \widehat{m}_\ell + \tau_\ell]$ satisfies the positivity conditions obeyed by the moments of every positive measure, $H_n \geq 0$ and, for one-sided $T = 0$ support, $H_n^{(1)} \geq 0$. Infeasibility would show that the estimated moments belong to no spectrum, pointing to the device, the mitigation or the estimator; it cannot refute a non-negative reconstruction, whose own moments $\bar{m}_k$ satisfy these conditions by construction. In the implementations reported here (the sealed blinded test and the interval battery below) the positivity condition is evaluated at $\bar{m}_k$ and therefore cannot reject; every rejection comes from a residual Δ_k . The operators, targets and intervals of each demonstration are in the reproduction materials.

The estimator also admits a device-native evaluation. We verify it in simulation because it settles how the off-diagonal (hopping) content of $\widehat{m}_1, \widehat{m}_2$ would be obtained on hardware. For a probe diagonal in the computational basis, the density probe $\rho_q = \sum_j \cos(qj) n_j$ at $q = \pi$, the moment operator is O_k of Appendix A with $J = \delta \rho_\pi \equiv \rho_\pi - \langle \rho_\pi \rangle$ and $e_0 = E_0$. It is a sum of Pauli strings, and $\widehat{m}_k = \sum_p c_p \langle P \rangle$, the expectation $\langle 0|O_k|0 \rangle$ written as a linear combination of Pauli expectations, is estimated from counts. The m_0 term is a pure computational-basis estimator. The hopping terms in m_1, m_2 need rotated-basis measurements, one circuit per qubit-wise-commuting group of Pauli strings, the standard grouping for Pauli observables; classical shadows are a randomized alternative [61]. This path measures a prepared ground state in rotated bases, with circuits separate from those whose samples define $\mathcal{S}$: it shares the device with the reconstruction but not the samples.

We verify it by finite-shot sampling of the exact ground state of an open $L = 6$ Hubbard chain at $U/t = 4$ (12 qubits), with 5×10^4 shots in each of the 357 qubit-wise-commuting bases that (m_0, m_1, m_2) require (1.8×10^7 shots per repetition). Over 8 repetitions the mean estimates $(\widehat{m}_0, \widehat{m}_1, \widehat{m}_2) = (2.730, 10.55, 47.01)$ lie 0.87, 0.72 and 0.66 standard errors (spread over repetitions divided by $\sqrt{8}$) from the exact $(2.737, 10.61, 47.64)$; for the $L = 4$ chain (8 qubits, 126 bases) every moment is recovered within 0.06 standard errors. The number of bases is set by the Pauli support of O_k , not by the moment values, so it can be counted without the statevector once the probe shift $\langle \rho_\pi \rangle$ is known. That shift vanishes by the reflection symmetry of the open chain: for even L , reflection maps ρ_π to $-\rho_\pi$, so $\langle \rho_\pi \rangle = 0$ in a reflection-symmetric ground state. The ground states are not half filled (2, 4 and 5 electrons at $L = 4, 6$ and 8, where $\langle \rho_\pi \rangle = 0$ was checked numerically; it is assumed at $L = 10$ and 12). The committed script `src/estimator_scaling.py` counts the bases this way and reproduces the count from the full eigendecomposition where affordable ($L \leq 8$). The full (m_0, m_1, m_2) battery needs 126, 357, 721, 1191, 1837 bases at $L = 4, 6, 8, 10, 12$ (8 to 24 qubits); the (m_0, m_1) pair

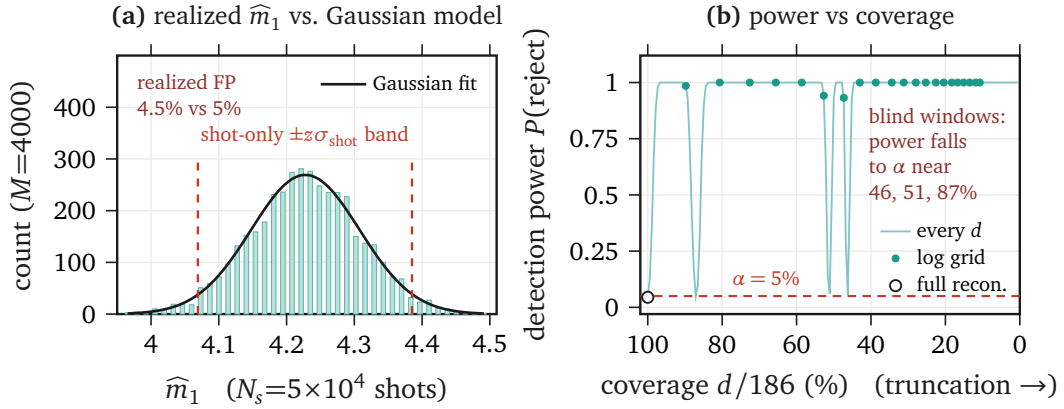


Figure 14: **Monte-Carlo calibration of the m_1 moment test** (simulation only, no device data: current probe, doped $L = 6$, $U/t = 4$ ring; $M = 4 \times 10^3$ replicas of $N_s = 5 \times 10^4$ shots from $|\langle x|0\rangle|^2$). **(a)** $\widehat{m}_1$ is near-Gaussian (curve: Gaussian fit, the draws' mean and s.d.); false-positive rate 4.5% (Wilson 95% interval $[3.9, 5.2]\%$), nominal 5%. Dashed: shot-only band $\pm z\sigma_{\text{shot}} = \pm 0.158$ about the exact m_1 , narrower than the deployed $\tau_1 = 0.229$ (2% bias budget). **(b)** Power of m_1 alone at the shot-only threshold versus coverage $d/186$ of the ground-state support (line: every d ; dots: 22-point log grid with Wilson 95% intervals; open marker at $d = 186$: full reconstruction, false-positive rate). Power reaches ≥ 0.95 at 161 of 185 truncations, falling to the false-positive level in windows near 46%, 51% and 87% coverage whose location depends on tie-breaking in degenerate $|\psi_0|^2$ shells. The shot term (exact-state variance) likely understates a device estimator's per-shot uncertainty.

alone needs 19, 35, 51, 67, 83, growing roughly linearly. The number of settings thus grows polynomially in this range, not exponentially. We do not turn this into a deployability claim: the counts-based recovery was validated only up to 12 qubits and the basis count against full diagonalization only up to $L = 8$; the shot budget and accuracy at 24 qubits remain to be characterized; and a device would have to prepare the ground state that the classical check takes as given.

The estimator, including its off-diagonal content, is therefore measurable from a fixed shot budget in principle, but not on present hardware; the noise forecast for the bond-basis estimate and its caveats are given once, in Sec. 6. The relation behind that forecast is simple: under global depolarizing noise of strength p the estimate of $\langle 0|O|0\rangle$ becomes $(1-p)\langle 0|O|0\rangle + p \text{Tr}(O)/\dim \mathcal{H}$, a bias $p[\text{Tr}(O)/\dim \mathcal{H} - \langle 0|O|0\rangle]$. For the electron-addition probe of the $L = 8$ device Hamiltonian, $O = O_1$ with $J = c_{0\uparrow}^\dagger$, that is $O = c_{0\uparrow}(H - E_0)c_{0\uparrow}^\dagger$ with $\langle 0|O|0\rangle = m_1$, one has $\text{Tr}(O)/\dim \mathcal{H} = 6.80$ against $m_1 = 2.10$ (ratio 3.24), so the bias is upward. No off-diagonal moment has been measured on a device in this work. A job script that builds the rotated-basis circuits and applies this estimator (`src/hardware_blind_job.py`) is in the repository; its state preparation is a placeholder, it has not been run on a quantum processor, and an off-diagonal test on hardware awaits lower gate error.

We close with a Monte-Carlo calibration of this test (Fig. 14) at $N_s = 5 \times 10^4$, the per-circuit budget of the sealed rule (that of the companion's `ibm_fez` run, and the largest of our later $L = 8$ budgets). It is a simulation: $M = 4 \times 10^3$ replicas of N_s shots are drawn from the exact ground-state distribution $|\langle x|0\rangle|^2$ of the doped $L = 6$, $U/t = 4$ ring, a system not run on hardware; $\widehat{m}_1$ is the local estimator above for its current probe, and each truncated reconstruction is built from the ground state of the truncated Hamiltonian. The Monte Carlo checks the analytic interval rather than replacing it. The realized estimator is close to Gaussian

(skewness -0.04 , excess kurtosis -0.05), its spread matches the analytic shot term to 1% (0.0799 against 0.0806), and its false-positive rate, 4.5% (Wilson 95% interval $[3.9, 5.2]\%$), is consistent with the nominal 5% (Fig. 14a). Two inputs are modeled: the shot term uses the exact-state local-estimator variance, which assumes known amplitudes and omits ratio-estimator and Pauli-grouping overhead, so it likely understates the per-shot uncertainty of a device estimator; the bias term b_k is assumed and swept, not measured. For this probe $\text{Var}(O_1) \approx 3.2 \times 10^2$ gives a shot standard error $\sqrt{\text{Var}(O_1)/N_s} \approx 0.081$ (1.9% of m_1); with the 2% bias budget $\tau_1 = 0.229$ (5.4% of m_1), close to the 5% relative line of Figs. 7 and 8.

Power is high but not uniform in the truncation depth. At the shot-only threshold m_1 alone reaches power ≥ 0.95 at 161 of the 185 truncations, already at $d = 181$ (97% coverage, power 0.98). The exceptions are the four mildest truncations ($d = 182$ –185, one to four determinants dropped) and three narrow windows, $d = 84$ –88, 93–98 and 158–166 (coverage 45–47%, 50–53%, 85–89%), at whose centers power falls to the false-positive level (5.5%, 6.1%, 5.0% at $d = 86, 95, 162$). The 22-point grid of Fig. 14b touches the first two windows only at their edges ($d = 88, 98$; power 0.93, 0.94) and misses the third. An earlier grid-only analysis, with ties broken by floating-point noise, showed a single low-power point near 53% coverage; the scan at every d supersedes it. The noise-free residuals agree: at the deployed 2% bias budget $|\widehat{m}_1 - \bar{m}_1| < \tau_1$ for 17 of the 185 truncations ($d = 183$ –185, 159–165, 94–97, 85–87), and at a 4% budget for 27.

Where the cut falls matters. The probabilities $|\psi_0|^2$ of this sector come in exactly degenerate shells of 6, 12 or 24 determinants, and which members of a shell a truncation keeps is a convention. All the window cuts, and 19 of the 21 truncated grid points, fall inside a shell. Our convention is deterministic (descending $|\psi_0|^2$, ties broken by sector index), but verdicts near the windows depend on it. At $d = 98$ it gives $|\widehat{m}_1 - \bar{m}_1| = 0.284 > \tau_1$, while 300 random choices of the 8 kept members of the 24-fold shell give 0.172–0.295 and rejection by m_1 alone in 57% of cases; an earlier run with ties broken by floating-point noise found 0.225, a miss. At the ten cuts that fall on a shell boundary ($d = 6, 18, 30, 42, 66, 90, 114, 138, 150, 174$), and so do not depend on the convention, the noise-free residual of m_1 alone exceeds τ_1 at the 2% budget.

Detection sets in at much higher coverage here than in the independence figures, whose idealized full-state and shared-state curves cross their 5% line near 7% and 18% retained determinants ($d \approx 1.6$ – 4.4×10^4 of 2.4×10^5 ; Figs. 7 and 8). This reflects a different test, not a sharper one: those curves score the large $L = 12$ sector by a *relative* discrepancy $|\widehat{m}_1 - \bar{m}_1|/\widehat{m}_1$ against a fixed line, this Monte Carlo the small $L = 6$ sector by an *absolute* threshold set by the shot budget. The coverage percentages thus come from different rulers on different-sized sectors (and, at $L = 12$, two reference states), not from one test giving several answers.

The direction of the residual uncertainty matters for a test that can only reject. Under-estimating δ_k shrinks τ_k : safe against missed errors (false negatives, acceptable for a necessary condition) but anti-conservative against false rejections (false positives, the costly error for such a test). The false-positive rate above relies on $\bar{m}_1$ being exact at full support ($|\widehat{m}_1 - \bar{m}_1| \sim 10^{-15}$), so that the null hypothesis holds exactly there. For m_2 it does not: the 186-determinant support contains all of φ_S but not $H\varphi_S$, so the reconstruction on it is wrong at second order, $\Delta_2 = 0.62$ ($0.48 \tau_2$ at the shot-only threshold, $0.34 \tau_2$ at the deployed one), the boundary-leakage term above. An m_2 rejection of it would be a correct detection, not a false positive; the Monte Carlo tests m_1 only. Away from full coverage the threshold should be set on $\text{Var}(\widehat{m}_k - \bar{m}_k)$, for which $\text{Var}(O_k)$ is a proxy valid only when the reference is trusted (same-sample statistics above; Sec. 9).

The second moment catches some lone- m_1 misses, not all. In the noise-free scan at the 2% budget, m_2 rejects 3 of the 17 truncations that m_1 alone passes ($d = 97, 96, 85$), and m_0 rejects two more ($d = 94$ and 159, residuals 0.044 and 0.037 against $\tau_0 = 0.029$). The clearest is

$d = 97$ (52% coverage): in the deterministic order $|\widehat{m}_1 - \bar{m}_1| = 0.186 < \tau_1 = 0.229$ but $|\widehat{m}_2 - \bar{m}_2| = 2.73 > \tau_2 = 1.85$, whatever the tie convention (m_1 alone rejects in none and the battery in all of 300 random shell choices). The full (m_0, m_1, m_2) battery nonetheless passes the remaining 12 truncated reconstructions at the 2% budget: $d = 183$ –185 (98–99.5% coverage), 160–165 (86–89%), and 86, 87, 95 (46–51%). Some of these passes are tie-convention effects (the battery rejects in 44%, 40% and 11% of random shell choices at $d = 87$, 160 and 95), but not those at $d = 86$ (1.3%), 161–165 (at most 0.3%) or 183–185 (none). Without a bias budget the battery catches the mid-coverage windows but still passes $d = 161$ –164, 184 and 185; with a 4% budget it passes 22 truncations. The joint battery is thus a partial remedy for the single-moment windows: with finite thresholds no battery of necessary conditions is free of blind spots, and at the deployed order this one has blind spots that we can list.

Two further boundaries frame the demonstration. First, the discrepancy is measured against the exact moments, which serve as an oracle: the demonstration calibrates the interval where classical ground truth exists, and a device-side version that propagates measured rather than assumed bias remains to be done. Second, the screen never certifies: moment-matrix certificates [82, 83] bound ground-state properties, whereas here two estimates of the same moments are compared and a discrepancy beyond threshold rejects. Its relation to sum-rule checks and other prior work is set out once, in Sec. 1 and Table 1.

C The Christoffel miss-distance: proof and evaluation

We re-derive the equality of the extremal cumulative-weight spread with the Christoffel function from three classical lemmas [48–50]; nothing in this derivation is new. Here x is a real frequency point in any affine frame (Fig. 13 uses the rescaled frame $x \in [-1, 1]$); the hopping t remains the energy unit. Write $\mathcal{P}_n$ for the real polynomials of degree at most n , $K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$ for the Christoffel–Darboux kernel and $\lambda_n(x) = 1/K_n(x, x)$ for the Christoffel function, and assume the moment vector is interior to the moment cone, $H_n = [m_{i+j}]_{i,j=0}^n \succ 0$ (for a finite N -atom Lehmann spectrum this holds for $n \leq N - 1$). The measures ν compared below are positive measures on $\mathbb{R}$. Restricting them to the Lehmann half-line $\omega \geq 0$ can only narrow the bracket, so W_n remains an upper bound there.

Lemma 1 (Variational identity). $\lambda_n(x) = \min\{\int |P|^2 d\mu : P \in \mathcal{P}_n, P(x) = 1\}$, with minimizer $P^* = K_n(\cdot, x)/K_n(x, x)$.

Proof. Expanding $P = \sum_{j=0}^n c_j p_j$ gives $\int |P|^2 d\mu = \sum_j c_j^2$, while by Cauchy–Schwarz

$$1 = P(x)^2 = \left(\sum_{j=0}^n c_j p_j(x)\right)^2 \leq \left(\sum_{j=0}^n c_j^2\right) K_n(x, x),$$

with equality at $c_j = p_j(x)/K_n(x, x)$; the minimum is $1/K_n(x, x) = \lambda_n(x)$. $\square$

Lemma 2 (Maximal atom). Among positive measures ν with $\int \omega^k d\nu = m_k$ for $k \leq 2n$, the supremum of the mass at a point x is $\sup_\nu \nu(\{x\}) = \lambda_n(x)$. It is attained if $p_n(x) \neq 0$, and approached but not attained at the n zeros of p_n .

Proof. The minimizer P^* of Lemma 1 has $P^*(x) = 1$, so

$$\nu(\{x\}) = P^*(x)^2 \nu(\{x\}) \leq \int (P^*)^2 d\nu = \int (P^*)^2 d\mu = \lambda_n(x),$$

the middle equality using $\deg(P^*)^2 = 2n$ and that ν shares those moments. If $p_n(x) \neq 0$ the bound is attained by the $(n+1)$ -point Gauss–Radau rule with a node prescribed at x : a positive

measure matching $m_0, \dots, m_{2n}$ whose weight at x is $\lambda_n(x)$. At the n zeros of p_n this rule does not exist (one node escapes to infinity), and the supremum is approached but not attained. $\square$

Lemma 3 (Ordinate gap). *Let $p_n(x) \neq 0$, and let $\xi_0 < \dots < \xi_n$ be the nodes of the Gauss–Radau rule of Lemma 2, one of them equal to x , with weights $\lambda_n(\xi_i) > 0$. For every positive ν sharing $m_0, \dots, m_{2n}$, with $F_\nu(x) = \nu((-\infty, x])$, the Chebyshev–Markov–Stieltjes inequalities [49–51, 101] hold:*

$$\sum_{\xi_i < x} \lambda_n(\xi_i) \leq F_\nu(x^-) \leq F_\nu(x) \leq \sum_{\xi_i \leq x} \lambda_n(\xi_i).$$

Consequently $\sup_{\nu_1, \nu_2} |F_{\nu_1}(x) - F_{\nu_2}(x)| = \lambda_n(x)$, and the same supremum holds at the zeros of p_n , so it holds for every real x .

Proof. The inequalities follow from polynomials $Q_\pm \in \mathcal{P}_{2n}$ with $Q_- \leq \mathbf{1}_{(-\infty, x]}$ and $\mathbf{1}_{(-\infty, x]} \leq Q_+$ that coincide with these indicators at the nodes, so that $\int Q_\pm d\nu = \int Q_\pm d\mu$ equal the two sums. The bracket has width $\lambda_n(x)$. Its upper end is attained by the Radau rule itself; its lower end is approached by Radau rules whose prescribed node x' decreases to x . At the zeros of p_n the same supremum follows from the continuity of λ_n and the right-continuity of F . $\square$

Combining the three lemmas, the extremal spread equals $\lambda_n(x) = W_n(x)$ for every real x , and any single reconstruction consistent with the moments obeys $|F_{\text{rec}}(x) - F_{\text{true}}(x)| \leq W_n(x)$. No convex-hull hypothesis is needed: the Radau rule has positive weights for every x with $p_n(x) \neq 0$, inside or outside the convex hull of the support. We checked this on the $U/t = 8$ current measure of Fig. 13: 24 Radau rules ($n = 1, \dots, 4$ at six values of x , 20 of them outside the convex hull of the weighted support) all have positive weights, place mass $W_n(x)$ at x to a relative error of 2×10^{-16} , and match $m_0, \dots, m_{2n}$ to 1.4×10^{-17} . Monotonicity $W_{n+1} \leq W_n$ follows from $K_{n+1} = K_n + p_{n+1}^2 \geq K_n$.

Pointwise vanishing $W_n(x) \rightarrow 0$ is a thermodynamic-limit statement. If the discrete spectrum densifies to a measure that is Stahl–Totik regular with compact support S and absolutely continuous with a density A obeying the local Szegő condition $\int_I \log A > -\infty$ on an interval $I \subset S$, then $n W_n(x) \rightarrow A(x)/\omega_S(x)$ for almost every $x \in I$, with ω_S the equilibrium density of S [97]; on a single interval this reduces to Máté–Nevai–Totik [98].

Evaluation of the bound from the moments. The Christoffel function is computed from the moments $\{m_k\}_{k=0}^{2n}$ by a short pipeline [48, 72]. With the inner product $\langle f, g \rangle_\mu = \int f(\omega)g(\omega)d\mu(\omega)$, so that $\langle \omega^a, \omega^b \rangle_\mu = m_{a+b}$, Gram–Schmidt on $\{1, \omega, \dots, \omega^n\}$ gives the orthonormal polynomials, with $p_0 = 1/\sqrt{m_0}$; they obey the three-term recurrence

$$\beta_{j+1} p_{j+1}(\omega) = (\omega - \alpha_j) p_j(\omega) - \beta_j p_{j-1}(\omega), \quad (\text{C.1})$$

with $\alpha_j = \langle \omega p_j, p_j \rangle_\mu$ and $\beta_j = \langle \omega p_j, p_{j-1} \rangle_\mu$, all fixed by $\{m_k\}$. The map from power moments to (α_j, β_j) becomes ill-conditioned as n grows, so high orders need high-precision arithmetic. The nodes and weights of the $(n+1)$ -point Gauss rule are the eigenpairs of the Jacobi matrix $\mathbf{J}_{n+1} = \text{tridiag}(\beta_j, \alpha_j, \beta_j)$ (Golub–Welsch), and $K_n(x, x) = \sum_{j=0}^n p_j(x)^2$, evaluated by the same recurrence, gives $W_n(x) = 1/K_n(x, x)$. Every quantity in the bound is thus an explicit function of the moments, readable at any frequency.

We evaluate it on the $L = 6$, $U/t = 8$ current measure of Fig. 13, normalized to $m_0 = 1$; five of its atoms carry weight. (Its ground level is a spin triplet; the current measure is the same for all three states, because J is a spin scalar.) This measure lies outside the blinded catalog ($U/t \in \{3, 4, 5, 6\}$, Sec. 5) and differs from the $U/t = 4$ spectrum of Appendix E; the committed record does not state why $U/t = 8$ was chosen for this section. The numbers below come from `src/export_christoffel_dat.py` and `src/christoffel_tolerance_lp.py` (data/2026-09-27_theory_numerics.json). We run the recurrence in 60-digit arithmetic on the atoms and weights of the measure, which fix the same coefficients as its moments, and

locate the maxima by root-finding: $\max_x W_n = 1, 0.969, 0.886, 0.882$ for $n = 1, \dots, 4$. The $n = 1$ maximum equals m_0 and lies at the mean frequency $\bar{\omega} = 11.10 t$, the zero of p_1 , so at the deployed order (m_0, m_1, m_2) the bracket is vacuous at the spectral peak. At the dominant atom ($\omega = 11.09 t$, residue 0.778), W_n reaches the residue at $n = N - 1 = 4$, the highest order at which H_n is positive definite, and for $n \leq 4$ it cannot go lower, because Lemma 2 with $\nu = \mu$ gives $W_n(x) \geq \mu(\{x\})$: within the range of the theorem no number of moments narrows the bracket at the peak below 0.778. For $n > 4$, H_n is singular and the theorem's hypothesis fails.

The rescaled abscissa $x \in [-1, 1]$ of Fig. 13 is the affine image of the frequency range $\omega \in [0.148, 52.21] t$, the positive excitation energies of the full Fock space, zero-weight states included. The weighted support $[2.49, 13.92] t$ maps to $[-0.910, -0.471]$, and 73.5% of $[-1, 1]$ lies above the last weighted pole. The affine map moves points along the axis but does not change W_n , which is a mass. The rescaled points $x = 0.3, 0.6, 0.9$ correspond to $\omega = 34.0, 41.8, 49.6 t$, far above the band. There $W_1 = 4.0 \times 10^{-3}, 2.2 \times 10^{-3}, 1.4 \times 10^{-3}$, and at $x = 0.6$ the higher orders give $W_2 = 1.4 \times 10^{-5}, W_3 = 2.7 \times 10^{-8}, W_4 = 3.6 \times 10^{-10}$. For $n = 1$ these small values are Cantelli's one-sided Chebyshev bound, $W_1(\omega) = m_0 \sigma^2 / [\sigma^2 + (\omega - \bar{\omega})^2]$ with σ^2 the variance of the normalized measure (it agrees with the orthogonal-polynomial evaluation to 1.6×10^{-15}). They reflect the distance from the mean, not a feature of this spectrum. Just above the top pole, at $\omega = 14 t$, $W_1 = 0.20$.

All of this assumes exact moments. If the moments are matched only within the deployed tolerances, $|\Delta_k| \leq \tau_k$ for $k \leq 2$, the largest atom a positive measure can place at ω is a linear program. With thresholds from the deployed rule applied to this state ($\tau_0 = 0.051$, $\tau_1 = 0.57$, $\tau_2 = 6.5$ before normalization; an illustration, since the rule was calibrated only on the catalog), it is $3.2 W_1$ (0.63 of m_0) at $\omega = 14 t$, $7.7 W_1$ at $20 t$, and 10.1 – $10.7 W_1$ at the rescaled points $x = 0.3$ – 0.9 . On the $U/t = 4$ measure of Appendix E, which is in the catalog, the same construction gives 3.6 – $5.7 W_1$ at $\omega = 12$ – $20 t$, so the loosening is not specific to $U/t = 8$. These τ_k use the exact-state local-estimator shot variance of Appendix B, which is expected to underestimate a realistic per-shot variance; wider tolerances would inflate the atom further. At the deployed order a passing reconstruction is therefore bracketed much more loosely than W_1 alone suggests.

Evaluation of the Markov–Krein window bound (Proposition 1). On the same $L = 6$, $U/t = 8$ current measure (committed script `src/markov_krein_window.py`), at exact moments: at $n = 4$ ($m_0, \dots, m_8$), a window with one edge at the 400-point-grid maximizer of W_4 , just above the dominant atom, gives $\mu \in [0.070, 0.945]$, width 0.875; the exact maximum of W_4 , at a nearby frequency, is 0.882. A window that contains the peak with both edges in weight-free frequency (rescaled $[-0.75, -0.35]$) gives $\mu \in [0.9732, 0.9758]$ around the exact 0.9744, and the empty window rescaled $[0.3, 0.9]$ ($\omega \approx 34$ – $50 t$) gives $\mu \in [0, 3.2 \times 10^{-9}]$. The peak window is tight here because W_4 is small at both of its edges (before intersection with $[0, m_0]$ the width is $W_4(c) + W_4(d)$); weight-free edges alone do not ensure this, as $W_1 = 0.20$ at $14 t$, just above the band, shows, so this is a property of this measure. At the deployed order $n = 1$ (m_0, m_1, m_2) the peak-containing window gives $\mu \in [0.848, 1]$ and the empty window $\mu \in [0, 0.004]$, while a window with an edge on the peak is uninformative (its bracket runs from 0 to within 2×10^{-4} of m_0). These are exact-moment brackets. Under the deployed tolerances the upper end for the empty window is at least 0.040, the largest atom that a measure with m_0, m_1, m_2 within tolerance can place at its lower edge ($\omega \approx 34 t$).

D Complementary state-level check: the Gauss law

The Gauss-law check mentioned in Secs. 1 and 3 is not a moment test; we record its construction and scope here. In the $U(1)$ lattice Schwinger model the generators G_x of gauge

transformations commute with the Hamiltonian, $[H, G_x] = 0$, and every physical state satisfies $G_x|\psi\rangle = 0$ at every site, so $\langle \sum_x G_x^2 \rangle = 0$ exactly. The operator $\sum_x G_x^2$ is diagonal in the computational basis, so on hardware it could be estimated from computational-basis samples of the prepared state, with no reconstruction and no moment. It tests the prepared state, not a reconstructed spectrum: it responds to leakage out of the physical sector, an error class the moment test does not target.

Figure 15 illustrates the identity on a small instance: the quantum-link form of the model with $N = 4$ staggered matter sites and 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$ ($[H, G_x] = 0$ to machine precision). The plotted state is $|\psi(\epsilon)\rangle \propto (1 - \epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ an unphysical state. The parameter ϵ is an amplitude, not a weight: the unphysical weight is $f_u(\epsilon) = \epsilon^2 / [(1 - \epsilon)^2 + \epsilon^2]$, 0.059 at $\epsilon = 0.2$ and 0.155 at $\epsilon = 0.3$. Because $\sum_x G_x^2|\psi_0\rangle = 0$ and $\langle\psi_0|\phi_u\rangle = 0$, the violation is $\langle \sum_x G_x^2 \rangle = g_u f_u(\epsilon)$ with $g_u = \langle\phi_u|\sum_x G_x^2|\phi_u\rangle$. The curve is therefore fixed by construction: zero for the physical state, rising monotonically with ϵ , with its scale set by g_u alone. It illustrates the identity and does not measure detection power. The values are computed exactly on the state vector; no sampling is simulated.

The admixed state is chosen by an explicit rule: $|\phi_u\rangle$ is the lowest-energy eigenstate of H among charge-neutral states with the smallest nonzero violation, $\sum_x G_x^2 = 2$ (one site with $G = +1$ and one with $G = -1$; a 27-dimensional subspace; generator `src/export_gauss_state.py`). Every state in that subspace has $g_u = 2$, so the plotted curve is $2f_u(\epsilon)$. The figure was first made with a different rule: the unphysical eigenstate of H with the largest overlap $|\langle k|M|\psi_0\rangle|$, where $M = \sum_n (-1)^n n_n$ is the staggered occupation. That rule is ill-posed. M is gauge-invariant, so every such overlap vanishes (the largest is 1.8×10^{-15}), and the pick, set by rounding noise, happened to be a state with $g_u = 2$. The explicit rule reproduces the plotted values to all five printed decimals (`data/2026-09-27_gauss_state.json`). Over the 124 unphysical eigenstates of this model g_u ranges from about 1 to 10, so the slope of the curve reflects this choice. An earlier sweep under an independent bit-flip channel on every qubit (`data/2026-08-18_gausslaw_falsifier.json`) also gives $\langle \sum_x G_x^2 \rangle > 0$ at every nonzero flip rate (0.25 at $p = 0.05$). It averages only 40 sampled flip patterns per rate and is not monotone in p at that sample size (3.05 at $p = 0.35$, 2.52 at $p = 0.40$), so we do not use it as a graded curve.

Gauss-law violation has been measured in a 71-site cold-atom quantum simulator [45] and analyzed theoretically under gauge-breaking errors [125]. The lattice Schwinger model is also the model of self-verifying variational simulation on a trapped-ion analog quantum simulator [47] and of early digital simulations of Schwinger-model dynamics [126, 127].

E Within-sector redistributions and the deployed battery

This appendix proves Theorem 2 and evaluates it on one spectrum. The independent estimator detects weight that leaks *outside* the retained determinant support. A redistribution strictly *within* it contributes no leak term; it is visible only through the moment shift $(V\delta)_k$, and not at all if it preserves the moments used (Sec. 4). We treat that residual in three steps: *injectivity* (part (a): in exact arithmetic, $m_0, \dots, m_{N-1}$ fix the weights on N distinct poles); *conditioning* (part (b) and the Corollary: a sufficient and a necessary condition for exposing a redistribution at finite shots); and *reach* (a linear program and explicit counterexamples on the doped $L = 6$, $U/t = 4$ current spectrum, with thresholds from Appendix B; Table 3). The result at the deployed order is negative: the battery (m_0, m_1, m_2) has no guaranteed power against within-sector redistributions of this spectrum, although its poles are well separated. We characterize this residual; we do not close it. As in Theorem 2, K denotes the highest

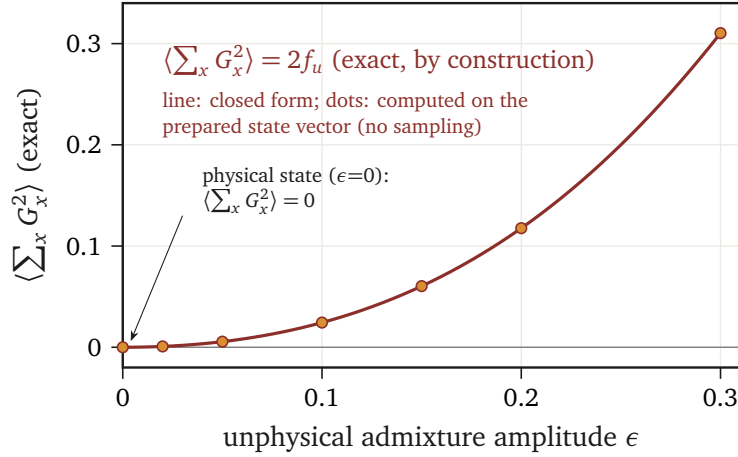


Figure 15: **A state-level check: the Gauss-law identity.** In the U(1) lattice Schwinger model physical states obey $G_x|\psi\rangle = 0$, so $\langle \sum_x G_x^2 \rangle = 0$; $[H, G_x] = 0$ keeps the dynamics in that sector. Plotted: $|\psi(\epsilon)\rangle \propto (1-\epsilon)|\psi_0\rangle + \epsilon|\phi_u\rangle$, with $|\psi_0\rangle$ the physical ground state and $|\phi_u\rangle$ the lowest-energy charge-neutral state with the smallest violation, $\sum_x G_x^2 = 2$. The amplitude ϵ gives unphysical weight $f_u = \epsilon^2/[(1-\epsilon)^2 + \epsilon^2]$, only 0.155 at $\epsilon = 0.3$. Line: the closed form $2f_u(\epsilon)$, which $\langle \sum_x G_x^2 \rangle$ equals by construction; dots: values computed exactly on the state vector (no sampling is simulated). It does not measure detection power. The check tests the prepared state, not a reconstructed spectrum, and targets gauge-symmetry breaking. $N = 4$ staggered sites, 3 spin- $\frac{1}{2}$ links, open boundaries, $w = 1$, $m = 0.6$.

Table 3: Within-sector redistributions of the $L = 6$, $U/t = 4$ current spectrum ($N = 5$ distinct poles). Each row keeps $m_0, \dots, m_K$ exactly and asks how far m_{K+1} can move. “LP range” is the range of the signed shift $\sum_n \delta_n \omega_n^{K+1}$ over all such redistributions with non-negative corrupted weights [Eq. (E.1)]; every range contains 0. The thresholds τ_{K+1} follow the deployed rule ($N_s = 5 \times 10^4$, an assumed 2% bias budget, $z_{0.975} = 1.96$; Appendix B); for m_3 and m_4 , unused in the deployed battery, they are the values the same rule would give. When $\dim \ker V_{0:K+1} > 0$, some finite redistributions leave m_{K+1} exactly unchanged, so the guaranteed power at that order is zero and the maximum describes only the best case.

preserved exactly	next moment	LP range of shift [Eq. (E.1)]	τ_{K+1}	max shift/ τ_{K+1}	$\dim \ker V_{0:K+1}$	in deployed battery?
m_0, m_1	m_2	$[-0.92, 7.09]$	1.848	3.8	2	yes
$m_0, \dots, m_2$	m_3	$[-0.14, 5.26]$	14.98	0.35	1	no
$m_0, \dots, m_3$	m_4	$[-0.30, 0.39]$	122.0	0.003	0	no

order a redistribution preserves; the deployed battery tests orders 0, 1 and 2. Unless stated otherwise, the numbers below are in data/2026-09-27_theory_numerics.json (script src/within_sector_lp.py).

Injectivity (proof of Theorem 2(a)). A $T = 0$ Lehmann current spectrum on a finite sector is a discrete measure on N distinct poles $\{\omega_n\}$,

$$A(\omega) = \sum_{n=1}^N w_n \delta(\omega - \omega_n), \quad w_n = \sum_{a: E_a - E_0 = \omega_n} |\langle a|J|0\rangle|^2 \geq 0,$$

where w_n is the total weight of the eigenstates at ω_n . Its moments are $m_k = \sum_n w_n \omega_n^k$, i.e.

$\mathbf{m} = V\mathbf{w}$ with $V_{kn} = \omega_n^k$ a Vandermonde matrix. Because the poles are distinct, the block $V_{0:N-1}$ of rows $0, \dots, N-1$ is invertible, so the weight-to-moment map is injective through order $N-1$: a redistribution $\mathbf{w} \rightarrow \mathbf{w} + \boldsymbol{\delta}$ with $\boldsymbol{\delta} \neq 0$ cannot preserve all of $m_0, \dots, m_{N-1}$. A redistribution that preserves $m_0, \dots, m_K$ ($K < N-1$) zeroes the first $K+1$ entries of $V_{0:N-1}\boldsymbol{\delta}$, so $\|(\Delta_{K+1}, \dots, \Delta_{N-1})\|_2 = \|V_{0:N-1}\boldsymbol{\delta}\|_2 \geq \sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2$; the largest of these $N-1-K$ shifts is at least this norm divided by $\sqrt{N-1-K}$, which is Eq. (5). If the poles may also move, an N -atom measure has $2N$ parameters and is fixed by $m_0, \dots, m_{2N-1}$ (Prony) [69, 102]. Weight traded among degenerate eigenstates at a single frequency changes no moment and leaves $A(\omega)$ unchanged; it is not an error.

For the $L = 6$, $U/t = 4$ spectrum the sixteen eigenstates with nonzero weight fall on $N = 5$ distinct frequencies, $\omega_n/t = 2.73, 5.57, 7.83, 8.70, 10.66$, with weights $0.040, 0.004, 0.393, 0.116, 0.001$ ($m_0 = 0.554$; the pole at $7.83t$ carries 71% of it). The map is therefore injective through m_4 : a redistribution among these poles that preserves $m_0, \dots, m_K$ changes some m_j with $K < j \leq 4$ (at most m_9 if the poles also move). In exact arithmetic the residual is thus resolved by m_4 . That is two orders beyond the deployed battery, and it says nothing about detection at finite shots. Equation (5) is valid here but loose: for the redistribution CE1 below, its right-hand side is 3.8×10^{-3} , while the larger of the shifts in m_3 and m_4 is 85. The sufficient condition of Theorem 2(b)(i) is far from met here: for CE1 and CE2 the right-hand side of (5) is below 4×10^{-3} , against $\max_j \tau_j = \tau_4 = 122$ in the same units.

Conditioning, and when detection is possible (proof of Theorem 2(b)). Invertibility of $V_{0:N-1}$ holds in exact arithmetic. At finite shots a shift counts only if it exceeds the threshold τ_j . Two bounds, in opposite directions, turn detectability into a statement about the size of $\boldsymbol{\delta}$ and the geometry of the poles it touches.

Sufficient condition. By Eq. (5), a redistribution preserving $m_0, \dots, m_K$ is exposed (its noise-free shift exceeds the threshold) at some order $j \leq N-1$ if

$$\frac{\sigma_{\min}(V_{0:N-1})\|\boldsymbol{\delta}\|_2}{\sqrt{N-1-K}} > \max_{K < j \leq N-1} \tau_j.$$

Since $\|\boldsymbol{\delta}\|_1/\sqrt{N} \leq \|\boldsymbol{\delta}\|_2 \leq \|\boldsymbol{\delta}\|_1$, this is a condition on $\rho = \|\boldsymbol{\delta}\|_1$ up to a factor $\sqrt{N}$. It is informative only when $\sigma_{\min}$ is not small, and it uses moments up to order $N-1$.

Necessary condition. Let $\boldsymbol{\delta}$ preserve $m_0, \dots, m_{s-2}$ and be supported on poles inside an interval of diameter D and midpoint ω_c . (Among s poles, preserving $m_0, \dots, m_{s-2}$ leaves a one-parameter family, and m_{s-1} is the first moment it changes.) Since $\sum_n \delta_n q(\omega_n) = 0$ for every polynomial q of degree at most $s-2$, the signed shift is $\sum_n \delta_n \omega_n^{s-1} = \sum_n \delta_n (\omega_n - \omega_c)^{s-1}$, hence

$$|\Delta_{s-1}| \leq \rho (D/2)^{s-1}.$$

Subtracting from ω^j its Taylor polynomial of degree $s-2$ at ω_c gives, for every $j \geq s-1$, the Lagrange-remainder bound

$$|\Delta_j| \leq \rho \binom{j}{s-1} (|\omega_c| + D/2)^{j-s+1} (D/2)^{s-1};$$

expanding instead $\omega^j = \sum_i \binom{j}{i} \omega_c^{j-i} (\omega - \omega_c)^i$, whose terms with $i \leq s-2$ are combinations of preserved moments and vanish, and bounding the rest by $|\omega_n - \omega_c| \leq D/2$ gives the term-by-term form of Theorem 2(b)(ii). Either way every shift vanishes as D^{s-1} . Exposure at order $s-1$ therefore requires $\rho (D/2)^{s-1} > \tau_{s-1}$. With $\tau_{s-1} = z_{1-\alpha/2} [\text{Var}(O_{s-1})/N_s + b_{s-1}^2]^{1/2}$ this reads

$$N_s > \frac{z_{1-\alpha/2}^2 \text{Var}(O_{s-1})}{\rho^2 (D/2)^{2(s-1)} - z_{1-\alpha/2}^2 b_{s-1}^2},$$

and no budget suffices once $\rho(D/2)^{s-1} \leq z_{1-\alpha/2} b_{s-1}$. The shot cost of detecting a redistribution of fixed size inside a collapsing cluster thus grows at least as $D^{-2(s-1)}$. Numerically, the identity holds to a relative error of 1.5×10^{-12} , and the bound holds in 11 structured cases (CE1 and CE2 below, the six optima of Eq. (E.1) and three committed random cases) and in 20 000 random null-space redistributions on random pole subsets (seed 20260927, no violation). The largest ratio $|\Delta_{s-1}|/[\rho(D/2)^{s-1}]$ is 1.000, 0.499 and 0.231 at $s = 2, 3, 4$, so the bound is tight at $s = 2$. If a cluster contracts uniformly, so that its minimal pole separation falls from $\Delta\omega_{\min}$ to $\Delta\omega$, D shrinks by the same factor and the lower bound (6) on the shot cost grows by at least $(\Delta\omega_{\min}/\Delta\omega)^{2(s-1)}$. A finite separation below which no budget suffices exists only with a bias floor, $b_{s-1} > 0$.

Conditioning numbers. The same collapse appears in $\sigma_{\min}$: for s nodes of fixed relative geometry and diameter $D \rightarrow 0$, $\sigma_{\min}$ of the $s \times s$ Vandermonde block scales as D^{s-1} . The sharp super-resolution results for this regime [103–106] are proved for Fourier (unit-circle) nodes, not for the real nodes used here, for which both directions follow from elementary bounds on the $s \times s$ block V of the cluster. Row n of V^{-1} holds the coefficients of the Lagrange polynomial $\prod_{j \neq n} (\omega - \omega_j)/(\omega_n - \omega_j)$, which gives Gautschi's bound [107, 108] and hence

$$\|V^{-1}\|_{\infty} \leq \max_n \prod_{j \neq n} \frac{1 + |\omega_j|}{|\omega_n - \omega_j|}, \quad \sigma_{\min}(V) \geq \left[\sqrt{s} \max_n \prod_{j \neq n} \frac{1 + |\omega_j|}{|\omega_n - \omega_j|} \right]^{-1},$$

a lower bound that scales as D^{s-1} at fixed relative geometry. Conversely, the redistribution that spans the null space of rows $0, \dots, s-2$ has $\|V\delta\|_2 = |\Delta_{s-1}| \leq \rho(D/2)^{s-1} \leq \sqrt{s} \|\delta\|_2 (D/2)^{s-1}$, so $\sigma_{\min}(V) \leq \sqrt{s} (D/2)^{s-1}$. A direct check with $s = 2, 3, 4$ equally spaced nodes near $\omega = 7.83t$ (spacing $0.02\text{--}0.2t$) gives fitted exponents 0.995, 1.98 and 2.96 (data/2026-08-24_within_sector_control.json). Condition numbers depend on the frequency unit, so we quote them only on nodes mapped affinely onto $[-1, 1]$ (outermost poles at ± 1): for the five poles above, $\sigma_{\min}(V_{0:4}) = 0.056$ and $\kappa = 50$. With a minimal separation of $0.87t$ these poles are well separated, yet good conditioning does not give the deployed battery guaranteed power: that is set by the dimension of a null space (next paragraph).

Reach of the deployed battery (instantiation of Theorem 2(b) and Corollary). The zero-power clause of part (b) is immediate: a vector in the null space of rows $0, \dots, K+1$ of V , scaled so that $w_n + \delta_n \geq 0$, preserves $m_0, \dots, m_{K+1}$; every smaller rescaling does too, so the smallest shift at order $K+1$ is zero. The deployed battery uses m_0, m_1, m_2 with the thresholds of Appendix B, $\tau_0 = 0.0287$, $\tau_1 = 0.229$ and $\tau_2 = 1.848$, at $N_s = 5 \times 10^4$ with an assumed, unmeasured 2% bias budget $b_k = 0.02 m_k$; each τ_k includes the factor $z_{0.975} = 1.96$ and is not a 1σ noise floor. The shot term uses the exact-state local-estimator variance, which is expected to underestimate the true per-shot variance, so realistic thresholds would be wider. For a redistribution that preserves $m_0, \dots, m_K$, the largest shift it can impose on the next moment is the linear program

$$\Delta_{K+1}^{\max} = \max_{\delta} \left| \sum_n \delta_n \omega_n^{K+1} \right| \quad \text{s.t.} \quad \sum_n \delta_n \omega_n^j = 0 \quad (j \leq K), \quad \delta_n \geq -w_n, \quad (\text{E.1})$$

the constraint $\delta_n \geq -w_n$ keeping the corrupted weights non-negative; together with $\sum_n \delta_n = 0$ it also bounds $\|\delta\|_1 \leq 2m_0$.

Table 3 gives the full range of the signed shift. A redistribution that preserves m_0, m_1 can move m_2 by up to $7.09 = 3.8 \tau_2$, so some such redistributions are exposed at m_2 . The guaranteed power, however, is set by the smallest shift a redistribution of a given size can produce, and that is zero. Rows $0, 1, 2$ of V have a null space of dimension $N - 3 = 2$: some redistributions preserve m_0, m_1 and m_2 exactly, and scaling one of them down gives an exactly

invisible redistribution of every smaller size. Three explicit cases, all positive measures that pass the deployed battery: (i) CE1 preserves m_0, m_1, m_2 exactly and moves 72% of the weight ($\|\delta\|_1/2 = 0.72 m_0$); it empties two of the five poles and lowers the dominant residue from 0.393 to 0.037. (ii) CE2 preserves m_0, m_1 exactly and shifts m_2 by exactly τ_2 , so it sits on the decision boundary and passes only because the rule rejects on the strict inequality $\Delta_k > \tau_k$; it moves 86% of the weight and lowers the dominant residue to 0.073. (iii) CE3 keeps $|\Delta_k| \leq \tau_k$ for $k \leq 2$, has $\|\delta\|_1 = 2.03 m_0$ (above $2m_0$ because m_0 itself moves by τ_0), and empties three poles, the dominant one included. The random $K = 1$ case committed in `src/within_sector_control.py` (seed 0) moves 25% of the weight ($\|\delta\|_1/2 = 0.25 m_0$; its output file lists $\|\delta\|_1/m_0 = 0.49$ as the moved fraction) and shifts m_2 by $0.044 \tau_2$; it passes as well. The guaranteed power of the deployed battery against within-sector redistributions of this spectrum is therefore zero. The reason is the dimension of the null space, not conditioning, and the same holds for any spectrum with four or more distinct poles.

Adding m_3 and m_4 would remove the null space (rows $0, \dots, 4$ have $\dim \ker = 0$), but at this shot budget the extended battery would still miss a redistribution that moves 72% of the weight. Under the same threshold rule, $\tau_3 = 14.98$ and $\tau_4 = 122.0$. Among redistributions that preserve m_0, m_1, m_2 exactly, the largest m_3 shift is $5.26 = 0.35 \tau_3$ (Table 3), and CE1, like every scaled-down copy of it, stays below both thresholds ($0.23 \tau_3$ and $0.70 \tau_4$). Some such redistributions would be caught at m_4 : the one that maximizes the m_3 shift moves m_4 by $145 = 1.19 \tau_4$, a marginal case, since the variance behind τ_4 is expected to underestimate a realistic one. CE2 and CE3, which use the tolerance on m_2 , would be caught ($|\Delta_3| = 3.3 \tau_3$; $|\Delta_4| = 7.1$ and $7.2 \tau_4$).

The joint battery does catch some errors that m_1 alone misses, but the example is a truncation, not a fixed-pole redistribution: truncating the determinant basis of the same system to the $d = 97$ determinants of largest $|\psi_0|^2$ (52% of the 186-determinant support; Appendix B) gives a reconstruction that m_1 alone passes and m_2 rejects. The cut falls inside a 24-fold degenerate shell, so the gaps depend on how ties are ordered. The committed smoke test `src/verify.py` (NumPy `argsort` order, which does not fix the order of ties; values from our machine) gives $|\bar{m}_1 - \hat{m}_1| = 0.155 < \tau_1$ and $|\bar{m}_2 - \hat{m}_2| = 2.560 > \tau_2$; the deterministic `lexsort` order of `src/interval_battery.py` gives 0.186 and 2.729; and for all 300 random tie-breaks within the shell m_1 alone rejects none and the battery rejects all (`data/2026-09-27_interval_battery.json`). This is one catch, not a guarantee: in the exhaustive scan of Appendix B ($d = 1, \dots, 185$, same thresholds, `lexsort` order) m_1 alone passes 17 truncated reconstructions, m_2 rejects three of them ($d = 97, 96, 85$) and m_0 two more ($d = 94, 159$), so the full (m_0, m_1, m_2) -Hankel battery passes 12 ($d = 86, 87, 95, 160$ – 165 and 183 – 185). Under random tie-breaks within the degenerate shells, ten of these pass in at least 89% of 300 draws, and $d = 87$ and $d = 160$ in 56% and 60%.

Positivity cannot help against within-sector redistributions: a redistribution with non-negative weights is itself a positive measure, so the Hankel and Stieltjes conditions hold by construction. Only independent moments of high enough order, estimated precisely enough, can expose it, and at the deployed order the guaranteed power is zero. We report this residual; we do not claim to close it.

F Device run: reference samplers, line shapes and the n_k channel

This appendix records the reference comparisons (ii) and (iii) and the line-shape comparisons of the $L = 8$ device run (Sec. 6) and the sealed on-device n_k test, which was not run. As in that section, Δ_0 measures support coverage and rewards support spreading; it is not a fidelity diagnostic, and the uniform in-sector sampler, which uses no device, scores lower than the

device at every budget.

(ii) *Noiseless sampling at the device's retained counts.* This comparison depends on how the retained counts are allocated. Matched circuit by circuit, the comparator puts 46,216 of its 82,697 shots at 50k on the $t = 0$ circuit, which is a single determinant without noise; its mean Δ_0 is 0.224, 0.279, 0.343 and 0.443 at 50/30/16/4k. With the same total split equally over the seven circuits it is 0.150, 0.205, 0.273 and 0.404. Under both allocations the device lies below every replica at 50, 30 and 16k; at 4k it lies at the 0.6th (per circuit) or 5.8th (equal split) percentile. The margin shrinks with the budget: the reference mean exceeds the device value 51-, 8.6-, 2.8- and 1.19-fold per circuit, and 34-, 6.3-, 2.2- and 1.08-fold with the equal split.

(iii) *Uniform noise.* Mixing each circuit's output distribution P_k with a fraction p of uniformly random bitstrings, $(1-p)P_k + p/2^{16}$, before post-selection lowers the expected Δ_0 at 50k monotonically, from 0.036 at $p = 0$ to 0.0056 at $p = 0.5$ and 0.0024 at $p = 1$. At 30k and 16k it is lowest near $p = 0.75$ –0.81 and rises again toward $p = 1$. At 4k it increases with p (0.270 $\rightarrow$ 0.326), because post-selection keeps only 6% of uniform shots. At the p that reproduces the device's retention ($p \approx 0.81$), the mixture's mean (0.0023, 0.017, 0.075, 0.295) is below the device at every budget. The device nevertheless lies inside the mixture distribution at 50–16k (85.7th, 91.6th and 88.7th percentiles of 1000 replicas) and at its far upper edge only at 4k (99.9th; one replica lies higher).

Line shape. On the line shape (the relative L_1 error of Sec. 6), too, noiseless sampling does worse at the two higher budgets. Its reconstructions give 0.566 ± 0.032 at 50k (all 80 replicas above the device) and 0.630 ± 0.043 at 30k (device at the 1.25th percentile). At 16k and 4k the device is worse (96th percentile, and above all 80 replicas). The uniform in-sector sampler gives 0.014 at 16k and 0.90 at 4k, below the device at every budget.

A superseded noise model. A simpler model draws shots from $|\varphi|^2$ at the pooled retained count; it is kept in the repository as a superseded record. It underestimates the spread of Δ_0 by factors of 34 to 195 relative to the raw-matched noiseless replicas, and under its own threshold a sampler drawing exactly from $|\varphi|^2$ would exceed that threshold at every budget (modeled false-positive rate 100%). We use neither that model nor any z -score derived from it. A non-parametric bootstrap of the retained counts is biased upward (its 50k interval [0.015, 0.101] excludes the point estimate 0.0044) and is not used as a test.

The n_k channel. For the momentum distribution n_k of Sec. 6: in the doped $L = 6$ ground state ($N_\uparrow = N_\downarrow = 2$: density 2/3 per site, i.e. 1/3 per spin orbital, the value a fully flattened n_k takes; committed script `src/nk_falsifier.py`) n_k has a finite-size step $\max_k n_k - \min_k n_k = 0.91$ at $U/t = 4$ and 0.80 at $U/t = 8$; in the thermodynamic limit the one-dimensional step softens to a Luttinger power-law cusp. A reconstruction that flattens the dispersion moves $\bar{n}_k$ toward this filling value, while the true n_k keeps the step. (At half filling the chain is a Mott insulator and n_k is analytic, so this discriminant targets doped sectors, not the half-filled spectrum of Fig. 3.) The occupation n_k is a number operator in the momentum basis, read out diagonally after a number-conserving fermionic Fourier transform (a Givens network of $L(L-1)/2$ rotations per spin sector). The particle-number and spin post-selection that protects the coverage channel therefore survives the basis change. The loss of post-selection and the full-space trace-to-value bias that affect the bond-basis rotations do not arise; depolarization acts within the sector instead, as described next. In an in-sector depolarizing model with post-selection at $\varepsilon = 2.5 \times 10^{-3}$, the ~ 60 -CZ transform adds a flattening bias of $\approx 9\%$ of the step; this assumes all-to-all connectivity, before the fermionic-SWAP routing a real device needs, which adds gates, so we expect it to underestimate the bias on hardware. The device estimate $\hat{n}_k$ is not immune to noise. Depolarization flattens it toward the same filling value that a dispersion corruption does, so it discriminates only corruptions whose flattening exceeds the device's own, and it inherits the depolarization of the state

preparation. Readout error and coherent or leakage noise, which need not flatten toward filling, are not modeled, and post-selection costs shots.

The sealed n_k gate. A pre-registered on-device n_k test (doped $L = 6$ ring, $U/t = 8$, 12 qubits) was sealed on 2026-09-02 (manifest SHA-256 4962...ae80) after a simulated Stage-0 gate, a noisy simulation of the sealed circuits and estimators run before any device time is spent, passed at the manifest's noise floors, $T_1 \geq 150\mu\text{s}$ and $T_2 \geq 100\mu\text{s}$ on every qubit of the chain. The gate passed only at its fourth step, after two revisions of the noise model and then a replication, all made the same day, before the seal (data/2026-09-02_nk_stage0_gate.json, _v2, _v3 and data/2026-09-02_nk_stage0_v3R2.json). Version 1 returned PROCEED, a verdict that an internal audit then ruled a model artifact (relaxation omitted, a finite-twirl residue and a transpilation bug). Version 2 added relaxation and finite (32-instance) gate twirling to the noise model and returned STOP on three of its seven pessimistic rows. For version 3 an adjudication changed the model: a heavy-hex instead of a line layout; relaxation scheduled at noise floors that first appear in this version, the manifest's $T_1 \geq 150\mu\text{s}$ and $T_2 \geq 100\mu\text{s}$, in place of the exposure multipliers behind two of the version-2 STOP rows; and a pulse-identical twin circuit for the post-mitigation residual check. Version 3 still stopped, on one of its nine verdict rows ($\varepsilon = 3.5 \times 10^{-3}$, twirled, second seed, at the floors), where the residual check gave 0.026674 against the threshold 0.026667. The fourth step, v3R2, was not a free revision but a replication of that row over ten seeds, the two already run in version 3 included, under a rule fixed before any replica ran (pass if the mean plus 1.645 standard errors lies below the threshold); it passed, with mean 0.0239 and upper confidence bound 0.0260, although two of the ten replicas individually exceeded the threshold (0.026674 and 0.0293). The manifest was sealed on that pass. Two candidate calibration estimators had also been tried and rejected before the seal (both would fail the binding simulated configuration), and the gate records themselves call the Stage-0 outcome a design result, not a pre-registered test pass; the pre-registered test is the device run alone. This is the kind of pre-seal iteration that a self-attested seal cannot exclude (Sec. 5); here it is documented in the committed records. The pass had little margin: at the manifest floors the residual check ran at 74–100% of its threshold.

The single permitted amendment would have relaxed these floors to $T_1 = 100\mu\text{s}$ and $T_2 = 70\mu\text{s}$, provided that three pre-committed simulated verdict rows all passed. Why the floors were to be relaxed, rather than the experiment run under the sealed manifest, is not stated in the committed record. The manifest's day-of rule forfeits any window in which no 12-qubit chain meets the floors on every qubit (with no chain CZ error above twice the device median), but no day-of chain check is committed, and the experiment was not run under the sealed manifest either. The worst-admissible row ($T_1 = 100\mu\text{s}$, $T_2 = 70\mu\text{s}$, two-qubit error parameter $\varepsilon = 3.5 \times 10^{-3}$) failed on 2026-09-05 at the tightest per-qubit readout cap of a pre-committed ladder (1.8%; the two looser rungs, 3.0% and 2.5%, are partial rows taken from run logs, with every recorded replica above the threshold): the upper confidence bound of the sealed post-mitigation residual check was 0.0287, against the sealed threshold 0.0267. A row identical except $\varepsilon = 2.85 \times 10^{-3}$ passed (0.0258). In this harness ε is a compound knob that moves the depolarizing and coherent-ZZ components together, so the pair of rows does not locate a physical two-qubit error rate. The forecasts above and in Sec. 6 assume $\varepsilon = 2.5 \times 10^{-3}$; the failed gate row used 3.5×10^{-3} . The amendment was denied, the original floors stand, and the experiment has not been run. An on-device n_k test, like the off-diagonal catch, awaits lower-error hardware.

The verdict file of the amendment gate, data/2026-09-05_nk_v4_verdict.json, was corrected and re-sealed on 2026-09-05, the day of the verdict (current SHA-256 7849...325a). Its first digest (f1b7...be3d) failed verification because it had been

1991 computed over LF line endings while the file was written with CRLF. In the same correction its
 1992 finding text, which had asserted that tightening the readout cap did not move the residual
 1993 check (the ladder in the same file shows a 14.2% reduction against the 18.4% needed), was
 1994 rewritten by hand, and caveats on attributing the failure to a single noise axis were added.
 1995 The verdict itself, amendment denied, is unchanged: the file's corrections record lists only
 1996 the text and the digest.

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